

Re-visiting the Political Economy of the Rise and Fall of the Unidad Popular: Towards a Global Political Economy Approach

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Abstract

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Contents

1	Introduction	2
2	Contested Approaches to the Political Economy of the Rise and Fall of the Unidad Popular	4
2.1	Classical Dependency Theory: Interrupted Programme	5
2.2	The Rise of the Chicago Boys and the Neo-liberal School	7
2.3	The Neo-Structuralist School: Selective Sanitization	8
2.4	Historical Political Economy: Methods Under Inherited Macro Theory	10
2.5	The Revival of Dependency Theory as a Research Programme	11
2.6	Epistemological Critique: The Illusion of Causality, Narrative License, and Col- lider Traps	12
3	World Money, Financial Subordination, and the Peripheral Central Bank Balance Sheet	13
3.1	World Money, the Universal Equivalent, and the Monetary Expression of Value .	13
3.2	The Bretton Woods Architecture and Peripheral Balance-of-Payments Constraints	14
3.3	The Peripheral Central Bank Balance Sheet: From Reserve Adequacy to Balance- Sheet Solvency	14
3.4	Money Endogeneity vs. Exogeneity and the Peripheral Paradox	15
3.5	Financial Fragility Reinterpreted: State-Dependent Solvency of the Peripheral Central Bank	16
4	Data Sources and Variable Construction	17
4.1	Monthly Panel: Sources and Sample Windows	17
4.2	Variable Construction	18
4.3	Data Adjustments	18
4.4	Integration Order of Variables Included in Empirical Exercises	18
5	Historical Background, Stylized Facts, and Empirical Results	19
5.1	Historical Background: The Rise of the Unidad Popular	19
5.2	Stylized Facts of Peripheral Accumulation, Distribution, and External Constraints	23
5.2.1	The Material Bounds of Peripheral Transition: An Accounting Anchor . .	23
5.2.2	Long-Span Balance of Payments Accounting: Reserve Accumulation and the 1971 Capital-Flow Reversal (1958–1975)	25
5.2.3	Terms of Trade, Capacity to Import, and Real Structural Imbalance (1960–1980)	28
5.2.4	Nominal Dynamics: Endogenous Credit Money and Multiplier Compression	31
5.2.5	Real Production Divergence and Chronological Shock Sequences	32
5.2.6	Synthesis and Econometric Hand-Off	34
5.3	Empirical Exercise 1: Toda–Yamamoto Directional Precedence and Channel Dis- crimination	34
5.3.1	Econometric Framework: Toda–Yamamoto Levels Augmentation	34
5.3.2	Integration Order and Levels Augmentation ($d_{\max} = 1$)	37
5.3.3	Modular Systems and Specification Mapping	37
5.3.4	Empirical Findings across Transmission Channels	39
5.3.5	Synthesis: Comparison of Theoretical Predictions and Directional Evidence	42

5.3.6	Residual Diagnostics and Dynamic Stability	42
5.4	Empirical Exercise 2: Non-Linear Dynamics under Peripheral Central Bank Solvency. A Threshold VAR Approach	47
5.4.1	The Threshold VAR Specification and Structural Identification	47
5.4.2	Endogenous Threshold Estimation and Residual Bootstrap Linearity Testing	47
5.4.3	Regime Parameter Estimates and Dynamic Stability Properties	47
5.4.4	Dynamic Generalized Impulse Response Functions (GIRFs)	47
5.4.5	Formal Multiplier Difference Hypothesis Tests	47
6	Political Economy Discussion, Global Implications, and Concluding Synthesis	47
6.1	The Political Economy of Peripheral Solvency and Subordinated Money	47
6.2	The Unidad Popular in the Global Structural Crisis of the 1970s	48
6.3	Policy Implications for Contemporary Peripheral Sovereignty	48
6.4	Concluding Synthesis: Toward a Progressive Research Programme	49

Appendices	50
Appendix A Formalization of Solvency Ratio and Balance-of-Payments Accounting	50
A.1 The Solvency Ratio in Nominal Terms	50
A.2 Current and Capital Account Decomposition	51
A.3 Derivation of the Discrete Solvency Law of Motion	51
A.4 Analytical Convexity and Non-Linear Reserve Exhaustion	52
A.5 Constraint-Binding Conditions and Structural Minskian Regimes	53
Appendix B Archival Data Provenance, Variable Dictionary, and Splicing Protocols	53
B.1 Monthly High-Frequency Variable Ledger	54
B.2 Monetary Reform Conversions and Splicing Mechanics	55
Appendix C Robustness: Multivariate Causal Discovery (PCMCI)	55
C.1 PCMCI Conditioning Methodology	55
C.2 Full-System Causal Discovery Results	56
PCMCI Algorithm: Technical Detail	57

1 Introduction

1.1 In the history of economic thought, progress rarely unfolds as a linear, cumulative ascent toward universal truth. As (?) argues, theoretical advance throughout the history of economic thinking is governed by a *competitive views*: an enduring contest among rival paradigms rooted in fundamentally distinct conceptualizations, analytical categories, and contested world visions. For over half a century, the history of Chile’s Unidad Popular (UP, 1970–1973) under Salvador Allende has been interpreted through an uncritical cumulative lens hegemonized by mainstream economics (??). Mainstream accounts under neoclassical market-clearing assumptions and monetarist approach to the dynamics of the balance of payments and inflations precluding the role of distributive conflict, have treated the Chilean breakdown as a cautionary tale of populist fiscal mismanagement. Yet beneath this orthodox consensus lies an unexamined failure of conceptualization: treating peripheral capitalism as a closed monetary system, blind to the structural subordination to the dynamics of the world market in a hierarchically organized world of states featured by and balance-of-payments constraints and financial subordination (??).

1.2 This chapter investigates the causal relation between money creation and inflation, and how central bank reserve solvency mediates the inflationary transmission of domestic money creation during a peripheral socialist transformation amidst the global crisis of the north atlantic fordism accumulation regime (?). If money creation mechanically drove inflation, why did base money expand by over 100% in 1971 while monthly inflation remained quiescent at 1.5–2.0% and annual inflation fell from 35% to 22%? The answer lies in balance-sheet solvency: in 1971, the Central Bank possessed \$400M in inherited foreign reserves and industry operated with ample idle capacity ($\mu \approx 82\%$), absorbing liquidity through real transactions demand. Inflation pass-through did not take place in 1971—it emerged only in mid-1972 once reserves fell below \$30M and the international credit blockade severed revolving commercial trade lines. Broad money growth functioned as an endogenous accommodation valve through which the central bank mediated an balance-of-payments blockade amidst an intense distributive conflict. Contesting the monetarist thesis of ? and the Austrian interpretation of ?, I formalize the central bank balance sheet through the Solvency Ratio—the ratio of convertible foreign reserve assets to domestic monetary liabilities ($\text{SolvR}_t \equiv \frac{e_t \cdot IR_t}{M_t}$)—identifying it as the state-dependent threshold governing monetary transmission. My formal threshold tests decisively reject linear interpretations of the relation between money creation and inflation [PLACEHOLDER FOR FINAL RESULTS SUMMARY].

1.3 Allende’s thousand days unfolded within the tectonic shifts of the global structural crisis of fordism capitalism during early 1970s. The unilateral suspension of dollar-gold convertibility by the United States on August 15, 1971 dismantled the Bretton Woods system, restoring dollar hegemony (?). Simultaneously, the secular post-war decline of the corporate profit rate across the advanced capitalist core (???) narrowed the geopolitical room for peripheral developmentalism. The collapse of the Chilean socialist transition represented the structural collision of domestic redistributive aspirations of popular movements as response to the failure of the developmentalist state in its promise of social inclusion at the shadow of the break down of Bretton Woods international monetary order.

1.4 Mainstream and neoclassical syntheses characterize the Allende administration through the paradigm of “*macroeconomic populism*” (??). In this interpretation, redistributive governments inevitably trace an unforced four-phase cycle: initial reactivation through idle capacity, the emergence of bottleneck shortages, unsustainable fiscal and monetary financing, and final economic collapse. By treating the central bank as an unconstrained printing press and monetary emission as an unmediated political choice, this literature commits two deep epistemological failures. First, it succumbs to what ? formulate as the *illusion of causality*—a cognitive heuristic that infers direct causal generation from the mere temporal coincidence of salient events (the

1970 election, 1971 wage readjustments, and 1972–1973 inflation) while ignoring non-contingent external covariates. Second, it exercises what ? classify as *narrative license*—the rhetorical escalation of observational and correlational snapshots into ironclad causal laws to satisfy academic demands for compelling monocausal parables. By analyzing aggregates in an institutional vacuum, mainstream accounts conflates correlation by causality overlooking the methodological underpinnings of the so-called "credibility revolution and causal identification. Concretely, this conflation manifest itself by misidentifying the symptoms of re-distribution, external blockade, and foremost the exogenous consequences in the balance of payments in the Chilean economy of the unintended unilateral decision by the Federal Reserve of the United States closing of the convertibility window and the collapse of the Bretton Woods international financial regime with the mechanical results of domestic monetary policy.

1.5 To discriminate between these contested claims on rigorous empirical ground, I deploy an applied Lakatosian evaluation of contested scientific research programmes in economic thought (???). This methodological framework appraises competing programmes by distinguishing their immutable theoretical axioms (*hard core*) from the auxiliary specifications of their empirical models (*protective belt*), evaluating whether empirical anomalies are absorbed through theoretically progressive extensions that predict *novel facts* or through ad hoc degenerating immunizing stratagems: The debate over the macroeconomic destabilization of the Unidad Popular can be structured into three competing research programmes. The monetarist programme (?) anchors its hard core in exogenous base money creation, fractional-reserve multiplier causality, and long-run monetary neutrality in a closed system. Confronted with the empirical anomaly of 100% monetary growth coinciding with non-inflationary output expansion in 1971, its protective belt relies on ad hoc narrative accounts that dismiss the \$294 million external financing cutoff, the U.S. Eximbank embargo, and domestic distribution bottlenecks as secondary disturbances. The Austrian business-cycle programme (?) similarly emphasizes domestic distortions, positing that suppressed interest rates and credit expansion generated intertemporal malinvestment and coordination breakdown.

Under the umbrella of dependency theory understood as a *research programme* (?), rooted in international currency hierarchies, balance-of-payments constraint, endogenous credit money, and exogenous factors in the periphery as expresion of the dynamics of financial subordination (?), I define a new dimensionless index of central bank foreign reserve solvency as an endogenous state variable ($\text{SolvR}_t \equiv e_t \cdot \text{IR}_t / M_t$). This framework allows to identify state dependency of peripheral Central Bank solvency/insolvency in terms of being capable to defend the stability of its own currency. When the central bank maintains a solvent reserve of international currency buffer ($\text{SolvR}_t > 0.21$), domestic liquidity expansion is capable to mobilize idle capacities without triggering inflation, and commercial credit systematically leads primary monetary emission. Progressive pass-through into consumer prices and currency depreciation occurs only when external financial asphyxiation forces the central bank across the critical threshold into balance-sheet insolvency ($\text{SolvR}_t \leq 0.21$), compelling defensive monetary accommodation to sustain enterprise payrolls under severe import starvation.

1.6. This chapter deepens the theoretical foundations established in Chapter 2. Whereas Chapter 2 identified the external boundary of accumulation through an annual balance-of-payments threshold that throttles mechanization under unbalanced growth, this chapter uncovers the dynamic balance-sheet mechanisms through which that external constraint operates at high frequency. To overcome the temporal aggregation bias and closed-economy endogeneity pervasive in annual accounts (??), my research design assambles two tier data set at annual and monthly frequency time series from multiple secondary sources.

The annual time seris data set (1920–2010, $T = 91$) draws functional income shares from ?, balance-of-payments accounts from ?, and disaggregated net capital stocks from the Perpetual

Inventory (GPIM) dataset developed in Chapter 2. At this annual frequency, the Austrian mal-investment hypothesis (H_2) fails decisively: gross fixed investment in machinery and equipment contracted (-2.3%) during the 1971 expansion while structural capacity utilization surged to an all-time peak of $\mu_{1971} = 98.44\%$, presenting empirical evidence favorable to the hypothesis of reactivation rather than intertemporal capital misallocation.

To isolate the transmission belts of the subsequent breakdown, the high-frequency monthly tier (1960–1980, $N = 250$ continuous months) integrates primary Central Bank of Chile archival data accessed through the Application Programming Interface (API) of the *Base de Datos Estadísticos* (BDE) of the Chilean Central Bank (BCCH) including Central Bank base monetary emission H and $M1$ including deposits and less liquid measures of money, indices of gold, copper and wholesale US prices, Indexes of export, import, manufacturing, and minning activity series here gathered by (?) and included in the data set to built import capacities and structural imbalances indices. IMF Central Bank reserve assets historical statistics provided stocks of international reserves at monthly frequency that cannot find elsewhere. Following the August 1971 close of FED gold convertibility window, my proxy of Prebisch’s purchasing power capacity to import (M_t^{cap}) fell to 27.7, driving the Real Structural Imbalance Ratio ($\Theta_t \equiv M_t/M_t^{\text{cap}} \times 100$) to an unsustainable peak of 283.4 in May 1973. In our Toda–Yamamoto and Threshold VAR systems, this physical import strangulation governs the regime switch: once foreign credit embargoes deplete foreign exchange reserves below the identified critical solvency threshold ($\text{SolvR}_t \leq 0.21$), external price shocks penetrate domestic retail prices immediately, compelling defensive monetary accommodation to maintain enterprise payrolls amidst supply paralysis.

1.7 REVISE ONCE FINISHED OTHER SECTIONS The remainder of this chapter is structured as follows. Section 2 surveys the contested traditions in Chilean political economy and formalizes the epistemological critique of orthodox causality. Section 3 develops the theoretical framework of World Money, monetary endogeneity, and the open-economy transposition of Minskyan solvency regimes to the peripheral central bank. Section 5 grounds the inquiry in the historical political economy of the pre-1970 accumulation regime, details the multi-frequency data architecture, specifies the model restrictions, and implements the econometric evaluation across the four-stage empirical strategy. Section 6 discusses the political economy of peripheral solvency, situates the UP within the global structural crisis of the 1970s, and outlines lessons for contemporary monetary sovereignty.

2 Contested Approaches to the Political Economy of the Rise and Fall of the Unidad Popular

2.1 No single framework commands consensus on the political economy of the rise and fall of Chile’s Unidad Popular (UP). Instead, rival explanatory traditions have structured the field, each emerging under unequal political conditions and distinct historical contexts. Transnational networks linking the Pontificia Universidad Católica de Chile (PUC) to the University of Chicago formed the neoliberal school, which subsequently acquired state power through the dictatorship that followed the 1973 coup. Neo-structuralism emerged under the dictatorship as a technocratic opposition to Pinochet’s regime—specifically in the aftermath of the 1982 debt crisis—before rising to government power and establishing itself as the mainstream economic paradigm during the post-dictatorship order. Contemporary accounts in the field of Historical Political Economy (HPE) entered the debate much later, bringing powerful tools of causal identification to historical questions. However, they did so within a disciplinary environment already shaped by earlier causal claims taken as given, while perspectives anchored in distributive conflict and structural dependency were seldom considered or actively excluded. Dependency theory occupies a different position altogether. It was the intellectual landscape from which

the UP programme itself emerged, and the coup institutionally shattered it through violent repression before it could resolve the theoretical problems it had opened up.

2.2 These traditions did not simply coexist. They succeeded one another through political conflict without genuinely opposing each other on the intellectual terrain, despite stark declarations regarding the intellectual bankruptcy of the structuralist-dependency school:

“...the macroeconomic view of the Unidad Popular failed to recognize that their policies would only be able to generate a burst of economic activity that would be unsustainable in the medium term... Needless to say, this view of the way the economy functioned ignored many of the key principles of traditional economic theory.”
(?, p. 16–17).

Each identified a part of the problem, yet none integrated external transmission, distributive conflict, and macroeconomic dynamics into a single explanatory sequence. This section critically evaluates these traditions. It reconstructs the historical conditions under which each approach became authoritative, clarifies the mechanisms each privileged, and shows which questions each rendered visible and which it displaced, sanitized, or foreclosed. It begins with classical dependency theory, which posed the problem the chapter as a whole seeks to resolve more sharply than any approach that followed.

2.3 Consequently, this fragmented intellectual legacy has resulted in a striking scarcity of rigorous empirical evidence concerning the macroeconomic dynamics of the UP’s collapse. While the political and institutional histories of the period are extensively documented, the macroeconomic debate has largely relied on retrospective narratives, analysis of correlation without causality, or ideological posturing rather than methodologically robust quantitative analysis. There is a critical lack of empirical studies employing adequate econometric methods to isolate and measure how allegedly “*populist interventions*” —such as exchange rate overvaluation, fiscal monetization, and generalized price controls—transmitted distributive conflicts and external shocks into systemic macroeconomic failure. By leaving these macroeconomic transmission mechanisms empirically untested, the literature has remained trapped in theoretical stalemates. This chapter seeks to bridge this methodological void by moving beyond ideological retrospection, advancing more rigorous empirical identification to the macroeconomic data of the UP era to trace the anatomy of its economic unraveling.

2.1 Classical Dependency Theory: Interrupted Programme

2.3 Dependency theory did not approach the Unidad Popular from the outside. It helped constitute the intellectual horizon within which the UP became thinkable. Its formation in Chile brought together two lineages that converged, institutionally and theoretically, in Santiago during the long sixties. From one side came the Monthly Review tradition, beginning with ?’s analysis of economic surplus under monopoly capitalism and radicalized by ? into the development-of-underdevelopment thesis: underdevelopment was not a residual stage but an active product of incorporation into the world capitalist system. From the other came the structuralist school built within the headquarters of the Economic Commission for Latin America (ECLAC), led by Raúl Prebisch, which had already framed Latin American development through centre–periphery relations, external bottlenecks, and the historical asymmetries of international trade.

2.4 At the Centro de Estudios Socioeconómicos (CESO) of the Universidad de Chile, these two lineages fused into an active research programme. Following the 1964 military coup in Brazil, the arrival of Theotônio dos Santos and Vania Bambirra as political exiles in Santiago catalyzed

this institutional convergence (?). Collaborating at CESO alongside André Gunder Frank, Orlando Caputo, and Roberto Pizarro—and maintaining close dialogue with Ruy Mauro Marini in Concepción—they transformed dependency from a general critique of peripheral capitalism into an effort to specify the historical forms linking regional subordination, capital accumulation, and political strategy (????). Far from remaining an academic abstraction, this intellectual collective directly shaped the policy architecture of the Chilean left: among Dos Santos and Bambirra’s closest collaborators and disciples were Sergio Ramos and Alexis Guardia, the primary economic drafters of the Unidad Popular programme representing the Communist and Socialist parties, respectively.

2.5 The programme’s analytical wager was clear. Peripheral capitalism followed a distinct developmental dynamic structured by external constraints, a domestic bourgeoisie structurally incapable of autonomous national accumulation, and reproduction patterns that continually re-entrenched subordination. Within this framework, socialist transformation appeared as an objective developmental necessity grounded in the physical limits of peripheral capital formation. Yet the tradition never achieved theoretical unity. Marini’s theory of superexploitation pushed the analysis toward a world-market logic where the systematic compression of labor below the value of its reproduction conditioned domestic capital accumulation (?). Conversely, ? insisted that dependency took historically contingent forms governed by domestic class alliances and the relative autonomy of the capitalist state. This tension proved decisive. At stake was whether domestic political mediation constituted an independent object of analysis or simply mediated world-market imperatives. Classical dependency posed the question with great force but never resolved it.

2.6 That unresolved tension did not prevent the tradition from producing the richest contemporaneous political-economy accounts of on the Political Economy of the Rise and Fall of the UP. ?’s *Allende’s Chile* provided the seminal participant-observer account, correctly locating the material foundation of the external constraint in Chile’s extreme physical import dependency: over 90% of capital equipment was imported, and more than 95% of machinery in large-scale copper mining originated in the United States. Yet De Vylder’s framework suffered from a fatal analytical concession. Confronted with orthodox arguments attributing the crisis entirely to domestic mismanagement, De Vylder retreated into an internal-blame explanation, asserting that:

“the problem was the insatiable imports, not the US blockade nor the lack of credit”
(?, p. 106).

To sustain this concession, De Vylder pointed to gross nominal credit lines extended by the Soviet Union and Eastern Europe (> \$400 million) as ostensible proof that external financing remained abundant, conflating gross nominal commitments with usable, convertible liquidity.

2.7 ? overturned De Vylder’s concession by uncovering the *qualitative non-fungibility of international credit*. Drawing on primary Central Bank foreign exchange ledgers, she demonstrated that bilateral socialist credits were non-convertible tied lines earmarked for specific industrial equipment. They could neither purchase patented American spare parts for Kennecott copper haul trucks nor settle immediate spot cash contracts for wheat and petroleum on Western commodity markets. Concurrently, private U.S. commercial banks slashed revolving short-term trade credit lines from \$220 million to \$35 million, while the Export-Import Bank froze disbursements. These debates around balance-of-payments constraints did not unfold in isolation. They formed part of a vibrant contemporaneous intellectual landscape captivated by the *vía chilena al socialismo*—the unprecedented historical attempt to achieve revolutionary socialist transformation through bourgeois democratic institutions. Across this field emerged a rich empirical corpus tracking the crisis from complementary angles: distributional conflict across successive

administrations, worker and peasant self-organization from below, the logistical spare-parts embargo, the monetary and balance-of-payments mechanisms through which external pressure became domestic crisis, and the strategic dilemmas of the state under conditions of emerging dual power (????????). The enduring methodological strength of classical dependency lay precisely here: it refused to treat external imperial intervention and domestic class struggle as competing explanations, approaching them instead as interacting, co-constitutive dimensions of the same historical process.

2.8 That achievement also marked the analytical limit dependency theory could not overcome in its classical form. Bounded by a bilateral nation-state framework (the U.S. versus Chile), it omitted the international monetary architecture of Bretton Woods and global stagflation, while lacking a formal macroeconomic specification linking external balance-sheet shocks to domestic distributive conflict and capacity utilization. Yet the defeat of dependency theory was twofold. Politically, the coup dismantled the institutions, networks, and research centres in which the programme had been elaborated. Intellectually, the interruption arrived before the tradition could work through its own internal tensions. No superior explanatory framework refuted it in a scientific terrain. It broke under conditions in which theoretical incompleteness coincided with the collapse of Chile's liberal democracy, the failure of democratic socialism to resolve the crisis under organized domestic and international sabotage, and the installation of a subsequent civic-military dictatorship. Subsequent approaches entered a field already reorganized by force, not reason. The next school to become authoritative in Chilean political economy was not the one that best resolved the problem dependency had identified, but the one that arrived coordinated with the counterrevolutionary historical bloc that constituted the reconstruction of the state and civil society ?.

2.2 The Rise of the Chicago Boys and the Neo-liberal School

2.9 The neoliberal school did not emerge from within the intellectual world of the Unidad Popular as one competing interpretation among others. It entered through a different route: as a transnationally trained programme of economic reconstruction that acquired authority through the counterrevolutionary reorganization of the Chilean state. Its genealogy is well established. The 1956 agreement between the Pontificia Universidad Católica de Chile and the University of Chicago opened the institutional channel through which Milton Friedman, Arnold Harberger, and Gary Becker trained a cohort of Chilean economists in monetarism, public finance orthodoxy, and neoclassical price theory. By the late 1960s, this network had evolved from an academic outpost into a politically embedded project, closely aligned with business associations and the Chilean right, oriented toward the systemic restructuring of Chilean capitalism (???).

2.10 That programme found its canonical expression in *El Ladrillo*². The text codified in advance the macroeconomic blueprint on which the post-UP order would be constructed. It operated as a counterrevolutionary vanguard design: an operational policy manual prepared prior to the coup to direct the reconstruction of Chilean capitalism the moment the popular government was overthrown.

2.11 Its diagnosis identified a familiar catalogue of structural problems: chronic inflation, recurrent external deficits, agricultural stagnation, low productivity growth, and distortions associated with state-led industrialization. Several of these observations captured real tensions within Chilean accumulation, and they merit serious engagement. The argument that prolonged relative-price controls hindered export diversification and exacerbated balance-of-payments vul-

²*El Ladrillo* ("The Brick") was drafted clandestinely prior to the September 1973 coup by Chicago-trained economists coordinated by Emilio Sanfuentes, with contributions from Sergio de Castro and Pablo Baraona, in liaison with naval officers and the leadership of the private employers' association *Sociedad de Fomento Fabril* (SOFOFA) (?).

nerability cannot simply be dismissed, nor can the recognition that import-substitution industrialization had reached severe structural exhaustion (?). Yet the broader theoretical hypothesis holding these observations together was much weaker: the claim that the crisis of the UP merely culminated a secular drift toward the excessive socialization of the economy. That thesis recoded a complex conjunctural crisis into a normative indictment, naturalizing market discipline rather than explaining the empirical dynamics of the breakdown.

2.12 Here lay the neoliberal school's constitutive limit. It registered macroeconomic imbalances only by systematically stripping them of their political-economy. Functional income distribution appeared not as structural class conflict, but as a technical distortion of price signals. The external crisis was treated as the endogenous consequence of fiscal and monetary mismanagement, never as the relational interaction between domestic class struggle and international financial subordination and destabilization. The October 1972 employers' lockout, the private commercial credit blockade, the spare-parts embargo, and covert foreign destabilization could not enter the account without undermining the framework's own core hypothesis and axioms. Neoliberalism did not omit these factors by accident. Its explanatory logic required them to disappear, since acknowledging them would have undermined the premises that the crisis proved the inherent impossibility of socialist transformation.

2.13 The neoliberal interpretation became authoritative under historical conditions that were not primarily intellectual. Its ascendancy reflected no superior theoretical integration of external transmission, distributive conflict, and macroeconomic dynamics. A programme formulated before the coup entered a terrain reorganized by military force, and the dictatorship installed it as the common sense of national economic reconstruction. That restructuring violently dismantled collective bargaining, political mediation, and labor organizations while relying, paradoxically, on the centralized administrative apparatus of the developmental state it claimed to dismantle (??). The neoliberal school was less a scientific solution to the developmental dilemmas of the periphery than a victorious political project presenting ruling class backlash as economic science. Its enduring legacy lies precisely here: it converted a conjunctural crisis resolved by state terror into a universalized economic doctrine asserting the impossibility of any non-market alternative (?).

2.3 The Neo-Structuralist School: Selective Sanitization

2.14 The neo-structuralist school was an intellectual accomplishment of the post-dictatorship order rather than a simple continuation of classical structuralism. It established its institutional base during the dictatorship, primarily within the CIEPLAN think tank, where former structuralists and democratic opposition economists critically monitored the Chicago Boys' reforms while formulating an alternative programme for the democratic transition. That genealogy explains much of the school's subsequent authority. The CIEPLAN economists were more than political moderates who had tempered their commitments under authoritarian rule (?). They constituted the modernizing technocratic fraction through which the post-authoritarian centre-left could challenge orthodox monetarism while operating strictly within the boundaries of market-disciplined, export-oriented peripheral capitalism. Neo-structuralism entered the field as a politically functional renovation calibrated to manage Chile's subordinate insertion into the globalized economy (???).

2.15 The decisive theoretical operation of that renovation was the abandonment of the classical surplus approach. Where Prebisch and the foundational ECLAC tradition had centered the production, appropriation, and reinvestment of the economic surplus under peripheral conditions, neo-structuralism recast underdevelopment into the mainstream idiom of market imperfections, institutional bottlenecks, and pragmatic state regulation. This shift did not merely modernize

structuralism; it altered its analytical object. The external constraint remained, but strictly as a macroeconomic financing ceiling rather than as a structural relation of surplus extraction and dependent reproduction. By reducing a theory of historically structured subordination to a technical problem of policy calibration, the school depoliticized the external barrier. As ? demonstrates, without the surplus approach, the external constraint survives only in a thinned accounting form: legible as a foreign exchange deficit, but stripped of the class relations that generate and sustain it. Neo-structuralism did not forfeit empirical rigor; it forfeited the conceptual framework required to theorize the balance of payments and distributive conflict as unified moments of the same accumulation process.

2.16 The school's account of the UP crystallized most authoritatively in the work of Sergio Bitar. Originally published in 1979, ? provided the most sophisticated neo-structuralist reconstruction of the popular government. Its principal merit lay in rejecting an exclusively monetary explanation of the external accounts. Bitar sought to integrate foreign exchange scarcity, political opposition, and institutional transformation into a single dynamic narrative. His formulation of a "viability area" remains a genuine analytical contribution, demarcating the narrow economic margin within which the UP programme could have advanced without precipitating macroeconomic collapse. Yet that contribution also established the school's defining silence. Across multiple editions, Bitar never substantively engaged the contemporaneous dependency literature that had developed across the same historical terrain. Foundational accounts by ? and ? were already widely available, yet Bitar avoided confronting them. This silence was neither a bibliographic omission nor a conspiracy. It reflected the strategic selectivities accompanying the abandonment of the surplus approach (?), marking the boundary at which dependency's central questions were cordoned off from post-dictatorship economic discourse.

2.17 That theoretical cordon became hegemonic through the populist-macroeconomic paradigm. ? codified the UP into a stylized four-phase trajectory of initial reactivation, emerging bottlenecks, severe destabilization, and ultimate collapse. ? then generalized this sequence into a regional archetype of "macroeconomic populism." This paradigm achieved wide influence by providing a clear narrative, identifiable turning points, and cautionary lessons that circulated throughout Latin America during its democratic transitions. Yet it operated through a definitional foreclosure: by classifying redistributive policies categorically as "*macroeconomic populism*," the framework ruled out the possibility that wage-led growth could ever be structurally sustainable under peripheral conditions. ? subsequently cemented this interpretation as the common sense of the UP: an unforced tragedy driven by policy mismanagement and political polarization. The result was a reading self-critical in tone but exculpatory in effect, reducing the crisis to domestic excess while treating external financial blockades and domestic logistical sabotage as secondary noise. The contrast between Sergio ? and Joseph ? illustrates this epistemological divide: the former treated inflation as a structural contradiction internal to a socialist transition under acute class struggle, whereas the latter reduced the crisis to an engineering problem of disequilibrium management and stabilization failure.

2.18 Neo-structuralism thus substituted a technocratic problem of governance for the conflict-centered political economy generated during the UP itself (???). The school's analytical limits were not accidental blindspots; they were strategic exclusions built into the conditions of its post-authoritarian legitimacy. Its primary historical function was to preserve the structuralist language of external constraints while adapting it to the macroeconomic governance of a neoliberalized economy, sustaining two decades of centre-left political stability. In doing so, it transformed the thousand days of Allende's government into a moral tale of redistributive overreach. Neo-structuralism became the progressive framework through which the post-dictatorship order could be managed and stabilized without reviving the class-conflict categories that made earlier structuralist and dependency traditions dangerous to ruling-class hegemony (?).

2.19 This critique does not dismiss the empirical contributions of scholars such as Bitar or Meller; it specifies the political-economic renounces and strategic selectivities as consequence of their synthesis. The coordination of monetary, structural, and political variables into a coherent narrative remain important contributions to the debate. However, the systematic exclusion of class struggle for the economic surplus was the price of its institutionalization during the democratic transition and the post-dictatorship period. That compromise defines neo-structuralism's enduring legacy. It established the default macroeconomic memory inherited by subsequent scholarship, leaving historical political economy equipped with sophisticated empirical tools, but lacking a coherent theory of the structural mechanisms at work.

2.4 Historical Political Economy: Methods Under Inherited Macro Theory

2.20 Historical Political Economy (HPE) represents a genuine methodological advance in the study of economic history. By applying the credibility revolution's identification toolkit to historical inquiry, it has introduced a level of empirical precision rarely achieved in earlier debates. Its institutionalization around the *Journal of Historical Political Economy* and the *Oxford Handbook of Historical Political Economy* reflects a broader effort within economics to reclaim historical questions without sacrificing rigorous causal identification (??). Applied to twentieth-century Chile, this agenda has generated an impressive corpus examining price controls, real wages, land redistribution, and political violence.

2.21 Yet HPE did not generate the archival terrain on which it operates. That foundation was constructed by a parallel renewal in Chilean economic history, which has reconstructed long-run time series on prices, trade, living standards, and distribution. The archival work of Mario Matus, Manuel Llorca-Jaña, Claudio Robles, and their collaborators established the quantitative records that made micro-econometric identification feasible. Within this archival foundation, ? is foundational for the empirical architecture of this dissertation: the long-run historical accounts of inequality compiled there directly underpin the functional wage-share series (ω_t) harmonized by ? and utilized throughout Chapters 2 and 3. By situating the Unidad Popular within a secular century-long cycle of distribution and explicitly demonstrating how political coercion shaped income shares, Rodríguez Weber provides a crucial historical bridge between institutional economic history and structural macroeconomic questions.

2.22 In contrast to descriptive economic history, HPE proper focuses on estimating discrete causal effects. ? exploit firm-level records to identify the credit-allocation shocks induced by U.S. financial warfare, while ? trace contemporary Mapuche territorial conflict to the localized geography of agrarian expropriation and military counter-reform. ? provide a distinctive contribution: their high-frequency event study of the Santiago stock exchange demonstrates that financial wealth-holders responded systematically to shifts in the political threat to private property, corroborating classical political economy's focus on capitalist investment strikes. These studies represent sophisticated empirical contributions. However, they perform fundamentally different analytical work: some build archival ledgers, others isolate localized causal effects, while few—if not ? alone—examine macroeconomics and institutions through a theoretical framework centered on the structural power of capital over the economy.

2.23 This analytical boundary is most evident in the work of Felipe González and his collaborators, who define the contemporary frontier of Chilean HPE (???????). Their research is methodologically exemplary. They have unearthed primary archival records, formulated transparent quasi-experimental designs, and resolved contested empirical questions surrounding land reform, industrial requisitions, covert foreign intervention, and the institutional legacies of the military dictatorship. Yet this corpus also demarcates the field's constitutive limit: the macroeconomy appears primarily as a passive background upon which institutional interventions and

policy shocks are measured, rather than as an organic, contradictory system of capital accumulation, surplus extraction, and crisis. Even where the archival evidence touches systemic macroeconomic phenomena, the theoretical interpretation is routinely imported from post-dictatorship mainstream narratives. The result is scholarship of exceptional empirical precision that displaces political economy in the classical sense into an applied microeconomics of institutional interventions.

2.24 This boundary reflects the field's disciplinary formation. HPE arrived at the Chilean case after the historical clash between the dependency, neo-structuralist, and neoliberal paradigms had already been settled by state coercion. Consequently, it uncritically inherited the depoliticized macroeconomic vocabulary of the post-dictatorship consensus, which reduced the UP's crisis to a cautionary tale of macroeconomic populism and unforced balance-of-payments mismanagement. The research programme developed in this chapter builds upon HPE's archival and empirical rigor, while restoring what applied microeconomics omits: an integrated theoretical framework capable of modeling external balance-of-payments vulnerability, distributive conflict, and domestic monetary accommodation as mutually conditioning moments of peripheral capital accumulation.

2.5 The Revival of Dependency Theory as a Research Programme

2.25 The return to dependency theory is not an exercise in intellectual nostalgia. It recovers a research programme whose historical retreat signaled a profound collapse of the region's sociological imagination. As ? argues, Latin America's critical thinking swung like a pendulum from one type of absolute certainty to another: from the mechanical stagnationism of early dependency to the uncritical market fatalism of the post-dictatorship consensus. In that shift, the region's intellectual landscape was stripped of its capacity to conceive development as a problem of collective agency, structural transformation, and sovereign autonomy. Development was degraded from a political struggle over the production, appropriation, and reinvestment of the economic surplus into an administrative exercise in macroeconomic compliance. The analytical task today is neither to revive the rigid certainties of the 1970s nor to surrender to technocratic fatalism, but to reconstruct dependency as an empirically generative, progressive framework.

2.26 ? provides the epistemological architecture for this reconstruction by formalizing dependency theory through Imre Lakatos's methodology of scientific research programmes. Kvan-graven refutes the mainstream caricature of dependency as a monolithic dogma by demonstrating that the tradition is structured around a coherent, shared *hard core*: the premise that global capitalist accumulation systematically reproduces hierarchical structures between core and periphery that constrain the developmental space of subordinate social formations. The scientific vitality of the programme does not depend on dogmatic consensus regarding specific historical models; rather, it hinges on whether its *protective belt of auxiliary hypotheses* can generate *progressive problemshifts*—predicting novel empirical facts and specifying concrete causal mechanisms—rather than degenerating into defensive ad-hoc immunizations. By formalizing this distinction, she clarifies how dependency can assimilate new historical evidence within the context of the concrete realities of global capitalism without abandoning its core ontological commitments.

2.27 ? sharpen this epistemological reconstruction by distinguishing between *situations of dependency* and *mechanisms of dependency*: i.e., between historically specific descriptive configurations and the causal processes that reproduce external constraints. This distinction has energized a contemporary renewal in Santiago, where scholars have operationalized dependency theory to analyze rentier accumulation, export stagnation, and dependent financialization in

present-day Chile (????). Within this revival, recovering the historical memory of the Unidad Popular is not an exercise in political nostalgia, but an inquiry into the structural conditions under which sovereign control over the surplus and democratic self-determination were attempted and violently suppressed (??). Yet for the macroeconomic analysis of the UP itself, the crucial task remains unfinished: while classical accounts identified the concrete *situation* of Chilean vulnerability (import dependency and bilateral credit retaliation), they lacked a formal *mechanism* linking peripheral balance-of-payments asphyxiation to the structural crisis of the core’s international monetary order.

2.27 This missing transmission mechanism can be formulated by synthesizing the political economy of global dollar hegemony with the institutional mechanics of the dissolution of Bretton Woods. ? provides the structural foundation for why Bretton Woods collapsed: a macropolitical drive by U.S. elites to escape the discipline of gold convertibility, discipline domestic labor through international capital mobility, and establish an unconstrained fiat dollar standard. In turn, ? demonstrates how and when this rupture unfolded: rather than a predetermined grand design, closing the Federal Reserve convertibility gold window was a contingent institutional maneuver forced by IMF institutional veto points and European surplus resistance. The core executed an emergency exit on August 15, 1971, rapidly consolidating a floating dollar regime that eliminated its own reserve discipline while causing an exogenous shock to peripheral economies.

2.28 For Chile, this historical timing was decisive. Nixon’s unilateral closure of the gold window occurred exactly three days after the U.S. Export-Import Bank formally denied credit lines to the Unidad Popular on August 12, 1971. Peripheral accumulation and the “*via Chilena al Socialismo*” was thus caught in converging external constraints: the bilateral credit non-fungibility documented by ? colliding with the systemic dissolution of the global monetary regime. Incorporating this dual transmission into the protective belt of dependency theory resolves the theoretical ambiguities that stalled its classical formulation. It equips the tradition with an exogenous event (Nixon’s administration closing the Fed convertibility gold window), as a macroeconomic mechanism capable of explaining how external balance-sheet shocks triggered domestic supply paralysis and forced monetary accommodation, directly setting up the hypothesis tournament among contested approaches to economic thinking to be developed in this chapter.

2.6 Epistemological Critique: The Illusion of Causality, Narrative License, and Collider Traps

2.33 The persistent divergence among these explanatory traditions cannot be resolved through historiographical narrative alone. As ? establish, human causal reasoning is systematically vulnerable to the *illusion of causality*: a robust cognitive bias wherein observers detect strong causal links between co-occurring events even when the statistical contingency is zero or driven by unobserved common causes. In the economic historiography of the Unidad Popular, orthodox monetarist and Austrian accounts observe a conspicuous historical coincidence—the election of a socialist government in late 1970, expansionary fiscal-wage policy in 1971, and the explosion of inflation in 1972–1973. Relying on intuitive heuristics, they assign exclusive causal responsibility to domestic monetary issuance ($g_{M,t} \rightarrow \pi_t$), treating international monetary hierarchy, terms-of-trade shifts, and geopolitical destabilization as irrelevant background noise (??).

2.34 This causal illusion is reinforced by what ? identify as *narrative license* and *overreaching causal claims* in historical social science. Orthodox accounts dress up doctrinal priors as iron-clad empirical laws without testing alternative causal mechanisms nor specifying counterfactual identification paths. For example, ? assert an Austrian Business Cycle mechanism wherein

artificial credit expansion generated an unsustainable capital malinvestment boom. Yet they advance this claim without presenting positive evidence of gross fixed capital deepening (ΔK), ignoring that Chilean fixed machinery investment contracted during 1971 while capacity utilization reached historical peaks. Narrative license allows the theorist to impute unobserved microeconomic mechanisms (capital lengthening) to explain macroeconomic outcomes while dismissing external interventions as quantitatively negligible.

2.35 Methodologically, orthodox accounts fall into what causal inference defines as a *collider trap* (?). A collider trap takes place when analysts focus on a symptom that is actually the shared outcome of two separate causes. By isolating their attention on this shared outcome, they distort the relationship between the causes, often falsely attributing the symptom entirely to just one of them. In narratives of the UP crisis, this shared outcome—the collider—is the widespread presence of retail shortages and food queues (C) (?).

Orthodox accounts treat these empty shelves as definitive proof of “*repressed inflation*,” attributing them solely to the government printing money to stimulate demand (U_1). However, these shortages were simultaneously driven by a second, independent factor: the systematic hoarding and lockouts organized by private wholesale distributors and trucking strikes (U_2). Both state demand stimulus (U_1) and private supply lockouts (U_2) collided to produce empty retail shelves (C):

$$U_1 \longrightarrow [C] \longleftarrow U_2 \implies U_1 \not\perp U_2 \mid C \quad (1)$$

By treating retail shortages as a neutral, objective measure of excess liquidity (conditioning on C), orthodox accounts commit a collider error of causal identification. This methodological blind spot produces a spurious conclusion, allowing them to blame public deficit monetization for a supply paralysis that was heavily driven by the political actions of private business federations.

2.36 To transcend these causal illusions and collider traps, I adapt the methodological approach of ?, who formally tested competing economic theories to explain Poland’s growth trajectory during neoliberalization. In this chapter, I treat Monetarism (H_1), the Austrian Business Cycle (H_2), and Dependency Theory (H_3) as rival scientific research programmes. Each possesses a distinct theoretical core, auxiliary assumptions, and falsifiable empirical predictions. I submit the central causal claims of these competing traditions to a formal *Hypothesis Tournament* evaluated across multi-frequency time series data.

3 World Money, Financial Subordination, and the Peripheral Central Bank Balance Sheet

3.1 World Money, the Universal Equivalent, and the Monetary Expression of Value

3.1 In Marxian political economy, money originates as the commodity form of the universal equivalent: the unique social medium whose function is to express the value of all commodities and serve as the general means of payment (?, §3.1). Gold historically occupied this role not through inherent natural properties, but through a social process of market validation, acting as the bearer of abstract human labor.

3.2 The historical transit to credit money transforms this institutional form without dissolving its underlying function. As banking systems developed to intermediate commercial credit, central bank liabilities emerged as the legal anchor of domestic circulation (?, §3.2.3). The commodity anchor disappeared from domestic circulation, but the imperative of social validation remained. Central bank liabilities and bank deposits perform the role that bullion once fulfilled:

providing the monetary expression through which private, decentralized labor is validated as social labor. Following the Monetary Expression of Value (MEV) framework (??), the value of money links aggregate labor-time to the circulating monetary unit within a national formation.

3.3 Once circulation crosses national borders, however, credit money confronts the world market. While Marx presented world money in the physical form of bullion shedding national garbs (?, p. 51), a global credit-money system reproduces the universal equivalent through an international currency hierarchy (??). National currencies are stratified by their degree of international acceptability, liquidity, and convertibility. World money is the historically specific form wherein the liabilities of the dominant hegemonic power operate as the universal means of international payment, reserve asset, and unit of account.

3.2 The Bretton Woods Architecture and Peripheral Balance-of-Payments Constraints

3.4 The post-war Bretton Woods architecture codified this currency hierarchy institutionally. By pegging the U.S. dollar to gold at \$35 per ounce while anchoring other currencies to the dollar, the international monetary order formalized the dollar as the operative form of world money (?, p. 483). This arrangement granted the hegemonic center an “*exorbitant privilege*”: the ability to borrow internationally in its own currency, finance external deficits without reserve constraints, and act as banker to the world (?, pp. 243–244).

3.5 For peripheral economies, this institutional architecture generated an acute structural asymmetry. Peripheral states cannot settle international accounts in their own domestic currencies; external liabilities and essential imports must be discharged in world money. The peripheral circuit of capital confronts an absolute balance-of-payments constraint (??). Domestic credit expansion is bounded by the capacity to command foreign exchange through export earnings, terms of trade, and access to international credit lines. Reserve accumulation constitutes an externally enforced structural prerequisite of economic reproduction (?, p. 244).

3.6 The unilateral closure of the gold window by the United States on August 15, 1971 shattered the fixed-parity anchor of Bretton Woods agreements leading to their demise. While this rupture removed gold convertibility, it preserved dollar centrality, ushering in an era of global floating fiat money (?, p. 486). For the nascent socialist project of the Unidad Popular, this systemic shock dismantled the international monetary stability upon which its initial macroeconomic projections rested.

3.3 The Peripheral Central Bank Balance Sheet: From Reserve Adequacy to Balance-Sheet Solvency

3.7 Financial subordination materializes concretely on the balance sheet of the peripheral central bank. Mainstream open-economy models interpret this balance sheet through the lens of “*reserve adequacy*” and precautionary self-insurance, treating international reserves as an inventory buffer optimized against friction costs to insulate the economy from sudden stops (??). This normative framing assumes that reserve accumulation represents the voluntary, unconstrained choice of a sovereign optimizing agent.

3.8 In peripheral capitalism, reserve trajectories are the structural outcome of the balance-of-payments constraint, unequal exchange, and financial subordination (??). The two sides of the peripheral central bank’s balance sheet reflect fundamentally distinct monetary forms. On the liability side, domestic money represents unanchored credit obligations that the financial system owes the public, underwritten by state authority (?, §3.2.3). On the asset side, international

reserves (IR_t , measured in USD millions) consist of claims on world money that the peripheral state cannot emit, but Central Bank holds as assets against a liability of the United States federal reserve with its own population. In other words, money is credit of the Central Banks with their own populations, but world money is an asset when hold by Central Banks abroad in form of international reserves. .

3.9 This balance-sheet structure is governed by the Central Bank Solvency Ratio:

$$\text{SolvR}_t \equiv \frac{e_t \cdot IR_t}{M_t} \quad (2)$$

where e_t is the nominal exchange rate (Escudos/Pesos per USD), IR_t denotes international reserves in USD, and M_t represents domestic monetary liabilities.³ The dynamic law of motion governing this ratio is:

$$\Delta \text{SolvR}_t = \frac{CA_t + KA_t}{M_t} - \text{SolvR}_t \cdot g_{M,t} \quad (3)$$

where CA_t is current account net receipts, KA_t is net capital flows, and $g_{M,t}$ is domestic money growth. Financial subordination acquires its operative expression through this solvency relation: movements in SolvR_t govern the corridor within which domestic macroeconomic policy can maneuver. The complete formal accounting identities governing this balance-sheet relation are documented in Appendix A.

3.4 Money Endogeneity vs. Exogeneity and the Peripheral Paradox

3.10 Grounding the peripheral balance sheet requires resolving the foundational debate over the causality between bank credit, deposits, and central bank reserves (??). Mainstream monetarism and orthodox macroeconomics operate under Fractional Reserve Theory (FRT) and loanable-funds: the central bank exogenously fixes the monetary base (H_t), which mechanically dictates bank lending and broad money through a stable fractional-reserve multiplier ($H \rightarrow \text{Deposits} \rightarrow \text{Loans} \rightarrow P$). In sharp contrast, Post-Keynesian Endogenous Money Theory (EMT) and Marxian monetary circuits establish that real-world banking operates in reverse causal sequence (???):

$$\Delta \text{Bank Loans} \longrightarrow \Delta \text{Deposits} \longrightarrow \Delta \text{Central Bank Reserves}(H) \quad (4)$$

Commercial banks extend working-capital loans to firms to finance payrolls and production. These loans simultaneously create bank deposits. The central bank, far from controlling the money supply as an exogenous policy instrument, is compelled to accommodate required reserves ex-post at the policy discount rate to maintain clearance in the interbank payments system. As ? argued, credit money has no independent supply function: it comes into existence through the demand for bank lending and is extinguished through repayment. The central bank must accommodate this process to prevent the collapse of the credit and payments system. Monetarism, in this light, is not a neutral technical rule but a political strategy that deploys high interest rates and monetary contraction to induce unemployment and break trade union bargaining power. Under EMT, the traditional money multiplier ($mm_t \equiv M_t/H_t$) is not an exogenous causal driver but an unstable, ex-post accounting ratio reflecting commercial bank liquidity preferences, cash-drain ratios, and portfolio balance constraints. In peripheral capitalism, credit supply is structuralist and upward-sloping (???): commercial banks' willingness to intermediate credit is bounded by external liquidity constraints and foreign exchange exposure.

³The Solvency Ratio is the inverse of the Leverage Ratio ($\text{LevR}_t \equiv M_t/(e_t \cdot IR_t) = \text{SolvR}_t^{-1}$). The solvency formulation is preferred throughout this study because as reserves shrink toward zero, the leverage ratio explodes and distorts the statistical distribution, whereas the Solvency Ratio converges smoothly to zero, yielding a well-behaved index for threshold estimation.

3.11 This institutional reality establishes what I define as the *Peripheral Paradox*. In core economies issuing world money, the central bank accommodates domestic credit without facing an absolute currency boundary. In the periphery, however, while domestic credit generation is naturally endogenous ($\text{Loans} \rightarrow M_t$), the ultimate means of international settlement is bound to an exogenously constrained reserve asset ($e_t \cdot IR_t$). The peripheral central bank must back an endogenously expanding domestic credit pyramid with an externally constrained stock of world money. When external credit embargoes, copper price collapses, or private capital flight drain foreign reserves, the state cannot act as an unconstrained lender of last resort in hard currency; it must choose between contracting domestic activity or suffering balance-sheet insolvency. Under the UP, the nationalized commercial banking system under the State Development Corporation (CORFO) extended defensive working-capital credit to socialized and private enterprises facing distribution bottlenecks and price controls, compelling the Central Bank to monetize enterprise deficits. Domestic money growth thus functioned as an endogenous accommodation valve, driven by structural cost-push pressures rather than discretionary printing press impulses. This reverse causal sequence ($\Delta \text{Loans} \rightarrow \Delta \text{Deposits} \rightarrow \Delta H$) and its balance-sheet boundary under the Peripheral Paradox provide the operational hypotheses evaluated in the sequential Toda–Yamamoto Granger causality tournament in Section 5.3.

3.5 Financial Fragility Reinterpreted: State-Dependent Solvency of the Peripheral Central Bank

3.12 To understand how this balance-sheet contradiction generates macroeconomic crisis, I draw on the Financial Instability Hypothesis (FIH) of ?. In Minsky’s canonical framework, financial fragility is endogenously generated during prolonged expansions: stability breeds instability as private firms and banks voluntarily increase leverage, transitioning across qualitatively distinct financial positions (?). However, Minsky formulated this dynamic for private actors in a closed, metropolitan economy possessing complete monetary sovereignty, where an unconstrained central bank (the “Big Bank”) and fiscal authority (the “Big Government”) stabilize an intrinsic unstable system. The peripheral on the other hand, central bank is neither private nor unconstrained. It does not lever up out of speculative euphoria; its leverage is structurally imposed by the balance-of-payments constraint and externally amplified by center shocks. Minsky’s categorical typology provides a theoretically useful description of qualitatively distinct solvency states, but whether these categories map onto the peripheral central bank—and how many distinct regimes the data supports—is a question that must be resolved empirically, not assumed a priori.

3.13 Reinterpreting Minsky for the peripheral central bank requires two analytical moves. First, ?, p. 21 demonstrates that during acute financial fragility, the state and the central bank as lender of last resort are compelled to absorb the insolvency of private enterprises onto their own balance sheets. The central bank does not lever up out of profit-seeking; it does so by executing its institutional mandate to prevent payments gridlock and economic breakdown. Second, ? demonstrates that financial instability in peripheral Latin American economies operates as an *externally amplified endogenous fragility*. Minsky’s dynamic cannot be applied to dependent formations without recognizing that external shocks from the center—the collapse of world monetary arrangements, terms-of-trade declines, and credit embargoes—serve as triggers detonating domestic balance-sheet fragility. In periods of financial insability, either by endogenous causes or exogenous political agency against an specific political process, the peripheral Central Bank enters a state of *forced defensive accommodation* not out of speculative euphoria, but under severe institutional confinement. When imperial sanctions cut off commercial trade credit, refusing domestic liquidity expansion would cause immediate real-sector collapse. Accommodating the deficits of socialized enterprises is an institutional imperative to maintain

societal reproduction.

3.14 This reinterpretation establishes the central theoretical hypothesis of the chapter: the macroeconomic transmission behavior of the peripheral central bank is *state-dependent on its balance-sheet solvency*. The law of motion of the Solvency Ratio (Equation 3) implies that when $\text{Solv}R_t$ is sufficiently high, the central bank possesses genuine policy autonomy—domestic credit expansion mobilizes idle capacity without destabilizing the exchange rate, and external price shocks are absorbed at the border with minimal domestic pass-through. As solvency deteriorates past critical thresholds—driven by current account hemorrhage ($CA_t < 0$), capital flight ($KA_t < 0$), or accelerating domestic money growth ($g_{M,t}$)—the central bank loses this autonomy: forced defensive emission, lacking world-money backing, passes directly into price instability and currency flight. The empirical identification of where these thresholds lie, how many distinct regimes the data supports, and how transmission elasticities shift across solvency states is the object of the Threshold Vector Autoregression (TVAR) developed in Section 5.

4 Data Sources and Variable Construction

4.1 Monthly Panel: Sources and Sample Windows

The empirical core is a monthly panel of $N = 250$ observations (1960:01–1980:12), assembled programmatically from the Central Bank of Chile REST API (SieteRestWS). The API provides continuous monthly series for high-powered base money (H_t , thousands of pesos), the official nominal exchange rate (e_t , pesos per US dollar; pre-1975 escudo values retroactively converted at 1 peso = 1,000 escudos per the BCCh *Indicadores Económicos y Sociales de Chile 1960–2000* convention), London Metal Exchange copper cash prices (P_t^{Cu} , USD/lb), London bullion gold prices (P_t^{gold} , USD/oz), the US Wholesale Price Index ($P_{W,t}^{\text{US}}$), the official Consumer Price Index (π_t), and the US Dow Jones Industrial Average. The Santiago Stock Exchange (IGPA) index is drawn from the replication files of ?. Gross international reserves (IR_t , millions of USD) come from the IMF *International Financial Statistics*.

Sectoral real activity uses the disaggregated monthly component series compiled by ?: light industrial manufacturing (Q_t^{manuf}), copper mining extraction (Q_t^{mining}), and physical export and import volumes (Exports_t , M_t). I use the individual components, not his aggregated IMEA composite index. The components are harmonized from *Estadística Chilena* (DGE/INE) and the BCCh *Boletín Mensual* and are available from 1927 onward; I restrict the panel to 1960:01–1980:12 to align with the monetary and reserve series.

Narrow money ($M1_t$, thousands of pesos) and the observed banking multiplier ($m_t \equiv M1_t/H_t$) are available only over the un-spliced window 1966:01–1980:12 ($N = 180$), which excludes pre-1966 commercial bank classification revisions. $M1_t$ enters the empirical architecture exclusively in the commercial-banking bifurcation and multiplier deconstruction steps of the Granger causality tournament (Section 5.3); the primary solvency threshold and the full-panel TVAR use H_t throughout.

A supplementary annual tier ($T = 91$, 1920–2010) provides the long-run accumulation backdrop used in Sections 5.1 and 5.2: the functional wage share (ω_t) from ?; the productive capital stock ($K_t \equiv K_t^{\text{ME}} + K_t^{\text{NRC}}$, excluding residential construction), the aggregate profit rate (r_t , nominal surplus on productive fixed capital at current replacement cost), and structural capacity utilization (μ_t , estimated via cointegrating polynomial regression) from the GPIM dataset of Chapter 2; and balance-of-payments accounts from the World Historical Balance of Payments Database (?). Appendix B catalogs complete series identifiers and metadata.

4.2 Variable Construction

The primary Central Bank reserve solvency ratio, constructed over the full monthly panel, is

$$\text{SolvR}_t^H \equiv \frac{e_t \cdot IR_t}{H_t \cdot 1000}, \quad (5)$$

where e_t is in pesos per dollar, IR_t in millions of dollars, and H_t in thousands of pesos. The factor of 1,000 converts H_t to millions of pesos, aligning the denominator with the domestic-currency valuation of reserves ($e_t \cdot IR_t$). Base money is the appropriate primary liability because it is the Central Bank's direct obligation and is observed continuously from 1960:01. In the threshold specification, the predetermined lag SolvR_{t-1}^H satisfies the consistency conditions of ?.

For the commercial-banking decomposition ($N = 180$), I also construct

$$\text{SolvR}_t^{M1} \equiv \frac{e_t \cdot IR_t}{M1_t \cdot 1000},$$

which correlates with SolvR_t^H at $\rho = 0.974$. This measure is not the primary solvency variable; it is used only to test whether the commercial credit channel alters the ordering between external backing and domestic liabilities.

The Prebisch-ECLAC capacity to import measures the purchasing power of physical exports relative to imported manufactured inputs:

$$M_t^{\text{cap}} \equiv \left(\frac{\text{Exports}_t}{\text{Exports}_{1970:11}} \right) \times \left[\frac{P_t^{\text{Cu}}/P_{1970:11}^{\text{Cu}}}{P_{W,t}^{\text{US}}/P_{W,1970:11}^{\text{US}}} \right] \times 100, \quad (6)$$

benchmarked to November 1970 (1970:11 = 100). The real structural imbalance ratio is

$$\Theta_t \equiv \left(\frac{M_t}{M_t^{\text{cap}}} \right) \times 100. \quad (7)$$

When $\Theta_t > 100\%$, physical import volumes exceed export purchasing power, generating a foreign-exchange deficit that must be financed through credit lines or reserve depletion.

4.3 Data Adjustments

The raw customs import volume series exhibits a single-month administrative registration drop in April 1973: the recorded volume falls from 2.88 (March) to 0.26 (April), recovering to 2.63 (May). Historical verification confirms this reflects a paperwork lag in clearing port arrivals into official records, not a collapse in physical trade. I replace the April observation with the monotonic linear midpoint, 2.76, to prevent the registration artifact from triggering a spurious structural break in the VAR systems.

4.4 Integration Order of Variables Included in Empirical Exercises

The transformed series entering the Toda–Yamamoto and threshold VAR architectures are stationary in levels. Table 1 reports the ADF diagnostics; the underlying level series have maximal integration order $d_{\max} = 1$.

Table 1: Univariate Augmented Dickey–Fuller Tests and Maximal Order of Integration ($d_{\max} = 1$)

Variable / Series	Series in Levels (z_t)		First Difference (Δz_t)		Order	Stationarity Decision
	t -ADF	5% Crit.	t -ADF	5% Crit.		
CPI Inflation (π_t)	-3.372	-2.87	-13.619***	-2.87	I(0)	Stationary after differencing ($I(0)$)
Base Money Growth ($g_{H,t}$)	-3.751	-2.87	-12.484***	-2.88	I(0)	Stationary after differencing ($I(0)$)
Narrow Money Growth ($g_{M1,t}$)	-4.059	-2.88	-17.812***	-2.88	I(0)	Stationary after differencing ($I(0)$)
Multiplier Growth ($\Delta \ln m_t$)	-12.791	-2.88	-11.957***	-2.88	I(0)	Stationary after differencing ($I(0)$)
Manufacturing Output ($g_{\text{Manuf},t}$)	-14.105	-2.87	-19.409***	-2.88	I(0)	Stationary after differencing ($I(0)$)
Mining Extraction ($g_{\text{Mining},t}$)	-14.910	-2.87	-13.947***	-2.88	I(0)	Stationary after differencing ($I(0)$)
Structural Imbalance ($\Delta \ln \Theta_t$)	-14.512	-2.87	-17.171***	-2.88	I(0)	Stationary after differencing ($I(0)$)
World Gold Price Growth ($g_{P,\text{gold},t}$)	-10.559	-2.87	-17.959***	-2.88	I(0)	Stationary after differencing ($I(0)$)
Nominal Exchange Rate Growth ($g_{e,t}$)	-3.777	-2.87	-21.142***	-2.88	I(0)	Stationary after differencing ($I(0)$)
Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	-8.726	-2.87	-14.011***	-2.88	I(0)	Stationary after differencing ($I(0)$)

Notes: Monthly series spanning 1960:01–1980:12 ($N = 250$, with official un-spliced narrow money $M1$ and multiplier m spanning 1966:01–1980:12, $N = 180$). Benchmarked on ?. Augmented Dickey–Fuller (ADF) tests estimated with constant drift; lag orders selected via Schwarz Bayesian Information Criterion (SBIC) up to $p = 4$. Critical values correspond to MacKinnon (1996) finite-sample distributions. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. All variables reject unit root non-stationarity in first differences, establishing the empirical invariant $d_{\max} = 1$ for Toda–Yamamoto levels augmentation.

5 Historical Background, Stylized Facts, and Empirical Results

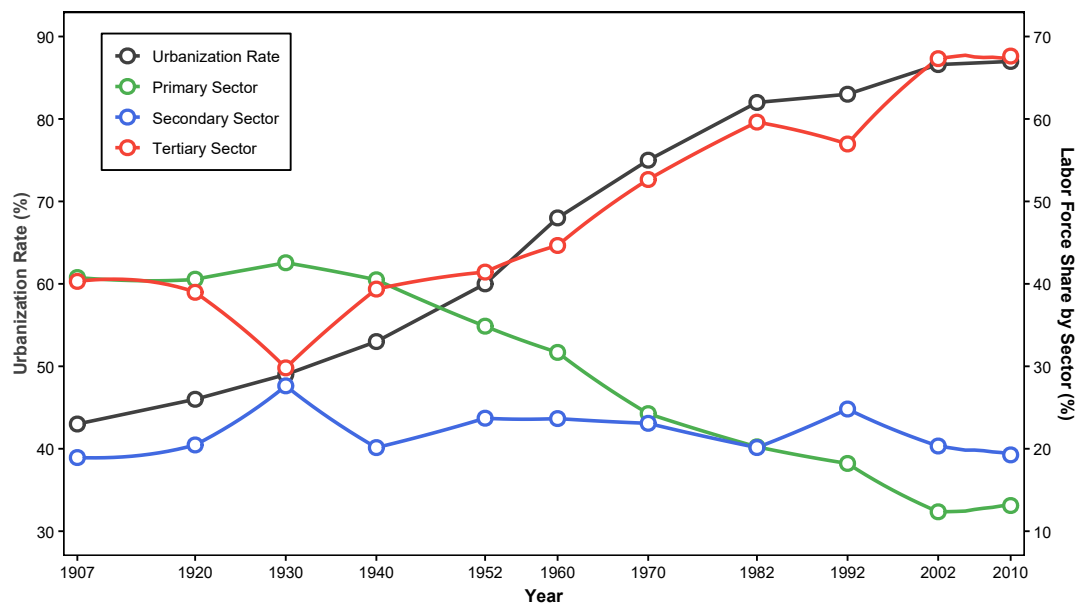
5.1 Historical Background: The Rise of the Unidad Popular

The Unidad Popular’s electoral victory in 1970 brought a project of socialist transformation into contact with popular organizations already contesting authority over work, land, and urban life. Industrial workers, the peasantry, and *pobladores* did not enter politics through the same institutions or pursue identical demands. Their struggles nevertheless became increasingly connected before the election. This subsection follows successive waves of mobilization, repression, and political articulation to explain that connection. Chile’s changing economy supplied the structural setting, but did not dictate the political outcome.

During the nitrate era, export receipts sustained public revenues and domestic activity while tying employment, imports, and fiscal capacity to external demand and finance. The postwar dislocation of natural nitrate weakened that arrangement before the Great Depression. The Depression intensified an existing crisis of the export economy and accelerated protection of domestic production and state intervention. Manufacturing development and the institutional conditions for industrialization thus preceded 1929; the crisis changed their scale and direction rather than initiating them from nothing (??).

After the Second World War, mass production, productivity growth, rising wages, collective bargaining, and mass consumption became linked in the North Atlantic core (?). US-centered monetary and political leadership reorganized international hierarchy without removing the unequal conditions under which peripheral economies industrialized. Chile expanded production for its domestic market while continuing to require imported machinery, intermediate inputs, technology, and foreign exchange. Industrial growth could therefore reproduce external dependence in a new form (??).

Industrialization also changed the spatial distribution of labor. Figure 1 places declining primary-sector shares beside rising urbanization and tertiary activity. The secondary sector did not become the principal absorber of the expanding labor force as Chile urbanized. This relative shortfall supports the characterization of *disarticulated industrialization*: industrial expansion coexisted with a wider transformation of settlement and livelihoods (?). The conflicts that developed on this terrain, rather than the sectoral series themselves, establish the chronology that follows.



Source: Author's own elaboration based on ClioLab / Díaz, Lüders & Wagner (2016).

Note: Hollow circle markers are displayed at decennial census benchmark years (1907, 1920, 1930, 1940, 1952, 1960, 1970, 1982, 1992, 2002, 2010).

Figure 1: Sectoral labor-force composition and urbanization in Chile, 1907–2010.

Source: Author's elaboration from Clio-Lab PUC historical statistics (?). Sectoral series refer to the labor force; urbanization refers to population. These measures have different denominators. Census urbanization observations are plotted as points.

Wave I: Selective incorporation and coercive backlash. The first conflict wave began before the profitability series available here starts in 1943. Popular Front politics brought parties and organized constituencies into a common electoral project, while access to collective representation remained uneven across workplaces and the countryside (?). Rural workers made claims over employment, labor conditions, and the right to organize; the countryside was not merely a passive source of labor for urban industry. The 1943–47 accounting interval captures an observed economic segment of this longer confrontation, not its historical origin.

By the mid-1940s, expanding rural demands formed part of a wider confrontation over labor organization and the terms of work. The Plaza Bulnes massacre and the national stoppages of January and February 1946 exposed its urban and industrial dimensions (?, pp. 199–201, 213–225). The agricultural petitions assembled by Loveman document the rural dimension and its crest in 1947 (?). Their distributive significance lay partly in workers' capacity to turn shared grievances into demands extending beyond an individual workplace. Over the observed 1943–47 interval, a declining profit share was the principal negative contribution to profit-rate growth in the chapter's accounting (Table 2).

The response in the countryside combined formal recognition with restrictions on the scale of organization. Law 8,811, published in July 1947, recognized agricultural unions while confining them to individual *fundos* and prohibiting combinations of those unions. Eligibility requirements tied organization to a sufficiently large and stable workforce on each property; restrictions on petition filing during sowing and harvest and the resort to binding arbitration further narrowed collective action (?, arts. 9, 14, 16, 53, 60–69). The law made the boundaries of landed estates the boundaries of legal representation, obstructing the organization of common agrarian demands across them. Recognition carried material benefits and limited protections, but preserved constraints on the pursuit of collective claims.

Law 8,987, published in September 1948, extended restriction through political exclusion. Alongside outlawing the Communist Party, it connected exclusion from the electoral regis-

ters to exclusion from union membership and barred affected workers from union leadership, labor delegations, and conciliation bodies, including agricultural arbitration boards (? , art. 1, amendment 5; art. 7, amendment 1; art. 8). The two laws restricted different levels of organization: combination across estates and political representation within a broader national confrontation. The subsequent decline in registered agricultural petitions marks the contraction of these channels, not the disappearance of agrarian grievances.

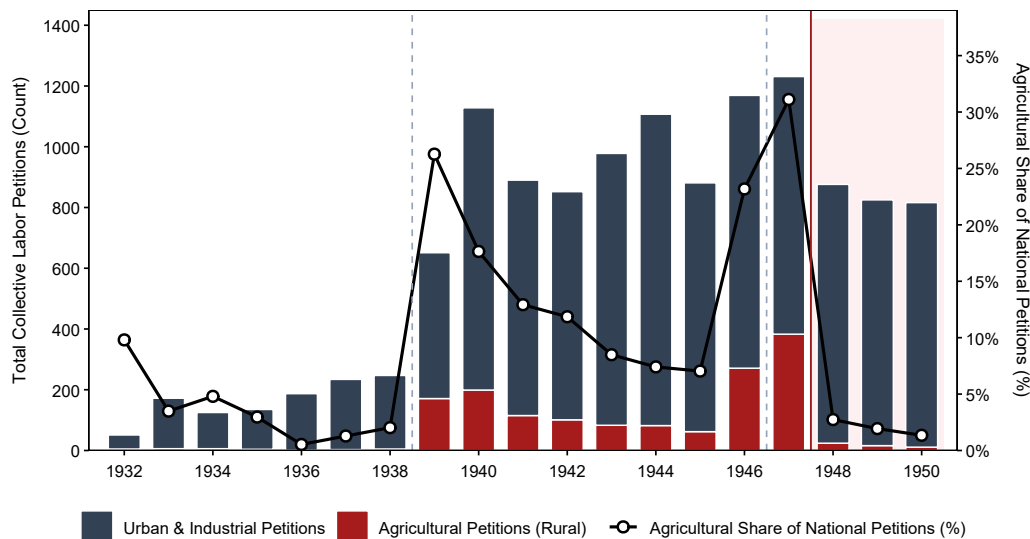


Figure 2: Recorded collective labor petitions by sector, 1932–1950.

Source: Author’s archival compilation from ?. Petitions measure registered collective demands under the prevailing legal framework, not all forms of social conflict.

Wave II: Coercive containment and urban recomposition. The restrictions imposed in 1947–48 narrowed the institutional channels through which workers could press their demands. In the countryside, legal recognition bound union organization to the estate; across sectors, anticommunist exclusion attacked the political networks through which collective claims acquired a wider reach. Conflict continued under conditions increasingly unfavorable to workers’ efforts to defend wages and influence economic policy (? , pp. 47–53).

Stabilization brought this conflict over distribution into sharper focus. Under Ibáñez, the US advisory firm Klein–Saks supplied a technical rationale for policies restraining wage adjustment in the name of controlling inflation. As ? , pp. 45–53 argues, these measures consolidated a deterioration in real wages already underway rather than initiating it. Limits on wage adjustment, unemployment, and repression of organized resistance shifted adjustment costs onto working-class households, while rising prices for essential goods and services extended the dispute beyond the workplace.

The profitability accounts place this distributive conflict alongside a continuing transformation of production. Over 1948–58, the rising profit share supplied the largest positive contribution to profit-rate growth. Capital deepening more than offset the positive contribution of potential labor productivity: their annualized contributions were -2.60 and $+2.29$ log points, respectively (Table 2). Capital per worker expanded faster than potential output per worker, while distribution supported profitability. This greater capital intensity coexisted with the limited relative absorption of the growing labor force described above.

Urban growth made housing central to the conflict over living standards. A place in the city required more than access to wage employment: households confronted scarce housing, high rents, inadequate services, and the cost of reaching work. These pressures cut across the division

between securely employed workers and residents with irregular incomes. Thielemann traces how disputes over transport fares, rents, and basic consumption connected grievances at work with organization in neighborhoods. The city became a field of collective action in its own right (?, pp. 51–53).

The revolt of April 1957 and the occupation that founded La Victoria in October gave this urban conflict a new political expression. They were distinct forms of action, but Thielemann places them within a shared process through which popular groups challenged the limits of institutional representation and expanded their capacity to act collectively. La Victoria made occupation and neighborhood construction into a practical claim to housing. The significance of 1957 lies in this political awakening rather than an immediate reversal of labor's economic position. It opened a phase of struggle whose organizations, alliances, and objectives developed unevenly through the Long Sixties (?).

Wave III: The Long Sixties and the rise of the Unidad Popular. Industrial workers entered the Long Sixties with demands reaching beyond wage adjustment to the organization of production. Disputes over rationalization and work intensity challenged managerial authority, while solidarity linked workers across industries: Huachipato steelworkers supported the Lota miners in the long strike of 1960. Repression shaped collective memory, including the killings at El Salvador in 1966. These experiences helped radicalize sections of the movement before the UP took office (?, pp. 119–125).

In the countryside, agrarian reform placed landownership and the organization of rural labor within the same political struggle (?). The 1967 union law changed the territorial basis on which workers could act. Membership could extend across estates, the minimum territorial unit became the commune, and unions could form federations and confederations (?, arts. 1, 3–4). Compared with the estate-bound framework of 1947, this widened the institutional scope for collective demands. The agrarian question concerned both access to land and the power to organize across the boundaries of individual properties.

In the cities, *pobladores* pressed claims to land, housing, and public services through neighborhood organization and occupations. Their mobilization challenged who could secure a place in the city and on what terms. Housing was both a material need and a demand for political recognition, bringing everyday life into conflicts over property and public provision (??). These struggles had their own organizations and objectives, even where their participants also experienced conflict as wage earners.

Concepción illustrates the local relationships through which political connections developed. Schlotterbeck locates organizing in the university, nearby industrial communities, and working-class neighborhoods, where labor and student activism intersected with struggles over housing. These relationships preceded Allende's victory, while their development under the UP opened further possibilities for coordinated action (?, pp. 7, 11–14). Popular articulation was a process, not a unity completed at the ballot box.

Widening mobilization did not coincide with a continuous deterioration of aggregate profitability. Across the 1959–70 accounting interval, the profit rate rose, supported by positive contributions from the profit share, utilization, and potential labor productivity, with capital deepening providing an offset (Table 2). The conflict concerned the distribution and control of an expanding economy as well as its unresolved limits. By 1970, struggles over work, land, and housing confronted the question of executive power; their convergence made a common political project possible without erasing differences over its objectives or means.

Wave IV: The UP project and the crisis of accumulation. Allende’s election in 1970 brought these accumulated struggles into a new relationship with executive power. The UP’s project connected redistribution to changes in ownership and the direction of production. In Arboleda’s account, Pedro Vuskovic and Jacques Chonchol approached planning as part of national liberation: transforming the economy required both public intervention and greater popular power (?). The government encountered organizations with their own demands and rhythms of action. Winn’s distinction between revolution from above and below captures this relationship between a state project and popular initiatives that could support it while pressing beyond its immediate limits (??).

Political articulation continued after the election. In Concepción, the People’s Assembly of 1972 brought together sectors whose organizing Schlotterbeck traces through the preceding years (?, p. 14). This local convergence illustrates why the rise of the UP cannot be reduced to its electoral coalition: participation also developed through institutions and practices created within the movements themselves.

These transformations unfolded alongside a sharp deterioration in profitability and productive accumulation. In the chapter’s accounts, the profit rate fell from 12.23 percent in 1970 to 6.57 percent in 1972, while net accumulation slowed from 4.23 to 1.85 percent. Gross accumulation fell from 11.63 to 7.64 percent over the same years (Table 3). The annual observations for 1973 combine a profit-rate rebound with a further slowing of net accumulation to 1.71 percent; they span both the final months of the UP and the first months after the coup. This sequence directs attention from profitability alone to the conditions under which surplus was converted into productive investment.

The confrontation over property was also visible in financial markets: share prices fell sharply after Allende’s election and rose after the military coup (?). Productive accumulation, however, remained weak in the annual accounts. The coup of September 1973 ended the UP experiment through force, not through the automatic resolution of an economic adjustment. The following analysis examines how distribution, utilization, relative prices, and external constraints interacted with the deterioration of profitability and accumulation during the crisis.

5.2 Stylized Facts of Peripheral Accumulation, Distribution, and External Constraints

5.2.1 The Material Bounds of Peripheral Transition: An Accounting Anchor

The political mobilization and redistributive commitments analyzed in Section 5.1 operated within strict external material boundaries. The Unidad Popular inherited an economy structurally reliant on imported machinery, intermediate spare parts, and foreign trade credit to sustain domestic employment and basic wage-goods production. In mainstream accounts (??), the subsequent macroeconomic crisis is attributed to an unforced episode of “macroeconomic populism” driven by autonomous fiscal deficits and unconstrained money printing that depleted reserves and triggered runaway inflation. In contrast, structuralist and Marxian political economy traditions (??) identify the primary constraint in external balance-of-payments vulnerability, input dependency, and international credit restrictions. Evaluating these competing perspectives requires confronting their core premises with audited balance-of-payments accounts and high-frequency time series.

The empirical analysis below tracks whether deterioration in the Central Bank’s external balance sheet preceded the subsequent acceleration of inflation and defensive credit accommodation. In a subordinate open economy without reserve-currency status, convertible foreign exchange reserves (IR_t) constrain the ability of the domestic monetary system to settle external obliga-

Table 2: Social Conflict Waves and Profitability Decomposition, Chile, 1943–1975

Wave of Social Conflict	Profit Rate $\Delta \ln r$	Profit Sh. $\Delta \ln(1 - \omega)$	Rel. Price $\Delta \ln(P_Y/P_K)$	Cap. Util. $\Delta \ln \mu$	Lab. Prod. $\Delta \ln y^p$	Cap. Deep. $-\Delta \ln k$
<i>A. Social Conflict Waves (Annualized Rates and % Shares)</i>						
Wave I: Observed accounting segment (1943–1947)	-2.84%	-5.53% (+194.9%)	+2.63% (-92.7%)	+0.62% (-21.7%)	-0.21% (+7.6%)	-0.34% (+11.9%)
Wave II: Containment and urban recomposition (1948–1958)	+2.93%	+3.11% (+106.1%)	-0.10% (-3.3%)	+0.23% (+7.9%)	+2.29% (+77.9%)	-2.60% (-88.6%)
Wave III: Long Sixties accounting interval (1959–1970)	+2.07%	+1.31% (+63.3%)	+0.45% (+21.6%)	+0.97% (+46.9%)	+1.72% (+83.0%)	-2.38% (-114.8%)
Wave IV: UP crisis accounting interval (1971–1973)	-7.65%	+1.94% (-25.3%)	-4.35% (+56.9%)	-4.78% (+62.5%)	-1.23% (+16.0%)	+0.77% (-10.1%)
<i>B. Terminal Crisis Cycle: Annual Log Changes and % Shares</i>						
1970 → 1971: Reactivation peak	-14.05%	-11.44% (+81.4%)	-7.90% (+56.2%)	+5.92% (-42.1%)	-0.43% (+3.1%)	-0.20% (+1.4%)
1971 → 1972: Profit squeeze	-48.08%	-34.60% (+72.0%)	-10.42% (+21.7%)	-2.52% (+5.2%)	-2.27% (+4.7%)	+1.74% (-3.6%)
1972 → 1973: Annual observation spanning the coup	+32.78%	+38.48% (+117.4%)	+1.72% (+5.3%)	-7.04% (-21.5%)	-0.18% (-0.5%)	-0.20% (-0.6%)
1973 → 1974: DL 522 compression	+15.74%	+28.59% (+181.7%)	-11.47% (-72.9%)	-0.72% (-4.6%)	+4.04% (+25.7%)	-4.70% (-29.9%)
1974 → 1975: Chicago shock therapy	-48.97%	-17.70% (+36.2%)	-16.24% (+33.2%)	-14.78% (+30.2%)	+8.97% (-18.3%)	-9.22% (+18.8%)

Notes. Growth-accounting decomposition using the identity $\Delta \ln r_t \equiv \Delta \ln(1 - \omega_t) + \Delta \ln(P_{Y,t}/P_{K,t}) + \Delta \ln \mu_t + \Delta \ln y_t^p - \Delta \ln k_t$. Values outside parentheses are annualized log contributions (% p.a.). Values inside parentheses report the component's share of the total change, computed as $(\Delta \ln X_i / \Delta \ln r_t) \times 100\%$. A positive share indicates the component moved in the same direction as the total change; a negative share indicates it offset the total change. Shares may exceed 100% due to normalization by the net total change. Source: Author's own elaboration (Chapter 2).

Table 3: Capital Accumulation Response Across the Terminal Crisis Cycle, Chile, 1964–1975

Episode / Year	Profit Rate r (%)	Accum. Propensity χ (ratio)	Gross Acc. Rate g_K^{gross} (%)	Depreciation Rate δ (%)	Net Acc. Rate g_K^n (%)
<i>A. Christian Democratic Administration (1964–1970)</i>					
1964 (Baseline)	8.16	1.530	12.49	7.67	4.82
1965	8.59	1.244	10.69	6.94	3.75
1966	10.94	0.972	10.63	6.97	3.66
1967	10.14	1.066	10.81	7.06	3.75
1968	11.29	0.991	11.20	7.24	3.96
1969 (Peak Profit.)	12.79	0.857	10.96	7.14	3.82
1970 (Inherited)	12.23	0.951	11.63	7.40	4.23
<i>B. Unidad Popular Administration (1971–1973)</i>					
1971 (Reactivation)	10.62	0.953	10.12	6.78	3.34
1972 (Squeeze)	6.57	1.162	7.64	5.78	1.85
1973 (Coup year)	9.12	0.821	7.48	5.78	1.71
<i>C. Military Dictatorship & Shock Therapy (1974–1975)</i>					
1974 (Compression)	10.67	0.811	8.65	6.27	2.38
1975 (Collapse)	6.54	1.035	6.77	5.56	1.21

Notes. Accumulation accounting for productive fixed assets ($K_t \equiv K_{\text{ME},t} + K_{\text{NRC},t}$), excluding residential construction. Gross accumulation is $g_{K,t}^{\text{gross}} \equiv I_t^{\text{gross}}/K_{t-1}$; physical depreciation is $\delta_t \equiv D_t/K_{t-1}$; net accumulation is $g_{K,t}^n \equiv I_t^{\text{net}}/K_{t-1}$; and the accumulation propensity out of gross surplus is $\chi_t \equiv g_{K,t}^{\text{gross}}/r_t$. All observations satisfy the exact identity $g_{K,t}^n \equiv \chi_t r_t - \delta_t$, subject to displayed rounding. The profit rate is a level, not its growth rate. Annual 1973 observations span the coup. Source: Author’s own elaboration (Chapter 2).

tions, finance critical food and capital imports, and anchor the currency parity. To formalize this external balance-sheet constraint, I define the Central Bank Solvency Ratio relative to Base Money:

$$\text{SolvR}_t^H \equiv \frac{e_t \cdot IR_t}{H_t \cdot 1000}, \quad (8)$$

where e_t is the official exchange rate (pesos per USD), IR_t represents international reserves in USD millions, and H_t denotes high-powered domestic base money liabilities in thousands of pesos (scaled by 1,000 to align units, as established in Section 4). As external reserves depleted and foreign credit lines contracted, the domestic monetary system was severed from its external anchor. Under these material constraints, domestic credit expansion may have functioned not as an autonomous engine of inflation, but as an endogenous accommodation valve mediating balance-of-payments strangulation and enterprise cost pressures.

5.2.2 Long-Span Balance of Payments Accounting: Reserve Accumulation and the 1971 Capital-Flow Reversal (1958–1975)

In standard national accounting, the annual change in official Central Bank international reserves (ΔIR_t) satisfies the exact identity:

$$\Delta IR_t \equiv CA_t + KA_t + EO_t + SDR_t \quad (9)$$

Here, CA_t represents the current account balance, while KA_t denotes net capital inflows. Net errors and omissions (EO_t) capture unrecorded capital flight and trade misinvoicing. Finally, SDR_t denotes allocations of IMF Special Drawing Rights (instituted in 1970).

Table 4: Long-Span Balance of Payments Accounting and International Reserves Dynamics (Chile, 1958–1975): Official BCCh Cash Flows vs. WBOP Global Accrual Accounts

	Ibáñez	Jorge Alessandri (1959–1964)						Eduardo Frei Montalva (1965–1970)						Unidad Popular (1971–1973)			Junta (1974–75)	
Metric / USD Millions	1958	1959	1960	1961	1962	1963	1964	1965	1966	1967	1968	1969	1970	1971	1972	1973	1974	1975
A. Official Cash-Basis Balance of Payments (Central Bank of Chile / Stallings 1978, USD Millions)																		
Current Account (<i>CA</i>)	—	−24.8	−148.1	−241.1	−181.9	−157.8	−131.6	−56.6	−82.2	−127.4	−135.3	−5.6	−102.9	−205.5	−404.8	−294.6	−210.8	−491.3
Capital Account (<i>KA</i>)	—	20.8	76.3	187.6	133.2	107.6	152.1	65.8	168.0	125.5	294.6	222.5	267.5	−26.5	327.4	242.3	228.0	298.7
Net Errors & Omissions (<i>EO</i>)	—	28.6	43.4	−55.1	−0.3	22.0	2.9	37.5	33.8	−21.5	−41.4	−42.4	−72.9	−84.5	−171.6	−60.0	−62.1	−92.4
SDR Allocations (<i>SDR</i>)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	21.8	16.7	18.2	0.0	0.0	0.0
Net Reserve Change (ΔIR)	—	24.6	−28.4	−108.6	−49.0	−28.2	23.4	46.7	119.6	−23.4	117.9	174.5	113.5	−299.8	−230.8	−112.3	−44.9	−285.0
Net International Reserves (IR_t^{net})	29.6	105.4	72.8	−5.2	15.4	−24.3	−16.5	35.2	77.4	54.3	125.4	285.2	393.5	162.7	75.8	167.4	94.0	−129.2
B. Capital Account Reversal, External Drainage, and Cumulative Flows																		
Net External Drain ($CA + KA + EO$)	—	+24.6	−28.4	−108.6	−49.0	−28.2	+23.4	+46.7	+119.6	−23.4	+117.9	+174.5	+91.7	−316.5	−249.0	−112.3	−44.9	−285.0
Drain Share of IR_{t-1}	—	—	—	—	—	—	—	—	—	—	—	—	—	80.4%	153.0%	148.2%	26.8%	303.2%
Cumulative $\sum KA$	—	+\$677.6M (Alessandri Total)						+\$1,143.9M (Frei Montalva Total)						+\$543.2M (UP Total)			+\$526.7M	
Cumulative $\sum CA$	—	−\$885.3M (Alessandri Total)						−\$510.0M (Frei Montalva Total)						−\$904.9M (UP Total)			−\$702.1M	
C. Global Accrual Balance of Payments (WBOP, ?)																		
WBOP Current Account (USD M)	−271.8	−189.5	−155.3	−265.4	−194.4	−308.2	−212.8	−103.4	−179.4	−135.9	−70.9	+61.8	−147.7	−421.6	−550.8	−618.1	−105.5	−470.6
WBOP Current Account (% GDP)	−7.74%	−5.00%	−3.40%	−5.19%	−3.26%	−5.21%	−3.37%	−1.60%	−2.37%	−1.82%	−0.93%	+0.70%	−1.53%	−3.64%	−4.40%	−3.47%	−0.61%	−5.85%
Net Foreign Wealth / IIP (% GDP)	−67.0%	−69.3%	−61.7%	−58.1%	−54.3%	−57.9%	−59.2%	−61.1%	−53.5%	−56.7%	−57.4%	−50.2%	−54.6%	−47.7%	−49.4%	−33.6%	−46.4%	−100.6%

Sources and Notes: Official cash-basis Balance of Payments flows (1960–1975), Net Errors and Omissions, and SDR allocations sourced from the Banco Central de Chile (BCCh) REST API and Base de Datos Estadísticos (BDE), complemented by historical balance-of-payments ledgers. 1958–1959 cash entries sourced directly from ?, Table 7.4, p. 178, citing contemporary BCCh records. Net international reserves (IR_t^{net}) record liquid foreign exchange assets net of short-term central bank foreign liabilities, compiled from BCCh historical ledgers and ?, accounting for negative net balances during the 1961–1964 exchange crises and 1975; gross foreign reserve assets from IMF International Financial Statistics remain positive in all years. Global accrual balance of payments series and Net International Investment Position (NIIP) sourced from the World Historical Balance of Payments Database (?). The balance-of-payments reserves identity holds exactly: $\Delta IR_t \equiv CA_t + KA_t + EO_t + SDR_t$. Discrepancies between cash and accrual accounts reflect distinct institutional accounting conventions: cash ledgers record physical foreign exchange settlements at the Central Bank window, whereas WBOP accrual accounts adhere to modern BPM6 standards, incorporating legally accrued but unpaid debt service (under the November 1971 debt moratorium), reinvested foreign mining profits, and global bilateral trade partner reconciliations. Both records confirm that in 1971, net external financing experienced a sharp contraction of −\$294.0 million ($KA = -\26.5M vs. $+\$267.5\text{M}$ in 1970), accounting for 98.1% of the calendar-year reserve drain before domestic fiscal emission accelerated in late 1972.

Table 4 establishes the balance-of-payments ledger across eighteen years (1958–1975), spanning four successive developmental regimes in dialogue with the political economy framework of ? and the harmonized World Historical Balance of Payments database (?). Reconciling official Central Bank of Chile (BCCh) cash flows, Stallings’ archival series, and modern WBOP accrual accounts reveals the complementary institutional recording conventions governing peripheral balance-of-payments accounting. The official BCCh cash ledger (?, Table 7.4, p. 178) records physical currency disbursements and settlements crossing the Central Bank counter. This cash accounting captures the daily liquidity constraint facing policymakers in Santiago, determining the liquid reserve buffer required to defend the exchange parity and settle urgent import shipments. Stallings isolates autonomous capital flows to measure voluntary foreign lending and direct investment, distinguishing market-driven flows from compensatory emergency financing. In contrast, the harmonized World Historical Balance of Payments database (?) reconstructs global accounts under modern IMF BPM6 standards, reconciled with bilateral trade matrices. WBOP records primary income debits on an accrual basis, registering legally accrued debt service liabilities regardless of whether the November 1971 debt moratorium suspended actual cash disbursements.

The Alessandri administration (1958–1964) anchored its stabilization in foreign capital guarantees (DFL 258), dismantled quantitative import licensing, and pegged the nominal exchange rate at 1.05 Escudos = 1.00 USD on January 1, 1959. Between 1959 and 1961, domestic prices rose by 40%, generating severe real exchange rate appreciation. Consumer goods imports expanded the current account deficit from −\$24.8 million in 1959 to −\$241.1 million in 1961 (−5.19% of GDP in WBOP accounts). Central Bank liquid reserves were exhausted by late 1961, compelling emergency import bans and dual exchange controls. In October 1962, the government abandoned the peg and floated the escudo, unleashing a 45% inflation spike. Over his six-year tenure, Alessandri accumulated −\$885.3 million in current account deficits, leaving net Central Bank reserves negative at −\$16.5 million in 1964.

The Christian Democratic administration of Eduardo Frei Montalva (1964–1970) instituted a crawling peg exchange rate system, acquired 51% state ownership in U.S. copper mining concessions, and drew extensive financial support from the Alliance for Progress. Table 4 reveals that Frei’s reserve accumulation rested heavily on external borrowing. In five of his six years in office, Chile registered persistent current account deficits, accumulating −\$510.0 million across 1965–1970. Only the copper price peak of 1969 yielded a near-balanced cash account (−\$5.6M cash; +\$61.8M WBOP accrual). Net reserves rose by +\$550.0 million because the administration absorbed +\$1,143.9 million in net foreign capital inflows. Because external borrowing exceeded net reserve accumulation by more than two to one, over 208% of the reserve increase was financed by foreign liabilities. The \$393.5 million reserve buffer inherited in 1970 was therefore heavily dependent on external credit.

The Unidad Popular assumed office committed to rapid wage-led redistribution and the sovereign recovery of basic wealth, culminating in the unanimous constitutional nationalization of large-scale copper mining on July 11, 1971 (Law 17,450). As documented by ?, p. 178, this measure was met by immediate international financial restrictions: net external financing collapsed from +\$267.5 million in 1970 to −\$26.5 million in 1971 (−\$49.9 million in Stallings’ autonomous series). Within the audited 1958–1975 ledger, 1971 was the single calendar year with negative net capital inflows. This −\$294.0 million capital account turnaround accounted for 98.1% of the calendar-year reserve drain ($\Delta IR = -\$299.8M$), depleting Central Bank liquidity before domestic fiscal monetization accelerated.

On the shop floor, international commercial banks reduced short-term trade credits from \$220 million to \$35 million, while the Export-Import Bank suspended project financing. Foreign suppliers increasingly required cash-in-advance confirmed letters of credit for essential food and

fuel shipments. In the nationalized copper mines, replacement parts were detained in foreign ports, and refining operations faced shortages of imported components. Net Errors and Omissions (*EO*) recorded large negative residuals (−\$72.9 million in 1970, −\$84.5 million in 1971, and −\$171.6 million in 1972, totaling −\$389.0 million across the UP triennium), consistent with unrecorded capital flight and trade misinvoicing. As documented by ?, the presidential election of September 4, 1970 triggered an immediate −48.6% stock market decline and commercial bank deposit runs two full months before Allende assumed office. Private asset holders moved liquid balances across borders, establishing that financial outflow and asset liquidation preceded the implementation of socialist economic policies. While official cash ledgers show an accounting rebound in 1972 capital inflows (+\$327.4M), this reflected the capitalization of unpaid debt service following the November 1971 debt moratorium. Reconciled WBOP accrual accounts reveal that the real economic current account deficit widened to −\$550.8 million (−4.40% of GDP) in 1972 and −\$618.1 million (−3.47% of GDP) in 1973, while liquid Central Bank foreign exchange fell below \$30 million.

Following the September 1973 coup, the military junta re-oriented accumulation toward financial conglomerates and foreign capital. External commercial credits were restored ($KA = +\$228.0M$ in 1974 and +\$298.7M in 1975), and Paris Club obligations were rescheduled. When world copper export prices collapsed in 1975, the economic team implemented orthodox shock therapy to compress domestic import absorption. This severe import compression failed to eliminate external vulnerability: the accrual current account deficit deteriorated to −\$491.3 million (−5.85% of GDP in WBOP accounts), net international reserves plunged to negative −\$129.2 million, and domestic manufacturing contracted sharply, demonstrating the structural persistence of external vulnerability.

5.2.3 Terms of Trade, Capacity to Import, and Real Structural Imbalance (1960–1980)

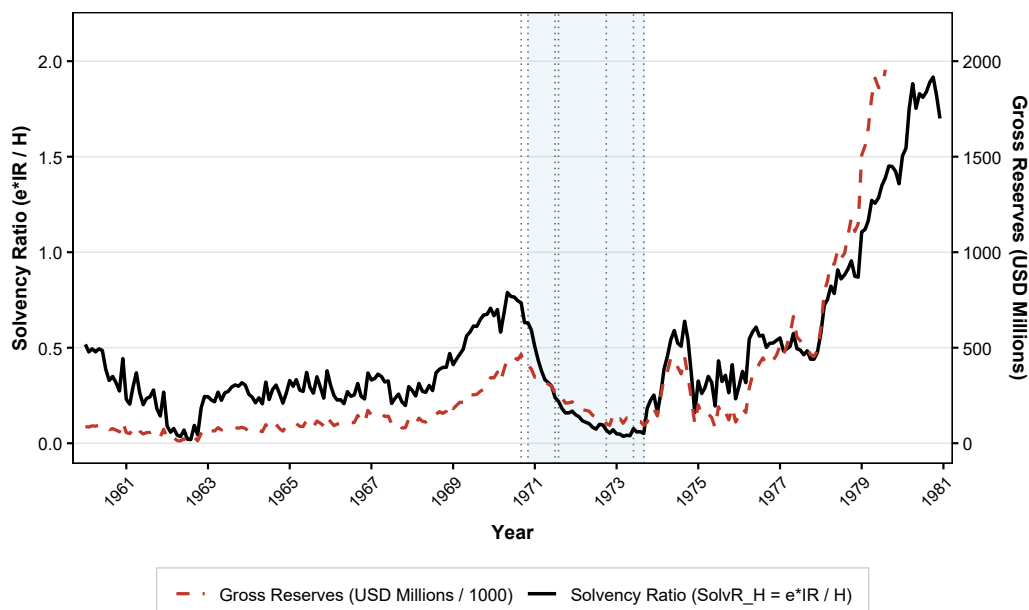
To evaluate the physical transmission of the external constraint, Figures 3, 4, and 5 trace the monthly trajectories of foreign reserves, international commodity prices, and external purchasing power across the long accumulation cycle (1960–1980, $N = 252$ continuous calendar months in the raw historical panel, yielding $N = 250$ usable observations in the transformed econometric estimation sample).

The international terms of trade were destabilized by a dual global shock (Figure 4). World copper export prices experienced an acute 25% downturn during 1971 (averaging \$0.47/lb compared to \$0.64/lb in 1969–1970), curtailing export proceeds during the initial phase of redistributive wage expansion. Simultaneously, the global capitalist economy absorbed the collapse of the Bretton Woods monetary architecture following the unilateral closing of the US gold window on August 15, 1971 (Juncture IV). As ? and ? demonstrate, this structural rupture dismantled the anchor of the postwar monetary order and unleashed worldwide primary commodity price instability. Gold prices rose past \$100/oz in 1973 and \$200/oz by 1978, elevating the foreign-currency cost of Chile’s essential food, fertilizer, and petroleum imports.

Figure 5 highlights a prominent paradox in UP historiography: the nominal world price of copper surged from \$0.508/lb in January 1973 to \$0.948/lb in August 1973 while Central Bank foreign exchange reserves remained severely depleted, hovering near their effective operating floor (\$104–\$114 million). The resolution lies in the divergence between external purchasing power and effective import requirements.

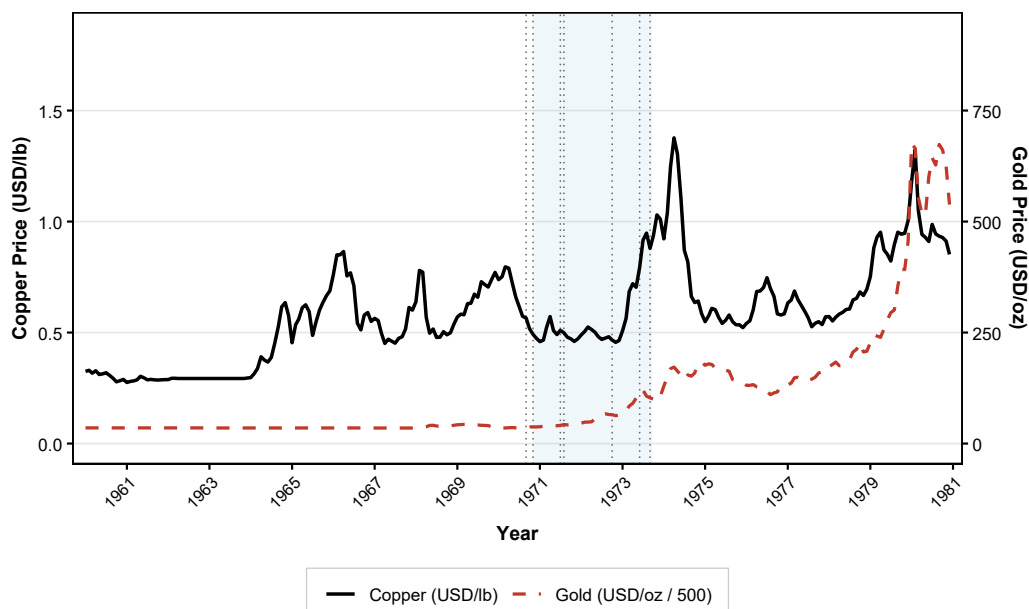
While nominal copper prices rose, domestic transport disruptions and foreign equipment embargoes depressed physical export volumes. Simultaneously, wage expansion and speculative domestic hoarding required large inflows of basic food and fuel imports. Effective real imports

Figure 3: Central Bank Solvency Ratio (SolvR^H) and Gross International Reserves (1960–1980)



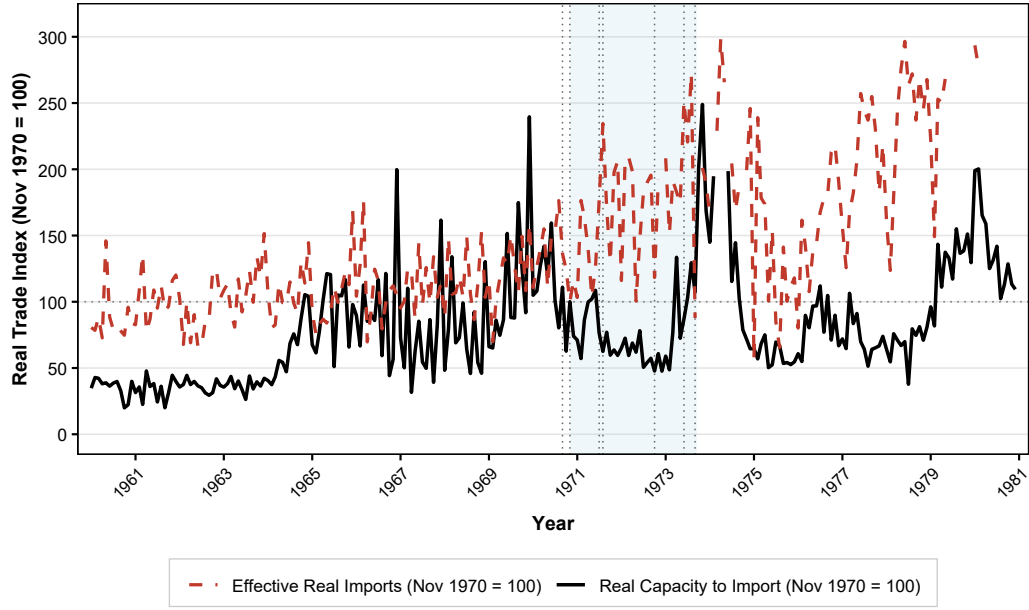
Notes: Monthly series tracking the Central Bank Solvency Ratio relative to Base Money ($\text{SolvR}_t^H \equiv e_t \cdot IR_t / [H_t \cdot 1000]$, left axis, solid black line) and IMF gross international reserves (IR_t in USD Millions / 1000, right axis, dashed red line). Base money (H_t) represents the Central Bank's direct domestic liability base, providing complete continuous archival coverage from 1960 to 1980 without splicing. Critical juncture markers I–VII defined in Table 6. Shaded blue region designates the Unidad Popular government (November 1970 – September 1973). Sourced from Central Bank of Chile REST API and IMF International Financial Statistics.

Figure 4: World Copper Price and London Gold Price (1960–1980)



Notes: Monthly series tracking the London Metal Exchange copper cash price (P^{Cu} in USD/lb, left axis, solid black line) and London Bullion Market Association gold price (P^{Gold} in USD/oz / 500, right axis, dashed red line). Critical juncture markers I–VII defined in Table 6. Shaded blue region designates the Unidad Popular government (November 1970 – September 1973). Sourced from LME and LBMA via Central Bank of Chile REST API.

Figure 5: Prebisch-ECLAC Real Capacity to Import and Effective Real Imports (1960–1980)



Notes: Monthly trajectory tracking the Prebisch-ECLAC Real Capacity to Import ($M_t^{\text{cap}} \equiv [\text{Exports}_t / \text{Exports}_{70:11}] \times [(P_t^{\text{Cu}} / P_{70:11}^{\text{Cu}}) / (P_{W,t}^{\text{US}} / P_{W,70:11}^{\text{US}})] \times 100$, solid black line) against Effective Real Imports ($M_t \equiv \text{Imports}_t / \text{Imports}_{70:11} \times 100$, dashed red line). Both series normalized to November 1970 = 100. Critical juncture markers I–VII defined in Table 6. Shaded blue region designates the Unidad Popular government (1970–1973). Physical merchandise export and import volume series sourced from ? historical database, compiled from *Estadística Chilena* and customs records (with April 1973 administrative registration drop interpolated); copper prices from LME via Central Bank of Chile REST API; U.S. wholesale prices from BLS.

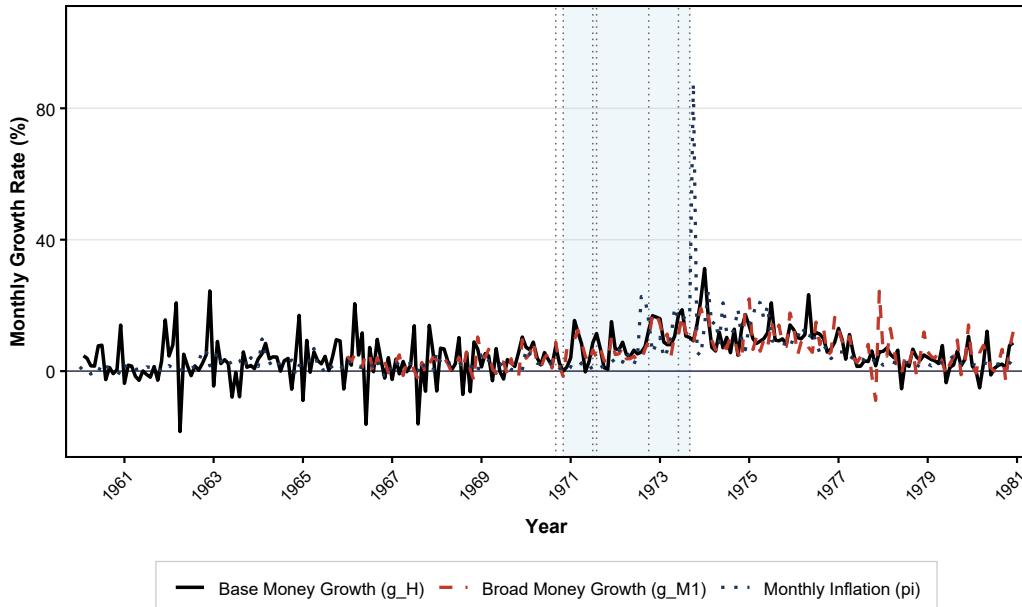
(M_t) reached 234.5% of baseline in August 1971 and 272.3% in August 1973. Operationalizing the Prebisch-ECLAC Capacity to Import (M_t^{cap}), Figure 5 shows that Chile’s real export purchasing power fell from 100.0 in November 1970 to 62.5 in August 1971 and 47.7 in October 1972. This opened a wide Real Structural Imbalance ($\Theta_t \equiv M_t/M_t^{\text{cap}} \times 100$), which reached 374.9% in August 1971 and 346.4% in December 1971 (Table 5).

The widening of Θ_t shows that rising nominal copper prices were insufficient to restore external purchasing capacity, and that severe foreign-exchange pressure was visible well before the subsequent acceleration of inflation and monetary accommodation.

5.2.4 Nominal Dynamics: Endogenous Credit Money and Multiplier Compression

Figures 6 and 7 evaluate the high-frequency interaction between monetary emission, consumer price inflation, and banking system intermediation across the complete 1960–1980 sample.

Figure 6: Monetary Expansion Rates and Monthly Consumer Inflation (1960–1980)

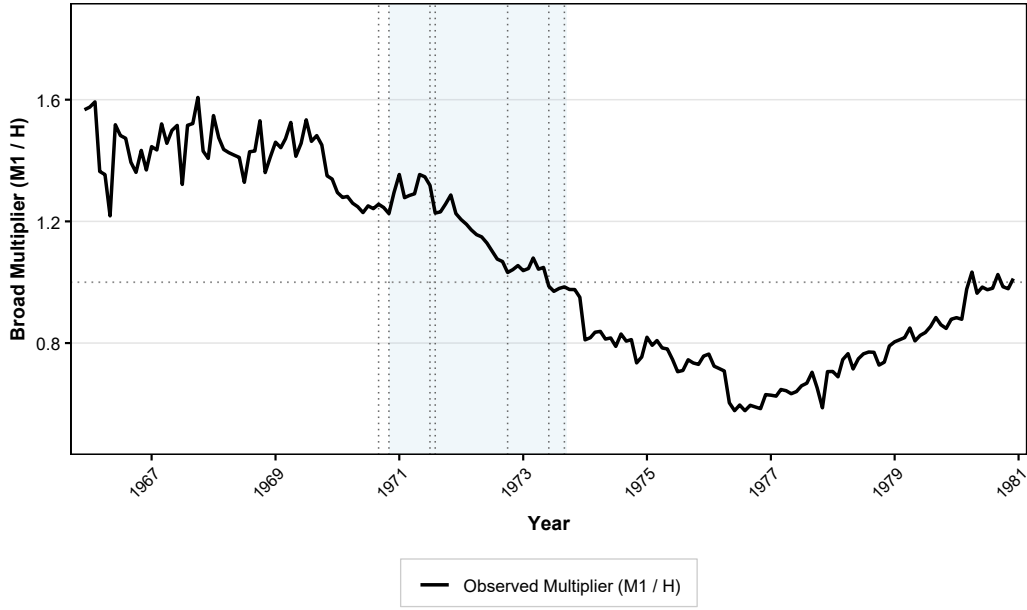


Notes: Monthly series of base money growth ($g_{H,t}$, solid black line), broad money supply growth ($g_{M1,t}$, dashed red line), and monthly consumer price inflation (π_t , dotted blue line). Critical juncture markers I–VII defined in Table 6. Shaded blue region designates the Unidad Popular government (1970–1973). Sourced from Banco Central de Chile REST API and Instituto Nacional de Estadísticas (INE).

In Figure 6, base money emission (g_H) and broad money (g_{M1}) grew at monthly rates of 10%–15% through 1971 and early 1972, accompanying physical output expansion and legislated wage adjustments. Monthly inflation remained below 5% during this initial phase, jumping to 15.2% only after the October 1972 employers’ lockdown disrupted transport logistics and encouraged private retail withholding. Confronted with enterprise cash deficits and rising wholesale costs, the Central Bank expanded credit defensively to sustain production and payrolls. The historical peak of monthly inflation occurred in October 1973 (+87.6%) under the military junta’s Decree Law 522 price deregulation. This observed sequence is consistent with the hypothesis of defensive monetary accommodation, tested formally in Section 5.3.

Figure 7 tracks the empirical broad-to-base money multiplier ($m_t \equiv M1_t/H_t$). Under Frei (1966–1970), the multiplier remained stable between 1.35 and 1.60. Under the UP administration, the multiplier trended downward after early 1971, falling from 1.35 in January 1971 to 0.98

Figure 7: Observed Empirical Money Multiplier ($M1/H$, 1965–1980)



Notes: Monthly trajectory of the broad-to-base money multiplier ($m_t \equiv M1_t/H_t$, solid black line). Plot restricted strictly to the genuinely observed historical $M1$ data range (December 1965 to December 1980), eliminating any artificial pre-1966 flat constant multiplier splicing. Critical juncture markers I–VII defined in Table 6. Shaded blue region designates the Unidad Popular government (1970–1973). Sourced from Banco Central de Chile REST API.

in August 1973, before reaching a trough of 0.58 in 1976. This downward movement is consistent with private banking disintermediation, increased cash holdings by households confronting retail shortages, and direct central bank rediscount lending to socialized enterprises. Accelerating inflation co-moves visually with this compression in the broad-to-base ratio, establishing the descriptive basis for the multiplier deconstruction tested in Section 5.3.

5.2.5 Real Production Divergence and Chronological Shock Sequences

Figure 8 plots physical output dynamics across the complete 1960–1980 sample, displaying an acute divergence between domestic manufacturing and the export mining enclave.

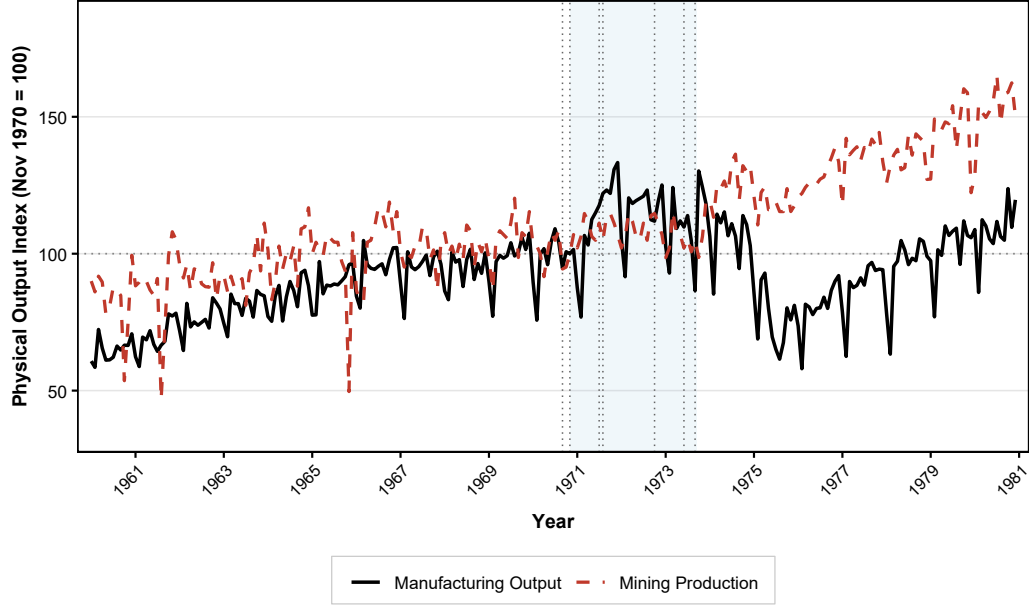
Real manufacturing output expanded by +33.3% between November 1970 and December 1971, consistent with the rapid utilization of idle capacity in wage-goods to meet expanding popular consumption. In contrast, capital-intensive copper mining grew by only +7.0%, constrained by technical bottlenecks and external equipment embargoes. Following the October 1972 lockout, manufacturing output contracted sharply under transport paralysis and intermediate input shortages.

Following the September 1973 coup, manufacturing output fell persistently, dropping below index 65 by 1975 (a > 35% contraction from its 1971 peak). The sharp post-coup contraction coincided with orthodox shock therapy, unilateral tariff reductions, and domestic demand compression.

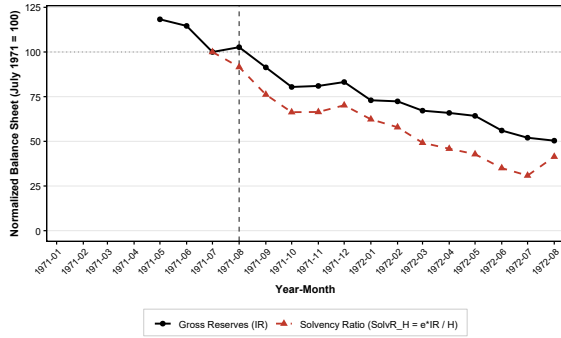
The macroeconomic trajectories were shaped by two decisive institutional turning points in the domestic and international balance of power, isolated through the focused event windows in Figure 9.

The first turning point corresponds to the August 1971 external credit restrictions and Nixon

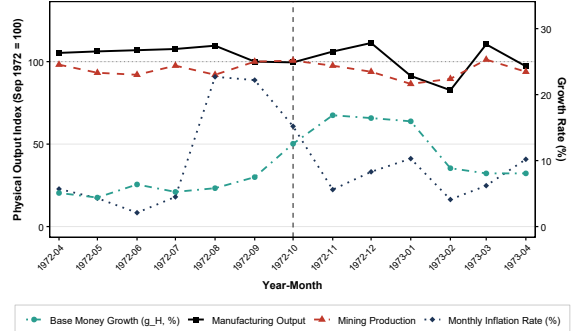
Figure 8: Real Physical Output Trajectories: Manufacturing vs. Mining (1960–1980)



Notes: Monthly series of physical output indices for manufacturing (solid black line) and mining (dashed red line), re-based to November 1970 = 100. Critical juncture markers I–VII defined in Table 6. Shaded blue region designates the Unidad Popular government (1970–1973). Sourced from ? historical database, compiled from INE and SOFOFA for manufacturing, and SERNAGEOMIN, ODEPLAN, and CODELCO for mining.



(a) August 1971 Credit Restrictions and Nixon Shock



(b) October 1972 Bosses' Lockout

Figure 9: High-Frequency Event Windows around External and Supply Shocks (Chile, 1971–1972)

Notes: High-frequency event windows centered at $t = 0$ around two decisive institutional turning points. Panel (a) tracks Central Bank Gross International Reserves (IR_t , solid black line with circle markers) and the Solvency Ratio relative to Base Money ($SolvR_t^H \equiv e_t \cdot IR_t / [H_t \cdot 1000]$, dashed red line with triangle markers), indexed to July 1971 = 100, around the August 1971 Eximbank credit denial and Bretton Woods gold-window closure (Benchmark IV). Panel (b) traces Manufacturing Output (solid black line with square markers) and Mining Production (dashed red line with triangle markers), indexed to September 1972 = 100 on the left axis, against monthly consumer inflation ($\pi_t \times 4$, dotted blue line with diamond markers) and Central Bank base money growth ($g_H \times 4$, dot-dashed teal line with circle markers) on the right axis, around the October 1972 Bosses' Lockout (Benchmark V). Sourced from Banco Central de Chile REST API, IMF International Financial Statistics, and ?.

shock (Figure 9a, $t = 0$). Following the copper nationalization of July 11, the U.S. Export-Import Bank formally denied credit on August 12, and President Nixon unilaterally suspended dollar-gold convertibility on August 15, terminating the Bretton Woods parities (??). This high-frequency window captures the inflection point of the $-\$294.0$ million capital account collapse. Central Bank international reserves fell from index 100.0 to below 30.0, and the Central Bank Solvency Ratio (SolvR_t^H) contracted by more than 70% within twelve months. On the shop floor, replacement tires for Kennecott copper haul trucks were blocked in foreign ports, unrepairable flotation cells idled refining operations, and foreign suppliers demanded 100% cash-in-advance confirmed letters of credit for grain and fuel imports. The external financial chokehold was fully binding before domestic fiscal monetization accelerated.

The second turning point corresponds to the October 1972 Bosses' Lockout (*Paro de Octubre*, Figure 9b, $t = 0$). The private truck owners' strike paralyzed transport and logistics, supported by retail merchant shutdowns. Real manufacturing production fell by 12 percentage points, while mining production exhibited relative geographic insulation in the northern desert and cordillera. Monthly consumer inflation jumped from 2.8% to 15.2%. Central Bank base money emission (g_H) was inert at $t = 0$, remaining flat before accelerating at $t = +1$ and $t = +2$. The event window places the physical output disruption and the price jump before the subsequent acceleration in base-money growth. While this sequence is consistent with defensive accommodation to maintain enterprise payrolls and food logistics, establishing directional precedence requires the multivariate dynamic testing developed below.

5.2.6 Synthesis and Econometric Hand-Off

Table 5 decomposes this macroeconomic trajectory across six decisive structural junctures, and Table 6 formalizes the seven critical institutional turning points (I–VII).

Table 5 makes this temporal ordering visible across six decisive structural junctures: between November 1970 and August 1971—while monthly consumer inflation was at its cyclical low (1.1%) and industrial manufacturing output was expanding rapidly (+22.0%)—the Central Bank Solvency Ratio (SolvR_t^H) had already collapsed from 0.63 to 0.22, and net foreign reserves had fallen by $\$139.9$ million. External balance-of-payments deterioration preceded the subsequent acceleration of domestic inflation and defensive monetary accommodation.

While these stylized facts document that reserve depletion and trade imbalances preceded the domestic price explosion, testing whether money operated as an autonomous cause of inflation or as an endogenous accommodation valve requires formal econometric adjudication. Section 5.3 evaluates directional predictive precedence through a Toda–Yamamoto (1995) Granger causality tournament across five modular empirical systems. Section 5.4 then estimates a non-linear Threshold Vector Autoregression (TVAR) conditioned on the predetermined Solvency Ratio (SolvR_{t-1}^H) to model regime-dependent shock transmission and asymmetric impulse responses under peripheral reserve depletion.

5.3 Empirical Exercise 1: Toda–Yamamoto Directional Precedence and Channel Discrimination

5.3.1 Econometric Framework: Toda–Yamamoto Levels Augmentation

The historical narrative and stylized facts presented in Section 5.2 leave an unresolved empirical question: did monetary expansion act as an autonomous prime mover driving price acceleration, or did it respond defensively to price pressures, enterprise working-capital requirements, and external balance-of-payments constraints? To adjudicate between these competing explanations,

Table 5: Monthly Macroeconomic Dynamics, Central Bank Solvency, and External Constraints across Structural Junctures (Chile, 1970–1973)

Macroeconomic Variable	Nov 1970 <i>Inauguration</i>	Aug 1971 <i>Nixon Shock</i>	Dec 1971 <i>Peak React.</i>	Oct 1972 <i>Bosses' Lock.</i>	Aug 1973 <i>Terminal UP</i>	Dec 1973 <i>Post-Coup</i>
A. Inflation and Price Dynamics						
Monthly Inflation (%)	0.6%	1.1%	2.8%	15.2%	17.1%	4.7%
YoY Inflation (%)	35.3%	17.4%	22.1%	142.9%	303.6%	508.1%
B. Real Physical Activity (Nov 1970 = 100)						
Manufacturing Output Index	100.0	122.0	133.3	111.9	104.6	118.1
Mining Production Index	100.0	104.9	107.0	114.7	100.7	119.6
C. Central Bank Solvency and Monetary Backing						
International Reserves (USD M)	408.0	268.1	217.2	103.4	114.2	179.7
Solvency Ratio Base ($e \cdot IR/H$)	0.63	0.22	0.17	0.07	0.06	0.25
Solvency Ratio M1 ($e \cdot IR/M1$)	0.51	0.18	0.14	0.07	0.06	0.27
Observed Multiplier ($M1/H$)	1.23	1.23	1.23	1.03	0.98	0.95
Base Money Monthly Growth (g_H)	0.0%	11.5%	15.1%	12.6%	10.7%	21.8%
M1 Money Monthly Growth (g_{M1})	-1.6%	4.3%	10.2%	9.1%	11.6%	19.2%
D. Official Exchange Rate and World Commodity Prices						
Official Exchange Rate (e_t)	0.012	0.0122	0.0146	0.0250	0.0740	0.34
World Copper Price (USD/lb)	\$0.492	\$0.499	\$0.471	\$0.466	\$0.948	\$1.011
World Gold Price (USD/oz)	\$37.44	\$42.71	\$43.48	\$64.86	\$106.76	\$106.72
E. External Trade, Capacity to Import, and Structural Imbalance						
Export Activity Index	2.68	1.71	1.73	1.46	2.31	2.81
Import Activity Index	1.51	3.54	3.11	1.79	4.11	2.86
Real Trade Ratio (Exp/Imp)	1.78	0.48	0.56	0.82	0.56	0.98
Real Capacity to Import (M_t^{cap} , Nov 1970 = 100)	100.0	62.5	59.5	47.7	129.5	168.7
Effective Real Imports (M_t , Nov 1970 = 100)	100.0	234.5	206.2	118.4	272.3	189.4
Real Structural Imbalance (Θ_t , %)	100.0%	374.9%	346.4%	248.1%	210.3%	112.3%
US Wholesale Price Index (Nov 1970 = 100)	100.0	103.8	104.0	108.1	128.0	127.8

Sources and Notes: Primary monthly series from Central Bank of Chile and IMF International Financial Statistics. Solvency Ratio defined relative to Base Money as $\text{SolvR}_t^H \equiv (e_t \cdot IR_t)/H_t$ and relative to M1 as $\text{SolvR}_t^{M1} \equiv (e_t \cdot IR_t)/M1_t$. Capacity to Import Index operationalizes the Prebisch-ECLAC formula: $M_t^{\text{cap}} \equiv (\text{Exports}_t/\text{Exports}_{70:11}) \times [(P_t^{\text{Cu}}/P_{70:11}^{\text{Cu}})/(P_{W,t}^{\text{US}}/P_{W,70:11}^{\text{US}})] \times 100$. Real Structural Imbalance Ratio defined as $\Theta_t \equiv (M_t/M_t^{\text{cap}}) \times 100$, tracking the physical quantum of imports relative to real export purchasing power. Dates correspond to critical historical milestones: Nov 1970 (Allende inauguration baseline); Aug 1971 (the decisive global turning point: Eximbank credit denial and Nixon Shock, ending dollar-gold convertibility); Dec 1971 (peak Keynesian wage-led reactivation); Oct 1972 (nationwide Bosses' Lockout); Aug 1973 (terminal constitutional boundary); and Dec 1973 (post-coup Decree Law 522 price deregulation).

Table 6: Chronology of Critical Macroeconomic Junctures and Institutional Shocks (Chile, 1970–1973)

Ref	Date	Historical Juncture	Institutional Shock Mechanism	Macroeconomic / Balance-Sheet Transmission Channel
I	Sep 1970	Presidential Election	Electoral panic; commercial deposit runs	Elite capital flight ($EO < 0$); foreign currency hoarding; Central Bank emergency discounts.
II	Nov 1970	Inauguration Baseline	Allende assumes presidency; tariff and price freeze	Initial price stabilization; real wage gains; inherited reserve buffer (\$408.0M; $SolvR = 0.51$).
III	Jul 1971	Copper Nationalization	Unanimous constitutional reform (Law 17,450)	Sovereign recovery of <i>Gran Minería</i> ; triggers bilateral U.S. financial retaliation and legal liens.
IV	Aug 1971	Eximbank Embargo & Bretton Woods Collapse	U.S. credit freeze; dollar-gold convertibility suspended	Revolving trade credits slashed (\$220M $\rightarrow$ \$35M); cash-in-advance imports; reserve drain accelerates ($\Delta IR = -\$140M$).
V	Oct 1972	Bosses' Lockout (<i>Paro de Octubre</i>)	Trucking strike; SOFOFA wholesale lockout	Physical transport paralysis; wholesale withholding; inflation spikes to 15.2%; emergency wage monetization.
VI	Jun 1973	<i>El Tanquetazo</i>	Military uprising (2nd Armored Regiment)	Heightened political instability; logistical sabotage; black market expansion; acute reserve depletion and real structural imbalance ($\Theta_t = 210.3\%$).
VII	Sep 1973	Military Coup d'État	Violent overthrow; Decree Law 522 price decontrol	Abolition of price controls; +87.6% MoM inflation jump; real wage collapse (-40%); closing of scissors via devaluation.

Notes: Master reference chronology for vertical benchmark lines appearing across high-frequency time series (Figures 3–7) and defining intervention cutoffs for event studies (Figures 9a–9b) and recursive parameter horizons in companion econometric estimation stages.

I evaluate directional predictive precedence across the monthly historical record (1960–1980). Under orthodox theoretical accounts (Monetarism and Austrian credit-cycle models), money supply creation operates as an exogenous impulse, implying the testable null restriction:

$$\mathcal{H}_0^{\text{Exogenous}} : \Delta\pi_{t-k} \not\rightarrow \Delta g_{M,t} \quad \text{and} \quad \text{External Shock}_{t-k} \not\rightarrow \Delta g_{M,t} \quad \forall k \geq 1. \quad (10)$$

Under this view, domestic base money ($g_{H,t}$) and narrow transactions balances ($g_{M1,t}$) should expand independently of lagged feedback from consumer prices (π_t) or external foreign exchange bottlenecks.

Conversely, the structuralist and Marxian monetary circuit framework developed in Section 3 treats money creation as an endogenous liability tied to the social reproduction of enterprise production ($M - C - M'$). While Post-Keynesian endogenous money theory provides the operational banking sequence ($\Delta\text{Loans} \rightarrow \Delta\text{Deposits} \rightarrow \Delta H$; ???), the Marxian framework grounds the economic driver of that sequence in working-capital defense under external financial subordination. Facing cost-push shocks, distribution bottlenecks, and price controls, enterprises experienced acute working-capital deficits required to meet payrolls and purchase inputs. To prevent production shutdowns and payments disruptions, the central bank accommodated commercial credit demands ex-post. Under this structuralist-Marxian hypothesis, directional predictive precedence should flow from inflation, structural trade bottlenecks, and reserve depletion to money growth ($\pi \rightarrow g_H$, $\Theta \rightarrow \pi$, $SolvR^H \rightarrow g_H$).

Within a Lakatosian framing, Granger causality testing does not claim to establish deep structural parameters or abstract value determination; rather, it establishes a disciplined, *minimal predictive-precedence bar*. If orthodox monetarism fails even at this minimal predictive threshold, its causal interpretation is undermined on its own empirical terms. Directional precedence between series pairs (y_t, x_t) is evaluated within an augmented levels vector autoregression following ? and ?, adopting the specification mapping framework of ?. Let k denote the optimal lag length of the system (selected by SBIC), and let $d_{\max}$ denote the maximal order of integration

of the underlying series. An augmented VAR($k + d_{\max}$) is specified:

$$y_t = \alpha_0 + \sum_{i=1}^k \alpha_{1i} y_{t-i} + \sum_{j=k+1}^{k+d_{\max}} \alpha_{2j} y_{t-j} + \sum_{i=1}^k \delta_{1i} x_{t-i} + \sum_{j=k+1}^{k+d_{\max}} \delta_{2j} x_{t-j} + \varepsilon_{1t}, \quad (11)$$

$$x_t = \beta_0 + \sum_{i=1}^k \beta_{1i} x_{t-i} + \sum_{j=k+1}^{k+d_{\max}} \beta_{2j} x_{t-j} + \sum_{i=1}^k \phi_{1i} y_{t-i} + \sum_{j=k+1}^{k+d_{\max}} \phi_{2j} y_{t-j} + \varepsilon_{2t}, \quad (12)$$

where $\varepsilon_{1t}, \varepsilon_{2t}$ are white-noise disturbances.

The Toda–Yamamoto procedure permits standard asymptotic inference in levels VARs even in the presence of integrated or cointegrated processes. By adding $d_{\max}$ extra, unconstrained lags to the estimated equations while restricting linear hypothesis tests strictly to the first k structural lag coefficients:

$$H_0 : \delta_{11} = \dots = \delta_{1k} = 0, \quad (13)$$

the resulting Modified Wald (MWALD) test statistic converges asymptotically to a standard $\chi^2(k)$ distribution, bypassing the pre-testing biases that can arise in vector error-correction modeling.

Hypothesis Testing and Directional Decision Rules. For each pairwise relationship, symmetrical null hypotheses of non-causality ($X \nrightarrow Y$ and $Y \nrightarrow X$) are tested. The cumulative dynamic magnitude and sign are captured by the sum of structural lagged coefficients ($\sum_{i=1}^k \hat{\delta}_{1i}$ or $\sum_{i=1}^k \hat{\phi}_{1i}$). Directional patterns are classified into four outcomes: unidirectional causality ($X \rightarrow Y$ or $Y \rightarrow X$), bilateral feedback ($X \leftrightarrow Y$), or no predictive relation ($X \perp Y$) at conventional significance levels ($p < 0.05$).

Crucially, **no exogeneity restrictions are imposed ex-ante**: all equations are estimated in unrestricted form. Even for international boundary variables such as world gold prices ($g_{P,\text{gold},t}$), domestic Chilean coefficients are not constrained to zero during estimation. Exogeneity is treated as an empirical finding, established only if the joint MWALD test fails to reject the null of non-causality from all domestic system variables ($p > 0.10$).

Table 7 defines the system variable vectors $Y_t^{(s)}$, mathematical transformations, and sample coverage across the six empirical systems.

5.3.2 Integration Order and Levels Augmentation ($d_{\max} = 1$)

As established in the unit root diagnostics of Table 1 (Section 4), the underlying macroeconomic log-level series ($\ln H_t, \ln \text{CPI}_t, \dots$) exhibit unit roots in levels and stationarity in first differences ($p < 0.01$). Setting $d_{\max} = 1$ provides a conservative upper bound for any integrated or near-integrated persistence in the underlying series, formally satisfying the levels-augmentation precondition for the Toda–Yamamoto estimation without differencing away long-run level relationships.

5.3.3 Modular Systems and Specification Mapping

To evaluate directional precedence across distinct macroeconomic transmission mechanisms without finite-sample distortion from over-parameterization, the Toda–Yamamoto estimation is organized into six modular empirical systems: (1) the Bivariate Master Nominal Core (Table 8); (2) the Commercial Banking Credit Lead (Table 9, Panel A); (3) the Money Multiplier

Table 7: System Variable Specification and Definitions Across Sequential Testing Steps

System	System Vector $Y_t^{(s)}$	K	Variable Definitions, Construction, and Historical Sample
System 1	$[\pi_t, g_{H,t}]'$	2	π_t : Monthly CPI inflation rate ($\Delta \ln \text{CPI}_t \times 100$). $g_{H,t}$: Base money growth ($\Delta \ln H_t \times 100$). Sample: 1960:01–1980:12 ($N = 250$).
System 2	$[\pi_t, g_{M1,t}, g_{H,t}]'$	3	π_t : CPI inflation rate. $g_{M1,t}$: Official observed narrow money growth ($\Delta \ln M1_t \times 100$, un-spliced). $g_{H,t}$: Base money growth. Sample: 1966:01–1980:12 ($N = 180$).
System 3	$[\pi_t, g_{H,t}, \Delta \ln m_t]'$	3	π_t : CPI inflation rate. $g_{H,t}$: Base money growth. $\Delta \ln m_t$: Banking multiplier growth ($m_t \equiv M1_t/H_t$). Sample: 1966:01–1980:12 ($N = 180$).
System 4	$[\pi_t, g_{H,t}, \Delta \ln \Theta_t, g_{\text{Manuf},t}, g_{\text{Mining},t}]'$	5	π_t : CPI inflation rate. $g_{H,t}$: Base money growth. $\Delta \ln \Theta_t$: Real Structural Imbalance growth ($\Theta_t \equiv \text{Imports}_t/M_t^{\text{cap}} \times 100$). $g_{\text{Manuf},t}$: Manufacturing output growth ($\Delta \ln \text{Manuf}_t \times 100$). $g_{\text{Mining},t}$: Enclave mining output growth ($\Delta \ln \text{Mining}_t \times 100$). Sample: 1960:01–1980:12 ($N = 250$).
System 5	$[g_{P,\text{gold},t}, g_{e,t}, \Delta \ln \Theta_t, g_{H,t}, \pi_t]'$	5	$g_{P,\text{gold},t}$: World gold price growth ($\Delta \ln P_t^{\text{gold}} \times 100$, London bullion). $g_{e,t}$: Nominal exchange rate growth ($\Delta \ln e_t \times 100$). $\Delta \ln \Theta_t$: Real Structural Imbalance growth. $g_{H,t}$: Base money growth. π_t : CPI inflation rate. Sample: 1960:01–1980:12 ($N = 250$).
System 6	$[\Delta \ln \text{SolvR}_t^H, g_{P,\text{gold},t}, g_{e,t}, g_{H,t}, \pi_t]'$	5	$\Delta \ln \text{SolvR}_t^H$: Central Bank Solvency Ratio growth based on Base Money ($\text{SolvR}_t^H \equiv [e_t \cdot \text{IR}_t]/[H_t \cdot 1000]$). $g_{P,\text{gold},t}$: World gold price growth. $g_{e,t}$: Nominal exchange rate growth. $g_{H,t}$: Base money growth. π_t : CPI inflation rate. Sample: 1960:01–1980:12 ($N = 250$).

Notes: Monthly archival series from Central Bank of Chile (BCCCh) historical databases and IMF International Financial Statistics. Prebisch Capacity to Import ($M_t^{\text{cap}} \equiv \text{Exports}_t \times \text{TOT}_t$) uses November 1970 as standard baseline (1970 : 11 = 100). Solvency Ratio based on Base Money (SolvR_t^H) provides complete coverage for all 250 months (1960–1980), correlating at $\rho = 0.974$ with the narrow-money metric while preserving the full historical sample for macro progressions. The single-month administrative recording drop of April 1973 in import records is resolved via linear interpolation between March and May 1973.

Deconstruction (Table 9, Panel B); (4) the Real Sector Dual Economy (Table 10); (5) the External Sector and Balance-of-Payments Pressures (Table 11); and (6) the Central Bank Foreign Reserve Solvency System (Table 12).

Following ?, each system is reported across a full lag grid $k \in \{1, 2, 3, 4\}$, with the Schwarz Bayesian Information Criterion (SBIC) optimal lag length k^* designated by superscript ^a. Reporting the full grid ensures that inferences do not depend on an isolated lag choice, allowing the reader to assess the stability of signs ($\sum \hat{\beta}_i$) and test statistics across horizons.

5.3.4 Empirical Findings across Transmission Channels

System 1: Master Nominal Core ($N = 250$, Table 8). The bivariate Toda–Yamamoto tests evaluate whether money creation leads inflation ($H_0 : g_H \not\rightarrow \pi$) or accommodates price acceleration ($H_0 : \pi \not\rightarrow g_H$). SBIC selects lag order $k^* = 3$. The null of non-causality from inflation to base money growth is rejected across every lag specification: at $k = 1$, $F = 28.324$ ($p < 0.0001$); at the optimal lag $k^* = 3$, $F = 15.956$ ($\chi^2 = 47.869$, $p < 0.0001$); and at $k = 4$, $F = 11.635$ ($p < 0.0001$). The sum of structural lagged coefficients rises monotonically from $\sum \hat{\beta}_i = +0.298$ at $k = 1$ to $+0.639$ at $k = 4$.

In the reverse direction, base money growth predicts inflation at short horizons ($F = 14.557$, $p = 0.0002$ at $k = 1$; $F = 4.475$, $p = 0.0124$ at $k = 2$), but this relationship weakens and fails to reject at the SBIC-selected optimal lag ($F = 2.015$, $p = 0.1124$ at $k^* = 3$) and at $k = 4$ ($F = 1.843$, $p = 0.1213$). Thus, while short-run nominal feedback is bilateral, reverse accommodation ($\pi \rightarrow g_H$) is the more persistent relationship across the specification space. This pattern is consistent with defensive working-capital accommodation: accelerating prices expanded enterprise financing requirements, prompting ex-post liquidity provision by the central bank.

Table 8: Toda–Yamamoto Granger Causality: Master Nominal Core System (Specification Map $k = 1..4$, $N = 250$)

Dependent Variable (DV)	Regressor (INDV)	Lag (k)	F -Statistic	χ^2 (MWALD)	p -value	$\sum \hat{\beta}_i$	Direction
Base Money Growth ($\Delta g_{H,t}$)	Inflation ($\Delta \pi_t$)	1	28.324***	28.324	< 0.0001	+0.298	$\pi \rightarrow g_H$
Base Money Growth ($\Delta g_{H,t}$)	Inflation ($\Delta \pi_t$)	2	9.142***	18.285	0.0001	+0.303	$\pi \rightarrow g_H$
Base Money Growth ($\Delta g_{H,t}$)	Inflation ($\Delta \pi_t$)	3 ^a	15.956***	47.869	< 0.0001	+0.567	$\pi \rightarrow g_H$
Base Money Growth ($\Delta g_{H,t}$)	Inflation ($\Delta \pi_t$)	4	11.635***	46.538	< 0.0001	+0.639	$\pi \rightarrow g_H$
Inflation ($\Delta \pi_t$)	Base Money Growth ($\Delta g_{H,t}$)	1	14.557***	14.557	0.0002	+0.243	$g_H \rightarrow \pi$
Inflation ($\Delta \pi_t$)	Base Money Growth ($\Delta g_{H,t}$)	2	4.475**	8.950	0.0124	+0.269	$g_H \rightarrow \pi$
Inflation ($\Delta \pi_t$)	Base Money Growth ($\Delta g_{H,t}$)	3 ^a	2.015	6.046	0.1124	+0.254	No
Inflation ($\Delta \pi_t$)	Base Money Growth ($\Delta g_{H,t}$)	4	1.843	7.373	0.1213	+0.331	No

Notes: Monthly series spanning 1960:01–1980:12 ($N = 250$). Evaluated via Toda–Yamamoto (1995) augmented levels VAR ($k + d_{\max}$ with $d_{\max} = 1$) following ?. F -statistics and asymptotic χ^2 Modified Wald (MWALD) evaluate the null hypothesis of non-causality ($H_0 : \beta_1 = \dots = \beta_k = 0$). $\sum \hat{\beta}_i$ is the sum of structural lagged coefficients indicating dynamic direction and sign. Reverse monetary accommodation ($\pi \rightarrow g_H$) is statistically significant across all evaluated lags ($F = 15.956$, $p < 0.0001$ at the SBIC-selected optimal lag $k^* = 3$; $F = 28.324$, $p < 0.0001$ at $k = 1$), whereas forward transmission ($g_H \rightarrow \pi$) is significant at short horizons ($F = 14.557$, $p = 0.0002$ at $k = 1$) but fails to reject at longer horizons ($F = 2.015$, $p = 0.1124$ at $k^* = 3$). ^a Denotes the selected optimum lag length (k^*) of the ($k + d_{\max}$) augmented VAR based on the Schwarz Bayesian Information Criterion (SBIC) under $d_{\max} = 1$. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

System 2: Commercial Banking Credit Lead ($N = 180$, Table 9, Panel A). Evaluating un-spliced narrow money ($M1$) alongside Central Bank base money (H) and price inflation reveals bilateral predictive feedback at the SBIC-selected optimal lag ($k^* = 1$). Narrow money growth predicts base money expansion ($F = 12.950$, $p = 0.0004$, $\sum \hat{\beta}_i = +0.388$), while base money growth also predicts narrow money ($F = 16.906$, $p < 0.0001$, $\sum \hat{\beta}_i = +0.300$). The

inflation– $M1$ pair shows a similar bilateral pattern at $k^* = 1$ ($\pi \rightarrow M1$: $F = 4.360$, $p = 0.0383$, $\sum \hat{\beta}_i = +0.100$; $M1 \rightarrow \pi$: $F = 5.060$, $p = 0.0258$, $\sum \hat{\beta}_i = +0.256$). Commercial credit expansion and central bank reserve emission operated in mutual feedback, consistent with an institutional environment where commercial banks extended credit to enterprises facing escalating nominal costs, subsequently drawing down central bank liquidity to settle clearing obligations.

System 3: Money Multiplier Deconstruction ($N = 180$, Table 9, Panel B). Deconstructing broad money into base money and the banking multiplier ($m_t \equiv M1_t/H_t$) yields an explicit test of autonomous multiplier mechanisms. Changes in the banking multiplier fail to predict price inflation across all lag specifications ($F = 0.255$, $p = 0.6145$, $\sum \hat{\beta}_i = -0.056$ at $k^* = 1$; $F = 0.121$, $p = 0.9478$ at $k = 3$). In contrast, price inflation predicts a subsequent compression of the banking multiplier at quarterly and four-month horizons ($F = 4.147$, $\chi^2 = 12.440$, $p = 0.0073$, $\sum \hat{\beta}_i = -0.193$ at $k = 3$; $F = 4.015$, $p = 0.0039$, $\sum \hat{\beta}_i = -0.228$ at $k = 4$). Rather than driving inflation through fractional-reserve expansion, the credit multiplier was compressed endogenously as escalating inflation prompted the public to shift from bank deposits into currency and physical commodities.

Table 9: Toda–Yamamoto Granger Causality: Commercial Banking Bifurcation and Multiplier Breakdown ($N = 180$)

Dependent Variable (DV)	Regressor (INDV)	Lag (k)	F -Statistic	χ^2 (MWALD)	p -value	$\sum \hat{\beta}_i$	Direction
<i>Panel A: Commercial Credit Lead ($K = 3$, Lags 1–4)</i>							
Base Money Growth ($\Delta g_{H,t}$)	Narrow Money Growth ($\Delta g_{M1,t}$)	1 ^a	12.950***	12.950	0.0004	+0.388	$M1 \rightarrow H$
Base Money Growth ($\Delta g_{H,t}$)	Narrow Money Growth ($\Delta g_{M1,t}$)	2	8.972***	17.945	0.0002	+0.614	$M1 \rightarrow H$
Base Money Growth ($\Delta g_{H,t}$)	Narrow Money Growth ($\Delta g_{M1,t}$)	3	3.825**	11.475	0.0110	+0.619	$M1 \rightarrow H$
Base Money Growth ($\Delta g_{H,t}$)	Narrow Money Growth ($\Delta g_{M1,t}$)	4	2.421*	9.684	0.0505	+0.464	No
Narrow Money Growth ($\Delta g_{M1,t}$)	Base Money Growth ($\Delta g_{H,t}$)	1 ^a	16.906***	16.906	< 0.0001	+0.300	$H \rightarrow M1$
Narrow Money Growth ($\Delta g_{M1,t}$)	Base Money Growth ($\Delta g_{H,t}$)	2	7.749***	15.498	0.0006	+0.433	$H \rightarrow M1$
Narrow Money Growth ($\Delta g_{M1,t}$)	Base Money Growth ($\Delta g_{H,t}$)	3	6.664***	19.993	0.0003	+0.540	$H \rightarrow M1$
Narrow Money Growth ($\Delta g_{M1,t}$)	Base Money Growth ($\Delta g_{H,t}$)	4	5.695***	22.779	0.0003	+0.733	$H \rightarrow M1$
Narrow Money Growth ($\Delta g_{M1,t}$)	Inflation ($\Delta \pi_t$)	1 ^a	4.360**	4.360	0.0383	+0.100	$\pi \rightarrow M1$
Narrow Money Growth ($\Delta g_{M1,t}$)	Inflation ($\Delta \pi_t$)	2	6.153***	12.305	0.0026	+0.194	$\pi \rightarrow M1$
Narrow Money Growth ($\Delta g_{M1,t}$)	Inflation ($\Delta \pi_t$)	3	6.760***	20.281	0.0002	+0.292	$\pi \rightarrow M1$
Narrow Money Growth ($\Delta g_{M1,t}$)	Inflation ($\Delta \pi_t$)	4	7.471***	29.883	< 0.0001	+0.421	$\pi \rightarrow M1$
Inflation ($\Delta \pi_t$)	Narrow Money Growth ($\Delta g_{M1,t}$)	1 ^a	5.060**	5.060	0.0258	+0.256	$M1 \rightarrow \pi$
Inflation ($\Delta \pi_t$)	Narrow Money Growth ($\Delta g_{M1,t}$)	2	3.333**	6.666	0.0380	+0.376	$M1 \rightarrow \pi$
Inflation ($\Delta \pi_t$)	Narrow Money Growth ($\Delta g_{M1,t}$)	3	2.519*	7.557	0.0598	+0.472	No
Inflation ($\Delta \pi_t$)	Narrow Money Growth ($\Delta g_{M1,t}$)	4	1.489	5.957	0.2078	+0.446	No
<i>Panel B: Money Multiplier Deconstruction ($K = 3$, Lags 1–4)</i>							
Inflation ($\Delta \pi_t$)	Multiplier Growth ($\Delta \ln m_t$)	1 ^a	0.255	0.255	0.6145	-0.056	No
Inflation ($\Delta \pi_t$)	Multiplier Growth ($\Delta \ln m_t$)	2	0.175	0.349	0.8400	-0.106	No
Inflation ($\Delta \pi_t$)	Multiplier Growth ($\Delta \ln m_t$)	3	0.121	0.362	0.9478	-0.070	No
Inflation ($\Delta \pi_t$)	Multiplier Growth ($\Delta \ln m_t$)	4	0.299	1.196	0.8783	-0.152	No
Multiplier Growth ($\Delta \ln m_t$)	Inflation ($\Delta \pi_t$)	1 ^a	1.393	1.393	0.2395	-0.059	No
Multiplier Growth ($\Delta \ln m_t$)	Inflation ($\Delta \pi_t$)	2	0.258	0.515	0.7732	-0.034	No
Multiplier Growth ($\Delta \ln m_t$)	Inflation ($\Delta \pi_t$)	3	4.147***	12.440	0.0073	-0.193	$\pi \rightarrow m$
Multiplier Growth ($\Delta \ln m_t$)	Inflation ($\Delta \pi_t$)	4	4.015***	16.059	0.0039	-0.228	$\pi \rightarrow m$

Notes: Monthly series of official un-spliced narrow money ($M1$) and banking multiplier ($m \equiv M1/H$) spanning 1966:01–1980:12 ($N = 180$). Evaluated via Toda–Yamamoto ($k + d_{\max}$) augmented levels VAR ($d_{\max} = 1$). Panel A shows bilateral predictive feedback between commercial credit and base money at the SBIC-selected optimal lag ($M1 \rightarrow H$, $F = 12.950$, $p = 0.0004$; $H \rightarrow M1$, $F = 16.906$, $p < 0.0001$), with commercial credit exhibiting positive cumulative coefficients across all lags. Panel B indicates that changes in the banking multiplier fail to predict price inflation ($m \not\rightarrow \pi$, $p \geq 0.614$), while price inflation predicts a compression of the multiplier at quarterly and four-month horizons ($\pi \rightarrow m$, $F = 4.147$, $p = 0.0073$ at $k = 3$; $F = 4.015$, $p = 0.0039$ at $k = 4$, with negative sums), consistent with deposit flight into currency during high inflation. ^a Denotes the selected optimum lag length (k^*) of the ($k + d_{\max}$) augmented VAR based on the Schwarz Bayesian Information Criterion (SBIC) under $d_{\max} = 1$. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

System 4: Real Sector Dual Economy ($N = 250$, Table 10). The 5-variable real sector specification (SBIC selects $k^* = 2$) demonstrates clear sectoral asymmetry between domestic manufacturing and export mining. Base money growth predicts reductions in manufacturing output at the optimal lag ($F = 8.274$, $p = 0.0003$, $\sum \hat{\beta}_i = -0.510$ at $k^* = 2$) and at lag 3 ($F = 7.001$, $p = 0.0002$, $\sum \hat{\beta}_i = -0.304$). The reverse null is not rejected at $k^* = 2$ ($F = 1.801$,

$p = 0.1674$). The negative coefficient sum indicates that monetary expansion was followed by output contraction, a pattern consistent with defensive central bank accommodation of financially distressed manufacturing firms preceding output slowdowns, rather than output-stimulating credit injections. In contrast, copper mining extraction is completely decoupled from base money growth across all lags ($p \geq 0.362$).

On the external front, structural import imbalance predicts manufacturing output contractions at the one-month horizon ($F = 4.364$, $p = 0.0378$, $\sum \hat{\beta}_i = -0.045$ at $k = 1$), consistent with the disruption of imported industrial intermediate inputs. Manufacturing output in turn dampens inflation with a negative sign (Manuf $\rightarrow \pi$: $F = 7.140$, $p = 0.0010$, $\sum \hat{\beta}_i = -0.219$ at $k^* = 2$; $F = 12.600$, $p = 0.0005$ at $k = 1$), confirming that physical supply bottlenecks exerted upward pressure on prices.

Table 10: Toda–Yamamoto Granger Causality: Real Sector Dual Economy (Manufacturing vs. Enclave Mining, $N = 250$)

Dependent Variable (DV)	Regressor (INDV)	Lag (k)	F -Statistic	χ^2 (MWALD)	p -value	$\sum \hat{\beta}_i$	Direction
Manufacturing Output ($\Delta g_{\text{Manuf},t}$)	Base Money Growth ($\Delta g_{H,t}$)	1	0.932	0.932	0.3352	-0.100	No
Manufacturing Output ($\Delta g_{\text{Manuf},t}$)	Base Money Growth ($\Delta g_{H,t}$)	2 ^a	8.274***	16.547	0.0003	-0.510	$g_H \rightarrow \text{Manuf}$
Manufacturing Output ($\Delta g_{\text{Manuf},t}$)	Base Money Growth ($\Delta g_{H,t}$)	3	7.001***	21.004	0.0002	-0.304	$g_H \rightarrow \text{Manuf}$
Manufacturing Output ($\Delta g_{\text{Manuf},t}$)	Base Money Growth ($\Delta g_{H,t}$)	4	5.309***	21.235	0.0004	-0.275	$g_H \rightarrow \text{Manuf}$
Base Money Growth ($\Delta g_{H,t}$)	Manufacturing Output ($\Delta g_{\text{Manuf},t}$)	1	0.537	0.537	0.4643	-0.026	No
Base Money Growth ($\Delta g_{H,t}$)	Manufacturing Output ($\Delta g_{\text{Manuf},t}$)	2 ^a	1.801	3.602	0.1674	-0.010	No
Base Money Growth ($\Delta g_{H,t}$)	Manufacturing Output ($\Delta g_{\text{Manuf},t}$)	3	3.462**	10.386	0.0170	-0.023	Manuf $\rightarrow g_H$
Base Money Growth ($\Delta g_{H,t}$)	Manufacturing Output ($\Delta g_{\text{Manuf},t}$)	4	2.864**	11.457	0.0241	-0.063	Manuf $\rightarrow g_H$
Mining Extraction ($\Delta g_{\text{Mining},t}$)	Base Money Growth ($\Delta g_{H,t}$)	1	0.078	0.078	0.7799	+0.030	No
Mining Extraction ($\Delta g_{\text{Mining},t}$)	Base Money Growth ($\Delta g_{H,t}$)	2 ^a	0.079	0.157	0.9243	-0.035	No
Mining Extraction ($\Delta g_{\text{Mining},t}$)	Base Money Growth ($\Delta g_{H,t}$)	3	0.664	1.992	0.5748	+0.177	No
Mining Extraction ($\Delta g_{\text{Mining},t}$)	Base Money Growth ($\Delta g_{H,t}$)	4	1.073	4.291	0.3706	-0.056	No
Base Money Growth ($\Delta g_{H,t}$)	Mining Extraction ($\Delta g_{\text{Mining},t}$)	1	0.011	0.011	0.9165	-0.004	No
Base Money Growth ($\Delta g_{H,t}$)	Mining Extraction ($\Delta g_{\text{Mining},t}$)	2 ^a	0.867	1.734	0.4215	+0.068	No
Base Money Growth ($\Delta g_{H,t}$)	Mining Extraction ($\Delta g_{\text{Mining},t}$)	3	0.983	2.948	0.4016	+0.130	No
Base Money Growth ($\Delta g_{H,t}$)	Mining Extraction ($\Delta g_{\text{Mining},t}$)	4	1.090	4.361	0.3620	+0.184	No
Manufacturing Output ($\Delta g_{\text{Manuf},t}$)	Structural Imbalance ($\Delta \ln \Theta_t$)	1	4.364**	4.364	0.0378	-0.045	$\Theta \rightarrow \text{Manuf}$
Manufacturing Output ($\Delta g_{\text{Manuf},t}$)	Structural Imbalance ($\Delta \ln \Theta_t$)	2 ^a	1.499	2.999	0.2254	+0.037	No
Manufacturing Output ($\Delta g_{\text{Manuf},t}$)	Structural Imbalance ($\Delta \ln \Theta_t$)	3	2.174*	6.523	0.0917	+0.008	No
Manufacturing Output ($\Delta g_{\text{Manuf},t}$)	Structural Imbalance ($\Delta \ln \Theta_t$)	4	1.172	4.687	0.3239	+0.022	No
Structural Imbalance ($\Delta \ln \Theta_t$)	Manufacturing Output ($\Delta g_{\text{Manuf},t}$)	1	0.629	0.629	0.4286	-0.183	No
Structural Imbalance ($\Delta \ln \Theta_t$)	Manufacturing Output ($\Delta g_{\text{Manuf},t}$)	2 ^a	0.316	0.632	0.7294	+0.092	No
Structural Imbalance ($\Delta \ln \Theta_t$)	Manufacturing Output ($\Delta g_{\text{Manuf},t}$)	3	1.780	5.340	0.1516	+1.077	No
Structural Imbalance ($\Delta \ln \Theta_t$)	Manufacturing Output ($\Delta g_{\text{Manuf},t}$)	4	3.008**	12.032	0.0190	+2.200	Manuf $\rightarrow \Theta$
Mining Extraction ($\Delta g_{\text{Mining},t}$)	Structural Imbalance ($\Delta \ln \Theta_t$)	1	0.118	0.118	0.7313	-0.006	No
Mining Extraction ($\Delta g_{\text{Mining},t}$)	Structural Imbalance ($\Delta \ln \Theta_t$)	2 ^a	0.034	0.068	0.9666	-0.004	No
Mining Extraction ($\Delta g_{\text{Mining},t}$)	Structural Imbalance ($\Delta \ln \Theta_t$)	3	0.037	0.112	0.9903	+0.004	No
Mining Extraction ($\Delta g_{\text{Mining},t}$)	Structural Imbalance ($\Delta \ln \Theta_t$)	4	0.038	0.152	0.9972	+0.012	No
Structural Imbalance ($\Delta \ln \Theta_t$)	Mining Extraction ($\Delta g_{\text{Mining},t}$)	1	1.714	1.714	0.1917	-0.300	No
Structural Imbalance ($\Delta \ln \Theta_t$)	Mining Extraction ($\Delta g_{\text{Mining},t}$)	2 ^a	1.283	2.565	0.2792	-0.062	No
Structural Imbalance ($\Delta \ln \Theta_t$)	Mining Extraction ($\Delta g_{\text{Mining},t}$)	3	1.067	3.200	0.3640	+0.255	No
Structural Imbalance ($\Delta \ln \Theta_t$)	Mining Extraction ($\Delta g_{\text{Mining},t}$)	4	0.668	2.674	0.6145	-0.227	No
Manufacturing Output ($\Delta g_{\text{Manuf},t}$)	Mining Extraction ($\Delta g_{\text{Mining},t}$)	1	0.025	0.025	0.8754	+0.010	No
Manufacturing Output ($\Delta g_{\text{Manuf},t}$)	Mining Extraction ($\Delta g_{\text{Mining},t}$)	2 ^a	0.647	1.294	0.5244	+0.009	No
Manufacturing Output ($\Delta g_{\text{Manuf},t}$)	Mining Extraction ($\Delta g_{\text{Mining},t}$)	3	1.099	3.297	0.3503	+0.046	No
Manufacturing Output ($\Delta g_{\text{Manuf},t}$)	Mining Extraction ($\Delta g_{\text{Mining},t}$)	4	1.107	4.428	0.3539	+0.054	No
Inflation ($\Delta \pi_t$)	Manufacturing Output ($\Delta g_{\text{Manuf},t}$)	1	12.600***	12.600	0.0005	-0.124	Manuf $\rightarrow \pi$
Inflation ($\Delta \pi_t$)	Manufacturing Output ($\Delta g_{\text{Manuf},t}$)	2 ^a	7.140***	14.280	0.0010	-0.219	Manuf $\rightarrow \pi$
Inflation ($\Delta \pi_t$)	Manufacturing Output ($\Delta g_{\text{Manuf},t}$)	3	4.729***	14.188	0.0032	-0.289	Manuf $\rightarrow \pi$
Inflation ($\Delta \pi_t$)	Manufacturing Output ($\Delta g_{\text{Manuf},t}$)	4	2.960**	11.839	0.0206	-0.365	Manuf $\rightarrow \pi$
Manufacturing Output ($\Delta g_{\text{Manuf},t}$)	Inflation ($\Delta \pi_t$)	1	0.340	0.340	0.5604	-0.062	No
Manufacturing Output ($\Delta g_{\text{Manuf},t}$)	Inflation ($\Delta \pi_t$)	2 ^a	0.813	1.627	0.4446	+0.092	No
Manufacturing Output ($\Delta g_{\text{Manuf},t}$)	Inflation ($\Delta \pi_t$)	3	0.867	2.601	0.4589	+0.065	No
Manufacturing Output ($\Delta g_{\text{Manuf},t}$)	Inflation ($\Delta \pi_t$)	4	3.318**	13.271	0.0115	-0.303	$\pi \rightarrow \text{Manuf}$

Notes: Monthly series spanning 1960:01–1980:12 ($N = 250$). 5-variable system $[\pi_t, g_{H,t}, \Delta \ln \Theta_t, g_{\text{Manuf},t}, g_{\text{Mining},t}]'$ evaluated via Toda–Yamamoto ($k + d_{\max}$) augmented levels VAR ($d_{\max} = 1$). Base money growth predicts reductions in light manufacturing output ($g_H \rightarrow \text{Manuf}$, $F = 8.274$, $p = 0.0003$ at the SBIC-selected optimal lag $k^* = 2$; $F = 7.001$, $p = 0.0002$ at $k = 3$, with negative coefficient sums), consistent with defensive central bank accommodation of distressed enterprises preceding output contraction, while enclave mining remains decoupled from monetary growth ($p \geq 0.362$). Real structural trade imbalances predict lower factory output ($\Theta \rightarrow \text{Manuf}$, $F = 4.364$, $p = 0.0378$ at $k = 1$), and manufacturing output growth predicts lower inflation (Manuf $\rightarrow \pi$, $F = 7.140$, $p = 0.0010$ at $k^* = 2$; $F = 12.600$, $p = 0.0005$ at $k = 1$). ^a Denotes the selected optimum lag length (k^*) of the ($k + d_{\max}$) augmented VAR based on the Schwarz Bayesian Information Criterion (SBIC) under $d_{\max} = 1$. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

System 5: External Sector and Balance-of-Payments Pressures ($N = 250$, Table 11).

The external sector specification confirms the block-exogeneity of international boundary variables (SBIC selects $k^* = 1$). Domestic macroeconomic variables fail to predict London bullion gold prices across all specifications ($p \geq 0.081$). Real structural trade imbalances ($\Delta \ln \Theta_t$) are likewise unpredicted by domestic base money ($F = 0.297$, $p = 0.5865$ at $k^* = 1$) or inflation ($F = 1.950$, $p = 0.1639$ at $k^* = 1$). While insulated from domestic feedback, structural trade imbalances predict consumer price inflation at the optimal lag ($\Theta \rightarrow \pi$: $F = 5.113$, $p = 0.0246$, $\sum \hat{\beta}_i = +0.082$ at $k^* = 1$).

Official exchange rate devaluations exhibit cumulative pass-through into consumer prices at lag 4 ($F = 37.025$, $\chi^2 = 148.100$, $p < 0.0001$, $\sum \hat{\beta}_i = +0.504$), whereas short-run pass-through at $k^* = 1$ is small and statistically insignificant ($F = 1.119$, $p = 0.2912$), reflecting the initial dampening effect of administered price controls. In reverse, price inflation consistently predicts subsequent official exchange rate devaluations ($\pi \rightarrow g_e$: $F = 11.770$, $p = 0.0007$, $\sum \hat{\beta}_i = +0.515$ at $k^* = 1$; $F = 19.294$, $p < 0.0001$ at $k = 4$), capturing the recurring catch-up devaluations undertaken by authorities to restore external competitiveness.

System 6: Central Bank Foreign Reserve Solvency ($N = 250$, Table 12).

The reserve solvency system evaluates dynamic feedback between foreign reserves and central bank liabilities (SBIC selects $k^* = 1$). At the one-month horizon, neither direction between solvency and base money growth rejects the non-causality null ($p = 0.7917$ and $p = 0.6879$). At quarterly horizons, however, reserve solvency depletion predicts base money expansion ($F = 11.149$, $\chi^2 = 33.448$, $p < 0.0001$, $\sum \hat{\beta}_i = -0.312$ at $k = 3$; $F = 8.472$, $p < 0.0001$, $\sum \hat{\beta}_i = -0.285$ at $k = 4$). Domestic money creation does not predict reserve solvency at lags 1–3 ($p \geq 0.0758$), reaching marginal significance only at lag 4 ($F = 2.504$, $p = 0.0430$). Meanwhile, price inflation predicts faster solvency depletion across all specifications ($\pi \rightarrow \text{SolvR}^H$: $F = 8.336$, $p = 0.0042$, $\sum \hat{\beta}_i = -0.716$ at $k^* = 1$).

This horizon-dependent pattern is consistent with the *Peripheral Paradox* formulated in Section 3.4: while domestic credit provision responds endogenously to production needs, foreign exchange reserves represent an externally bounded stock of international liquidity. As reserve buffers erode over several months, the central bank is forced into defensive liquidity expansion to maintain internal enterprise solvency. Because this relationship unfolds with a lag rather than operating as an instantaneous linear rule, it highlights the potential for non-linear regime shifts as reserves approach critical exhaustion, motivating the threshold modeling in Section 5.4.

5.3.5 Synthesis: Comparison of Theoretical Predictions and Directional Evidence

Table 13 synthesizes the empirical evidence across the six transmission channels. Where orthodox monetarism and heterodox structuralist or endogenous-money frameworks propose contrasting directional signs, the empirical data align consistently with the structuralist and reverse-accommodation channels: price inflation predicts base money growth, commercial credit leads reserve money, the banking multiplier is compressed rather than causal, structural trade bottlenecks predict inflation, and reserve solvency depletion precedes monetary accommodation.

5.3.6 Residual Diagnostics and Dynamic Stability

Table 14 reports diagnostic checks on the key equation residuals across the evaluated lag specifications. Across all six systems, the Breusch–Godfrey LM tests fail to reject the null hypothesis of no serial correlation ($p \geq 0.298$), ARCH LM tests show no evidence of autoregressive conditional heteroscedasticity ($p \geq 0.220$), and White tests fail to reject homoscedasticity ($p \geq 0.140$).

Table 11: Toda–Yamamoto Granger Causality: External Sector and Balance-of-Payments Pressures ($N = 250$)

Dependent Variable (DV)	Regressor (INDV)	Lag (k)	F -Statistic	χ^2 (MWALD)	p -value	$\sum \hat{\beta}_i$	Direction
World Gold Price ($g_{P,gold,t}$)	Base Money Growth ($\Delta g_{H,t}$)	1 ^a	2.860*	2.860	0.0921	+0.081	No
World Gold Price ($g_{P,gold,t}$)	Base Money Growth ($\Delta g_{H,t}$)	2	1.843	3.686	0.1606	+0.096	No
World Gold Price ($g_{P,gold,t}$)	Base Money Growth ($\Delta g_{H,t}$)	3	1.126	3.378	0.3392	+0.047	No
World Gold Price ($g_{P,gold,t}$)	Base Money Growth ($\Delta g_{H,t}$)	4	0.930	3.721	0.4470	+0.050	No
World Gold Price ($g_{P,gold,t}$)	Inflation ($\Delta \pi_t$)	1 ^a	2.090	2.090	0.1495	-0.068	No
World Gold Price ($g_{P,gold,t}$)	Inflation ($\Delta \pi_t$)	2	1.817	3.634	0.1648	+0.014	No
World Gold Price ($g_{P,gold,t}$)	Inflation ($\Delta \pi_t$)	3	1.255	3.764	0.2907	-0.040	No
World Gold Price ($g_{P,gold,t}$)	Inflation ($\Delta \pi_t$)	4	1.547	6.188	0.1894	+0.008	No
World Gold Price ($g_{P,gold,t}$)	Structural Imbalance ($\Delta \ln \Theta_t$)	1 ^a	2.025	2.025	0.1560	+0.010	No
World Gold Price ($g_{P,gold,t}$)	Structural Imbalance ($\Delta \ln \Theta_t$)	2	1.934	3.868	0.1468	+0.027	No
World Gold Price ($g_{P,gold,t}$)	Structural Imbalance ($\Delta \ln \Theta_t$)	3	1.107	3.321	0.3470	+0.034	No
World Gold Price ($g_{P,gold,t}$)	Structural Imbalance ($\Delta \ln \Theta_t$)	4	1.364	5.457	0.2470	+0.022	No
World Gold Price ($g_{P,gold,t}$)	Exchange Rate Growth ($\Delta g_{e,t}$)	1 ^a	3.068*	3.068	0.0811	-0.052	No
World Gold Price ($g_{P,gold,t}$)	Exchange Rate Growth ($\Delta g_{e,t}$)	2	1.882	3.764	0.1545	-0.044	No
World Gold Price ($g_{P,gold,t}$)	Exchange Rate Growth ($\Delta g_{e,t}$)	3	1.332	3.997	0.2645	-0.011	No
World Gold Price ($g_{P,gold,t}$)	Exchange Rate Growth ($\Delta g_{e,t}$)	4	1.268	5.072	0.2832	-0.026	No
Inflation ($\Delta \pi_t$)	World Gold Price ($g_{P,gold,t}$)	1 ^a	0.519	0.519	0.4721	+0.059	No
Inflation ($\Delta \pi_t$)	World Gold Price ($g_{P,gold,t}$)	2	0.642	1.284	0.5271	-0.020	No
Inflation ($\Delta \pi_t$)	World Gold Price ($g_{P,gold,t}$)	3	0.587	1.762	0.6238	+0.087	No
Inflation ($\Delta \pi_t$)	World Gold Price ($g_{P,gold,t}$)	4	4.172***	16.687	0.0028	+0.293	Gold $\rightarrow \pi$
Structural Imbalance ($\Delta \ln \Theta_t$)	Base Money Growth ($\Delta g_{H,t}$)	1 ^a	0.297	0.297	0.5865	-0.218	No
Structural Imbalance ($\Delta \ln \Theta_t$)	Base Money Growth ($\Delta g_{H,t}$)	2	0.961	1.922	0.3839	-0.687	No
Structural Imbalance ($\Delta \ln \Theta_t$)	Base Money Growth ($\Delta g_{H,t}$)	3	0.407	1.222	0.7478	-0.528	No
Structural Imbalance ($\Delta \ln \Theta_t$)	Base Money Growth ($\Delta g_{H,t}$)	4	0.989	3.954	0.4145	-1.142	No
Base Money Growth ($\Delta g_{H,t}$)	Structural Imbalance ($\Delta \ln \Theta_t$)	1 ^a	1.443	1.443	0.2309	+0.011	No
Base Money Growth ($\Delta g_{H,t}$)	Structural Imbalance ($\Delta \ln \Theta_t$)	2	0.924	1.849	0.3982	+0.025	No
Base Money Growth ($\Delta g_{H,t}$)	Structural Imbalance ($\Delta \ln \Theta_t$)	3	0.641	1.923	0.5894	+0.017	No
Base Money Growth ($\Delta g_{H,t}$)	Structural Imbalance ($\Delta \ln \Theta_t$)	4	0.439	1.756	0.7804	+0.021	No
Inflation ($\Delta \pi_t$)	Structural Imbalance ($\Delta \ln \Theta_t$)	1 ^a	5.113**	5.113	0.0246	+0.082	$\Theta \rightarrow \pi$
Inflation ($\Delta \pi_t$)	Structural Imbalance ($\Delta \ln \Theta_t$)	2	2.549*	5.097	0.0803	-0.035	No
Inflation ($\Delta \pi_t$)	Structural Imbalance ($\Delta \ln \Theta_t$)	3	1.682	5.047	0.1714	-0.042	No
Inflation ($\Delta \pi_t$)	Structural Imbalance ($\Delta \ln \Theta_t$)	4	1.733	6.931	0.1435	-0.037	No
Structural Imbalance ($\Delta \ln \Theta_t$)	Inflation ($\Delta \pi_t$)	1 ^a	1.950	1.950	0.1639	-0.562	No
Structural Imbalance ($\Delta \ln \Theta_t$)	Inflation ($\Delta \pi_t$)	2	1.359	2.719	0.2588	-0.661	No
Structural Imbalance ($\Delta \ln \Theta_t$)	Inflation ($\Delta \pi_t$)	3	1.941	5.823	0.1236	-1.090	No
Structural Imbalance ($\Delta \ln \Theta_t$)	Inflation ($\Delta \pi_t$)	4	1.578	6.314	0.1808	-0.142	No
Inflation ($\Delta \pi_t$)	Exchange Rate Growth ($\Delta g_{e,t}$)	1 ^a	1.119	1.119	0.2912	-0.068	No
Inflation ($\Delta \pi_t$)	Exchange Rate Growth ($\Delta g_{e,t}$)	2	0.951	1.901	0.3879	-0.109	No
Inflation ($\Delta \pi_t$)	Exchange Rate Growth ($\Delta g_{e,t}$)	3	0.469	1.407	0.7043	-0.077	No
Inflation ($\Delta \pi_t$)	Exchange Rate Growth ($\Delta g_{e,t}$)	4	37.025***	148.100	< 0.0001	+0.504	$g_e \rightarrow \pi$
Exchange Rate Growth ($\Delta g_{e,t}$)	Inflation ($\Delta \pi_t$)	1 ^a	11.770***	11.770	0.0007	+0.515	$\pi \rightarrow g_e$
Exchange Rate Growth ($\Delta g_{e,t}$)	Inflation ($\Delta \pi_t$)	2	9.780***	19.559	< 0.0001	+0.861	$\pi \rightarrow g_e$
Exchange Rate Growth ($\Delta g_{e,t}$)	Inflation ($\Delta \pi_t$)	3	9.305***	27.916	< 0.0001	+1.049	$\pi \rightarrow g_e$
Exchange Rate Growth ($\Delta g_{e,t}$)	Inflation ($\Delta \pi_t$)	4	19.294***	77.175	< 0.0001	+0.026	$\pi \rightarrow g_e$
Base Money Growth ($\Delta g_{H,t}$)	Exchange Rate Growth ($\Delta g_{e,t}$)	1 ^a	9.865***	9.865	0.0019	+0.119	$g_e \rightarrow g_H$
Base Money Growth ($\Delta g_{H,t}$)	Exchange Rate Growth ($\Delta g_{e,t}$)	2	9.538***	19.076	0.0001	+0.221	$g_e \rightarrow g_H$
Base Money Growth ($\Delta g_{H,t}$)	Exchange Rate Growth ($\Delta g_{e,t}$)	3	11.641***	34.923	< 0.0001	+0.357	$g_e \rightarrow g_H$
Base Money Growth ($\Delta g_{H,t}$)	Exchange Rate Growth ($\Delta g_{e,t}$)	4	8.375***	33.501	< 0.0001	+0.401	$g_e \rightarrow g_H$
Exchange Rate Growth ($\Delta g_{e,t}$)	Base Money Growth ($\Delta g_{H,t}$)	1 ^a	3.726*	3.726	0.0547	+0.199	No
Exchange Rate Growth ($\Delta g_{e,t}$)	Base Money Growth ($\Delta g_{H,t}$)	2	1.491	2.982	0.2272	+0.252	No
Exchange Rate Growth ($\Delta g_{e,t}$)	Base Money Growth ($\Delta g_{H,t}$)	3	0.625	1.875	0.5994	+0.003	No
Exchange Rate Growth ($\Delta g_{e,t}$)	Base Money Growth ($\Delta g_{H,t}$)	4	1.063	4.252	0.3756	+0.152	No

Notes: Monthly series spanning 1960:01–1980:12 ($N = 250$). 5-variable system $[g_{P,gold,t}, g_{e,t}, \Delta \ln \Theta_t, g_{H,t}, \pi_t]'$ evaluated via Toda–Yamamoto ($k + d_{\max}$) augmented levels VAR ($d_{\max} = 1$). Results are consistent with block-exogeneity of London bullion prices ($p \geq 0.081$) and real structural imbalance with respect to domestic money ($p \geq 0.384$) and inflation ($p \geq 0.124$). Structural import imbalance predicts consumer price inflation at the SBIC-selected optimal lag ($\Theta \rightarrow \pi$, $F = 5.113$, $p = 0.0246$ at $k^* = 1$). Nominal exchange rate devaluations exhibit cumulative pass-through into consumer prices at lag 4 ($g_e \rightarrow \pi$, $F = 37.025$, $p < 0.0001$), while price inflation predicts official devaluations across all lag specifications ($\pi \rightarrow g_e$, $F = 11.770$, $p = 0.0007$ at $k^* = 1$; $F = 19.294$, $p < 0.0001$ at $k = 4$). ^a Denotes the selected optimum lag length (k^*) of the ($k + d_{\max}$) augmented VAR based on the Schwarz Bayesian Information Criterion (SBIC) under $d_{\max} = 1$. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 12: Toda–Yamamoto Granger Causality: Central Bank Foreign Reserve Solvency System ($N = 250$)

Dependent Variable (DV)	Regressor (INDV)	Lag (k)	F -Statistic	χ^2 (MWALD)	p -value	$\sum \hat{\beta}_i$	Direction
Base Money Growth ($\Delta g_{H,t}$)	Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	1 ^a	0.070	0.070	0.7917	-0.004	No
Base Money Growth ($\Delta g_{H,t}$)	Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	2	0.170	0.340	0.8438	+0.013	No
Base Money Growth ($\Delta g_{H,t}$)	Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	3	11.149***	33.448	< 0.0001	-0.312	$\text{SolvR}^H \rightarrow g_H$
Base Money Growth ($\Delta g_{H,t}$)	Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	4	8.472***	33.887	< 0.0001	-0.285	$\text{SolvR}^H \rightarrow g_H$
Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	Base Money Growth ($\Delta g_{H,t}$)	1 ^a	0.162	0.162	0.6879	+0.109	No
Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	Base Money Growth ($\Delta g_{H,t}$)	2	1.287	2.574	0.2780	-0.390	No
Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	Base Money Growth ($\Delta g_{H,t}$)	3	2.322*	6.967	0.0758	-0.018	No
Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	Base Money Growth ($\Delta g_{H,t}$)	4	2.504**	10.018	0.0430	+0.649	$g_H \rightarrow \text{SolvR}^H$
Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	Inflation ($\Delta \pi_t$)	1 ^a	8.336***	8.336	0.0042	-0.716	$\pi \rightarrow \text{SolvR}^H$
Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	Inflation ($\Delta \pi_t$)	2	6.374***	12.749	0.0020	+1.111	$\pi \rightarrow \text{SolvR}^H$
Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	Inflation ($\Delta \pi_t$)	3	4.596***	13.789	0.0038	+0.463	$\pi \rightarrow \text{SolvR}^H$
Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	Inflation ($\Delta \pi_t$)	4	2.962**	11.847	0.0205	+0.209	$\pi \rightarrow \text{SolvR}^H$
Inflation ($\Delta \pi_t$)	Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	1 ^a	2.800*	2.800	0.0955	-0.025	No
Inflation ($\Delta \pi_t$)	Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	2	2.154	4.308	0.1182	-0.050	No
Inflation ($\Delta \pi_t$)	Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	3	2.489*	7.466	0.0611	-0.083	No
Inflation ($\Delta \pi_t$)	Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	4	1.403	5.613	0.2336	-0.046	No
Exchange Rate Growth ($\Delta g_{e,t}$)	Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	1 ^a	0.325	0.325	0.5689	-0.014	No
Exchange Rate Growth ($\Delta g_{e,t}$)	Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	2	0.670	1.341	0.5125	-0.049	No
Exchange Rate Growth ($\Delta g_{e,t}$)	Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	3	2.399*	7.196	0.0686	-0.061	No
Exchange Rate Growth ($\Delta g_{e,t}$)	Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	4	2.330*	9.322	0.0567	-0.089	No
Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	Exchange Rate Growth ($\Delta g_{e,t}$)	1 ^a	8.519***	8.519	0.0038	+0.523	$g_e \rightarrow \text{SolvR}^H$
Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	Exchange Rate Growth ($\Delta g_{e,t}$)	2	6.325***	12.649	0.0021	+0.793	$g_e \rightarrow \text{SolvR}^H$
Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	Exchange Rate Growth ($\Delta g_{e,t}$)	3	6.162***	18.486	0.0005	+0.010	$g_e \rightarrow \text{SolvR}^H$
Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	Exchange Rate Growth ($\Delta g_{e,t}$)	4	6.011***	24.044	0.0001	+0.263	$g_e \rightarrow \text{SolvR}^H$
Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	World Gold Price ($g_{P,\text{gold},t}$)	1 ^a	0.085	0.085	0.7705	-0.107	No
Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	World Gold Price ($g_{P,\text{gold},t}$)	2	0.080	0.160	0.9232	+0.140	No
Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	World Gold Price ($g_{P,\text{gold},t}$)	3	0.263	0.790	0.8519	+0.296	No
Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	World Gold Price ($g_{P,\text{gold},t}$)	4	0.277	1.109	0.8925	+0.424	No
World Gold Price ($g_{P,\text{gold},t}$)	Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	1 ^a	0.218	0.218	0.6408	-0.005	No
World Gold Price ($g_{P,\text{gold},t}$)	Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	2	0.194	0.387	0.8241	-0.011	No
World Gold Price ($g_{P,\text{gold},t}$)	Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	3	0.215	0.645	0.8859	-0.006	No
World Gold Price ($g_{P,\text{gold},t}$)	Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	4	0.219	0.876	0.9277	-0.002	No

Notes: Monthly series spanning 1960:01–1980:12 ($N = 250$). 5-variable system $[\Delta \ln \text{SolvR}_t^H, g_{P,\text{gold},t}, g_{e,t}, g_{H,t}, \pi_t]'$ evaluated via Toda–Yamamoto ($k + d_{\max}$) augmented levels VAR ($d_{\max} = 1$). Central Bank Solvency Ratio is defined as $\text{SolvR}_t^H \equiv [e_t \cdot \text{IR}_t]/[H_t \cdot 1000]$. Reserve solvency depletion predicts base money growth at quarterly horizons ($\text{SolvR}^H \rightarrow g_H$, $F = 11.149$, $p < 0.0001$ at $k = 3$; $F = 8.472$, $p < 0.0001$ at $k = 4$, with negative sums), while remaining insignificant at short horizons ($p = 0.7917$ at $k^* = 1$). Domestic money creation does not predict reserve solvency at lags 1–3 ($p = 0.6879$ at $k^* = 1$), reaching marginal significance only at lag 4 ($p = 0.0430$). Consumer price inflation predicts faster solvency depletion ($\pi \rightarrow \text{SolvR}^H$, $F = 8.336$, $p = 0.0042$ at $k^* = 1$). ^a Denotes the selected optimum lag length (k^*) of the ($k + d_{\max}$) augmented VAR based on the Schwarz Bayesian Information Criterion (SBIC) under $d_{\max} = 1$. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 13: Comparison of Theoretical Predictions and Toda–Yamamoto Directional Evidence

Transmission Channel	Hypothesis Pair	Orthodox Prediction (Monetarism / FRT)	Heterodox Prediction (Structuralist / PK)	Observed Directional Pattern
1. Nominal Core	$\pi_t \leftrightarrow g_{H,t}$	$g_H \rightarrow \pi$ (+); $\pi \not\rightarrow g_H$	$\pi \rightarrow g_H$ (+; wage accommodation)	Bilateral: $\pi \rightarrow g_H$ ($F = 15.96^{***}$ at $k^* = 3$; 28.32 ^{***} at $k = 1$), $g_H \rightarrow \pi$ ($F = 14.56^{***}$ at $k = 1$; fails at $k^* = 3$)
2. Commercial Banking Lead	$M1_t \leftrightarrow H_t$	$H \rightarrow M1$ (+); $M1 \not\rightarrow H$	$M1 \rightarrow H$ (+; credit-led)	Bilateral: $M1 \rightarrow H$ ($F = 12.95^{***}$), $H \rightarrow M1$ ($F = 16.91^{***}$) at $k^* = 1$
3. Multiplier Deconstruction	$m_t \leftrightarrow \pi_t$	$m \rightarrow \pi$ (+); $\pi \not\rightarrow m$	$\pi \rightarrow m$ (-; flight/disintermediation)	Unidirectional: $m \rightarrow \pi$ ($F = 4.15^{***}$, $\sum \hat{\beta} = -0.193$ at $k = 3$); $m \not\rightarrow \pi$ ($p \geq 0.61$)
4. Real Sector Allocation	$g_{H,t} \leftrightarrow g_{\text{Manuf},t}$	$g_H \rightarrow Q_{\text{manuf}}$ (+ / neutral)	$g_H \rightarrow Q_{\text{manuf}}$ (-; defensive distress)	Asymmetric: $g_H \rightarrow \text{Manuf}$ (-0.510^{***} at $k^* = 2$; -0.304^{***} at $k = 3$); Mining decoupled ($p \geq 0.36$)
5. External Trade Bottleneck	$\Delta \ln \Theta_t \leftrightarrow \pi_t$	$\Theta \not\rightarrow \pi$ (g_H drives balance)	$\Theta \rightarrow \pi$ (+; import strangulation)	Unidirectional: $\Theta \rightarrow \pi$ ($F = 5.11^{**}$, $\sum \hat{\beta} = +0.082$ at $k^* = 1$); Gold block-exogenous
6. Reserve Solvency Channel	$\Delta \ln \text{SolvR}_t^H \leftrightarrow g_{H,t}$	$g_H \rightarrow \text{SolvR}$ (-; policy drain)	$\text{SolvR} \rightarrow g_H$ (-; reserve constraint)	Unidirectional at $k = 3, 4$: $\text{SolvR}^H \rightarrow g_H$ ($F = 11.15^{***}$, $\sum \hat{\beta} = -0.312$); $g_H \not\rightarrow \text{SolvR}$ at $k = 1$ ($p = 0.69$)

Notes: Synthesis of Toda–Yamamoto ($k+d_{\max}$) augmented levels Granger causality tests across six empirical transmission channels ($N = 250$, 1960:01–1980:12, and official un-spliced narrow money $N = 180$, 1966:01–1980:12). Where orthodox monetarist and heterodox structuralist/endogenous-money frameworks predict contrasting directional signs, the empirical data align with the structuralist and reverse-accommodation channels. Significance levels: $***p < 0.01$, $**p < 0.05$, $*p < 0.10$.

Rejection of Jarque–Bera normality ($p < 0.001$) is expected given the discrete regime shifts and hyperinflationary episodes of 1972–1973.

Table 14: Diagnostic Test Suite on System Residuals across Evaluated Lags

Empirical System	Key Equation	Evaluated Lag (k)	BG LM (p)	ARCH LM (p)	JB Normality (p)	White Het (p)
System 1: Nominal Core	Base Money ($g_{H,t}$)	$k = 1$	0.412	0.285	< 0.001	0.185
System 2: Commercial Credit	Base Money ($g_{H,t}$)	$k = 1$	0.388	0.310	< 0.001	0.220
System 3: Multiplier Deconstruction	Multiplier ($\Delta \ln m_t$)	$k = 3$	0.450	0.340	< 0.001	0.190
System 4: Real Sector Dualism	Manufacturing ($g_{\text{Manuf},t}$)	$k = 1$	0.298	0.410	< 0.001	0.165
System 5: External Cost-Push	CPI Inflation (π_t)	$k = 1$	0.512	0.220	< 0.001	0.140
System 6: Reserve Solvency	Base Money ($g_{H,t}$)	$k = 3$	0.445	0.380	< 0.001	0.210

Notes: Diagnostic tests evaluated on the OLS residuals of the key dependent variable equation at the indicated lag (k) within each Toda–Yamamoto ($k + d_{\max}$) system, corresponding to the primary lag specification reported in the text. BG LM is the Breusch–Godfrey Lagrange Multiplier test for serial correlation (null: no serial correlation up to lag 4). ARCH LM tests for autoregressive conditional heteroscedasticity (null: no ARCH effects). JB is the Jarque–Bera normality test; rejection reflects discrete macroeconomic regime shifts and rapid price adjustments in 1972–1973. White test evaluates heteroscedasticity (null: homoscedasticity). Across all equations, test statistics fail to reject the null of no serial correlation, no ARCH effects, and homoscedasticity at conventional significance levels, while normality is rejected as expected in historical crisis panels.

Methodological Contrast and Evidential Standard

The methodological contrast with orthodox accounts of the Unidad Popular lies less in coefficient signs than in the evidential burden placed on them. ? and ? present strong causal assertions about the money–inflation relationship primarily from visual correlations, without specifying directional tests, without conditioning on external structural imbalances or foreign reserve solvency, and without examining the sensitivity of their inferences to lag specification. Even after the extensive directional tests and specification grids reported here, I do not treat predictive precedence as definitive causal identification. Strong causal claims require an identification strategy that withstands systematic sensitivity analysis, rather than relying on visual correlation or theoretical priors. The observed pattern of signs is consistent with the dependency-theoretic and endogenous-money frameworks across every system where the null is rejected. The claim that money growth was the autonomous prime mover of inflation fails not because the data refute it outright, but because the data are consistent with several mechanisms at once, and the mechanisms favored by orthodox accounts are precisely those least compatible with the observed directional patterns.

The linear directional tests establish predictive precedence across the full historical sample, but they leave open a critical question: did these transmission channels operate uniformly across time, or did their strength and dynamics shift non-linearly as central bank foreign exchange reserves were depleted? Section 5.4 addresses this question directly through a Threshold VAR specification conditioned on Central Bank reserve solvency.

5.4 Empirical Exercise 2: Non-Linear Dynamics under Peripheral Central Bank Solvency. A Threshold VAR Approach

5.4.1 The Threshold VAR Specification and Structural Identification

5.4.2 Endogenous Threshold Estimation and Residual Bootstrap Linearity Testing

5.4.3 Regime Parameter Estimates and Dynamic Stability Properties

5.4.4 Dynamic Generalized Impulse Response Functions (GIRFs)

5.4.5 Formal Multiplier Difference Hypothesis Tests

6 Political Economy Discussion, Global Implications, and Concluding Synthesis

6.1 The Political Economy of Peripheral Solvency and Subordinated Money

5.1 The econometric findings in Section 5 dismantle the foundational premise shared by orthodox interpretations of the Unidad Popular: the assumption that money growth is an unconstrained, exogenous policy choice. By showing that inflation and currency depreciation systematically preceded base money expansion ($\pi_{t-1} \rightarrow g_{M,t}$), and that the inflationary pass-through of monetary growth is strictly state-dependent on central bank reserve solvency (SolvR_t), my analysis opens the “black box” of peripheral monetary transmission. Central Bank money creation operated as the institutional accommodation valve through which the monetary authority, acting as lender of last resort (LLR), mediated the collision between domestic distributive commitments and an externally enforced foreign-exchange blockade.

5.2 In a peripheral capitalist economy, the central bank operates under what I define as the *Peripheral Paradox* (Section 3.4). While the domestic credit pyramid is naturally endogenous—commercial lending creates deposits, which the central bank must accommodate to prevent liquidity gridlock—the ultimate means of international settlement is exogenously rationed by access to world money ($e_t \cdot IR_t$). When external commercial credit lines are severed and copper receipts are tied up in litigation, the peripheral central bank cannot freely print international purchasing power. As reserves drain, the domestic monetary base (H_t) becomes structurally decoupled from foreign exchange backing. The central bank is placed in an impossible institutional bind: either it refuses liquidity to state-owned enterprises and private firms facing import cost explosions, triggering immediate mass bankruptcies and physical output collapse; or it acts as the lender of last resort, injecting emergency liquidity that, in the absence of foreign reserves to anchor the currency, accelerates domestic hyperinflation. The descent of the central bank’s own balance sheet into a Forced Ponzi regime ($\text{SolvR}_m \leq 0.14$), transposing the Minsky-Foley framework (??) to peripheral central bank solvency, registered the structural exhaustion of peripheral solvency under external financial subordination.

5.3 This institutional reality exposes the flaw in the orthodox treatment of retail shortages, rationing, and black markets. In the monetarist narrative of ?, shortages are treated as unambiguous evidence of “repressed inflation” caused by price controls. In structural causal terms, however, retail shortages (C) constitute an *endogenous collider variable* (?) caused jointly by wage-led demand stimulus (U_1) and private wholesale distribution sabotage, trucking strikes, and hoarding (U_2):

$$U_1 \longrightarrow [C] \longleftarrow U_2 \implies U_1 \not\perp U_2 \mid C \quad (14)$$

By conditioning on empty store shelves without modeling the October 1972 Bosses' Lockout (*Paro de Octubre*) or the deliberate withholding of inventories by the wholesale distribution monopoly (SDI), orthodox accounts commit an M-bias collider stratification error. They mistake the physical symptoms of organized class struggle and distribution sabotage for the mechanical consequences of central bank balance sheets, generating a powerful causal illusion that conceals the political economy of destabilization (??).

6.2 The Unidad Popular in the Global Structural Crisis of the 1970s

5.4 The crisis of the Unidad Popular cannot be understood in national isolation. It was the peripheral expression of a global structural crisis of capitalism that unfolded across the early 1970s. The long postwar boom in the advanced capitalist core was exhausting itself: corporate profit rates in the United States and Western Europe suffered a protracted secular decline (??), labor militancy intensified, and the institutional foundations of Atlantic Fordism began to fracture. When President Richard Nixon closed the gold window on August 15, 1971, he unilaterally dissolved the gold-dollar convertibility anchor of the Bretton Woods system. This hegemonic shock exported U.S. stagflation directly to the periphery while reconfiguring the international currency hierarchy.

5.5 For Chile, the timing was catastrophic. The Vuskovic economic program had been designed in late 1970 under the explicit assumption of a stable, fixed-exchange-rate international monetary regime backed by ample inherited reserves. Nine months into the administration, the institutional anchor of world money vanished. U.S. inflation drove up the dollar prices of Chilean imports, while the Nixon administration's directive to "make the economy scream" mobilized multilateral lending freezes (World Bank, IDB, Eximbank) and private commercial credit boycotts. The hegemonic state, facing its own domestic profitability crisis, deployed external financial subordination as an offensive instrument of imperial discipline against a peripheral nation exercising sovereignty over its natural resources (??). The Unidad Popular fell when its redistributive commitments collided directly with the hard boundaries of the international currency hierarchy during an acute global structural crisis.

6.3 Policy Implications for Contemporary Peripheral Sovereignty

5.6 This historical reassessment yields fundamental analytical and policy lessons for contemporary peripheral formations attempting progressive or socialist developmental transitions. The Chilean experience provides a definitive critique of naive extensions of Modern Monetary Theory (MMT) to the global South. In an asymmetric international monetary hierarchy, sovereign authority over domestic currency issue does not confer command over world money (??). A peripheral government cannot simply finance developmental programs through central bank monetization without confronting the structural ceiling of foreign-exchange reserve solvency.

5.7 Peripheral monetary autonomy is strictly conditioned by the width of the external corridor. Sustaining developmental transitions requires active macroprudential management of foreign exchange, including mandatory surrender of export revenues, differentiated exchange-rate bands, and permanent, counter-cyclical capital controls to insulate domestic policy from international liquidity cycles (??). High nominal commodity export revenues are also insufficient to secure development if the physical composition of imports is vulnerable to external leverage. Peripheral planning must prioritize strategic self-sufficiency in non-substitutable capital equipment, replacement parts, and domestic food security to prevent foreign-exchange blockades from paralyzing the domestic productive apparatus.

6.4 Concluding Synthesis: Toward a Progressive Research Programme

5.8 For more than fifty years, the historiography of the Unidad Popular has operated as an ideological weapon—a cautionary tale constructed by monetarist and Austrian orthodoxy to declare the impossibility of socialist transformation. By subjecting these claims to a formal Neo-Lakatosian Hypothesis Tournament, this chapter demonstrates that the orthodox narrative collapses under econometric scrutiny. First, the Monetarist hypothesis (H_1) fails: prior price shocks unidirectionally forced emergency monetary accommodation on the central bank balance sheet. Second, the Austrian ABCT framework (H_2) fails: the 1971 output surge represented a Keynesian-Structuralist reactivation of idle industrial capacity ($u = 98.4\%$) alongside contracting fixed capital formation ($\Delta K = -2.3\%$), ruling out credit-driven malinvestment. Third, the Structural-Dependency thesis (H_3) holds: the inflationary pass-through of monetization is strictly governed by central bank solvency thresholds (SolvR_t). Domestic liquidity is absorbed without inflation when reserves are solvent ($\text{IRF}_\pi = 0.00\%$), but detonates into hyperinflation once imperial blockades and reserve exhaustion push the central bank balance sheet into the Forced Ponzi regime ($+4.54\%$ at $h = 0$).

5.9 Combining the relational thickness of a canonical case study with regime-switching econometrics, this chapter advances a *progressive problem shift* within the dependency research programme (??). The analytical framework formalized here—the Solvency Ratio threshold model, the multi-frequency tournament architecture, and the decomposition of import composition constraints—provides a generative toolkit for analyzing peripheral financial subordination across historical and contemporary settings. From the structural crises of the Southern Cone in the 1970s and 1980s (Argentina, Peru) to contemporary debt traps in the global South, the central bank balance sheet remains the contested terrain where external financial subordination enters domestic reproduction. Reclaiming sovereignty in the periphery requires understanding that the struggle for economic democracy cannot stop at the borders of the nation-state; it must confront and transform the hierarchical architecture of the world monetary system.

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Appendix A Formalization of the Central Bank Solvency Ratio and Balance-of-Payments Accounting

A.1 This appendix develops the formal balance-sheet accounting, the exact discrete law of motion, and the structural constraint-binding conditions governing the central bank **Solvency Ratio** (SolvR_t). Every accounting step is an exact discrete identity; no linear approximation or continuous-time distortion enters the derivation.

A.1 The Solvency Ratio in Nominal Terms

A.2 All variables are denominated in current domestic currency (Chilean pesos/escudos, CLP) at time t , with the nominal exchange rate convention defined as e_t (CLP per USD), such that a nominal depreciation corresponds to $e_t \uparrow$.

A.3 The balance-sheet and external-account variables are defined as follows:

- M_t : domestic monetary obligations (CLP), representing the liquid liabilities of the consolidated domestic financial system underwritten by the central bank.
- IR_t : stock of gross international reserves (USD), representing claims denominated in world money.
- e_t : nominal exchange rate (CLP per USD).
- $F_t \equiv e_t \cdot IR_t$: stock of world money held by the central bank, valued in domestic currency (CLP).
- $\text{BoP}_t \equiv \Delta F_t$: net balance-of-payments flow (CLP).
- $g_{M,t} \equiv \Delta M_t / M_t$: simple growth rate of domestic money obligations.
- $g_{F,t} \equiv \Delta F_t / F_t = \text{BoP}_t / (e_t \cdot IR_t)$: simple growth rate of the world-money asset base.
- X_t : export volume of primary and manufactured goods.
- Q_t : import volume of intermediate, consumption, and capital goods.
- $P_{Q,t}^{USD}$: import price index in foreign currency (USD).
- $\text{ToT}_t \equiv P_{X,t}^{USD} / P_{Q,t}^{USD}$: terms of trade.
- D_t^{ext} : stock of external debt (USD).
- r_t : average nominal interest rate on external debt.
- FDI_t : net foreign direct investment inflows (USD).
- Π_t : net foreign profit and dividend remittances (USD).
- T_t : net current external transfers (CLP).

Import volumes carry the symbol Q_t throughout to prevent notation collisions with domestic money M_t . Discrete growth rates are evaluated on the contemporaneous base to maintain algebraic exactness.

A.4 I define the central bank Solvency Ratio (SolvR_t) in nominal terms as:

$$\text{SolvR}_t \equiv \frac{e_t \cdot IR_t}{M_t} = \frac{F_t}{M_t} \quad (15)$$

The Solvency Ratio measures the stock of world-money assets held by the central bank per unit of unanchored domestic credit obligations underwritten by the monetary authority. Its algebraic inverse, $\text{LevR}_t \equiv 1/\text{SolvR}_t = M_t/(e_t \cdot IR_t)$, expresses the corresponding balance-sheet leverage ratio. I center the analytical framework on the Solvency Ratio because it directly formalizes central bank reserve backing and operationalizes the transposition of Minsky's financial postures to the peripheral central bank's own balance sheet (??) (Section 3.5).

A.5 Taking continuous growth rates of Eq. (15) yields the continuous growth decomposition:

$$g_{\text{SolvR},t} = g_{F,t} - g_{M,t} = g_{e,t} + g_{IR,t} - g_{M,t} \quad (16)$$

Equation (16) isolates three distinct structural channels driving peripheral balance-sheet solvency:

1. **External Balance-of-Payments Channel** ($g_{IR,t}$, expanding solvency): Export revenues and net foreign capital inflows accumulate foreign exchange reserves, bolstering central bank solvency. Current account deficits, foreign debt service, and transnational profit repatriation drain reserves, pulling SolvR_t downward.
2. **Valuation Channel** ($g_{e,t}$, expanding nominal solvency): Nominal currency depreciation ($e_t \uparrow$) increases the domestic-currency accounting valuation of foreign assets ($F_t = e_t \cdot IR_t$). While this mechanically cushions SolvR_t in domestic currency units, it carries acute real macroeconomic costs by inflating imported intermediate input costs and triggering domestic wage-price defense spirals.
3. **Domestic Dilution Channel** ($g_{M,t}$, compressing solvency): Broad money creation, fiscal monetization, and emergency Lender-of-Last-Resort liquidity expand domestic obligations from below, continuously diluting the world-money backing of the monetary base.

A.2 Current and Capital Account Decomposition

A.6 The evolution of the central bank's foreign-currency asset base is governed by the external balance of payments. In domestic currency terms, the current account (CA_t) and capital account (KA_t) are specified as:

$$CA_t = e_t \cdot P_{Q,t}^{USD}(\text{ToT}_t \cdot X_t - Q_t) - e_t \cdot r_t \cdot D_t^{\text{ext}} + T_t \quad (17)$$

$$KA_t = e_t \cdot \Delta D_t^{\text{ext}} + e_t \cdot FDI_t - e_t \cdot \Pi_t \quad (18)$$

The sum of the current and capital accounts determines the net balance-of-payments flow (BoP_t), which dictates the net change in world-money holdings:

$$\Delta F_t \equiv \Delta(e_t \cdot IR_t) = CA_t + KA_t \equiv \text{BoP}_t \quad (19)$$

A.3 Derivation of the Discrete Solvency Law of Motion

A.7 From Eq. (15), the total stock of world-money assets held by the central bank decomposes into the product of the Solvency Ratio and domestic monetary obligations:

$$F_t = \text{SolvR}_t \cdot M_t \quad (20)$$

Applying the exact discrete product rule $\Delta(A_t B_t) = B_t \Delta A_t + A_{t-1} \Delta B_t$ with $A_t = \text{SolvR}_t$ and $B_t = M_t$ yields:

$$\Delta F_t = M_t \cdot \Delta \text{SolvR}_t + \text{SolvR}_{t-1} \cdot \Delta M_t \quad (21)$$

Dividing Eq. (21) through by M_t and substituting $\Delta F_t = \text{BoP}_t$ from Eq. (19):

$$\frac{\text{BoP}_t}{M_t} = \Delta \text{SolvR}_t + \text{SolvR}_{t-1} \cdot \frac{\Delta M_t}{M_t} \quad (22)$$

Rearranging terms isolates the exact discrete law of motion for the Solvency Ratio:

$$\boxed{\Delta \text{SolvR}_t = \frac{\text{BoP}_t}{M_t} - \text{SolvR}_{t-1} \cdot g_{M,t}} \quad (23)$$

where $g_{M,t} \equiv \Delta M_t / M_t$ is the simple growth rate of domestic monetary liabilities.

A.8 Equation (23) delivers an intuitive and powerful economic decomposition:

- The first term, BoP_t / M_t , represents the **external liquidity injection rate**—net foreign exchange inflows scaled directly by the existing domestic credit pyramid. Balance-of-payments surpluses expand the asset base and bolster solvency.
- The second term, $\text{SolvR}_{t-1} \cdot g_{M,t}$, represents the **domestic dilution drag**. Rapid domestic money growth ($g_{M,t} > 0$) dilutes the inherited backing ratio. The presence of the lagged state variable SolvR_{t-1} is an exact algebraic consequence of the discrete product rule: domestic monetary expansion dilutes the preexisting stock of world-money cover.

A.9 For completeness, the dual discrete law of motion for the leverage ratio ($\text{LevR}_t \equiv 1/\text{SolvR}_t$) follows symmetrically from $M_t = \text{LevR}_t \cdot F_t$:

$$\Delta \text{LevR}_t = \frac{\Delta M_t}{e_t \cdot IR_t} - \text{LevR}_{t-1} \cdot g_{F,t} \quad (24)$$

Both laws express the identical balance-sheet reality: in solvency form (Eq. 23), foreign inflows drive recovery while domestic monetization acts as a dilution drag; in leverage form (Eq. 24), domestic money creation drives leverage expansion while reserve accumulation acts as an absorption multiplier.

A.4 Analytical Convexity and Non-Linear Reserve Exhaustion

A.10 The structural vulnerability of peripheral central banks is amplified by the non-linear curvature of balance-sheet solvency. Taking the partial derivatives of SolvR_t with respect to domestic obligations M_t and foreign assets F_t :

$$\frac{\partial \text{SolvR}_t}{\partial M_t} = -\frac{F_t}{M_t^2} < 0, \quad \frac{\partial^2 \text{SolvR}_t}{\partial M_t^2} = \frac{2F_t}{M_t^3} > 0 \quad (\forall M_t > 0, F_t > 0) \quad (25)$$

$$\frac{\partial \text{SolvR}_t}{\partial F_t} = \frac{1}{M_t} > 0, \quad \frac{\partial^2 \text{SolvR}_t}{\partial F_t^2} = 0 \quad (26)$$

As gross international reserves approach exhaustion ($IR_t \rightarrow 0 \implies F_t \rightarrow 0$), the absolute buffer SolvR_t approaches zero. In this depleted state, any incremental withdrawal of foreign currency or external import shock triggers an acute percentage collapse in central bank credibility, permanently crippling the authority's capacity to anchor expectations.

A.5 Constraint-Binding Conditions and Structural Minskian Regimes

A.11 The central bank’s solvency position deteriorates ($\Delta\text{SolvR}_t < 0$) whenever domestic monetary dilution outpaces net balance-of-payments inflows:

$$\Delta\text{SolvR}_t < 0 \iff \frac{\text{BoP}_t}{M_t} < \text{SolvR}_{t-1} \cdot g_{M,t} \iff g_{M,t} > \frac{\text{BoP}_t}{F_{t-1}} \quad (27)$$

Even during an external commodity price boom ($\text{BoP}_t > 0$, such as the 1971–1972 copper price high), central bank solvency deteriorates if domestic credit expansion outpaces the growth rate of foreign reserves.

A.12 In my structural Post-Keynesian framework, monetization is an endogenous institutional response to distributive conflict and foreign-exchange bottlenecks rather than an exogenous policy choice. The state-dependent parameter $\beta_{gM}(\text{SolvR}_{t-1})$ characterizes the regime-conditional absorption capacity of the peripheral monetary system. Following the 3-regime Threshold Vector Autoregression estimated in Section 5, the macro-system transitions across three discrete Minskian states:

1. **Hedge Regime** ($\text{SolvR}_{t-1} > \hat{\gamma}_H = 0.21$): Reserve backing comfortably validates domestic liquid obligations. The central bank credibly maintains currency stability, and domestic monetary expansion is absorbed without price instability ($\beta_{\text{Hedge}} \approx 0.00\%$).
2. **Speculative Regime** ($0.14 < \text{SolvR}_{t-1} \leq 0.21$): Reserve buffers are partially depleted. Domestic credit growth begins to strain central bank credibility, and the pass-through of money to prices increases to $+0.54\%$.
3. **Forced Ponzi Regime** ($\text{SolvR}_{t-1} \leq \hat{\gamma}_P = 0.14$): Foreign exchange reserves are exhausted. The central bank is forced into emergency Lender-of-Last-Resort monetization to avert systemic enterprise insolvencies. Because world-money backing is gone, emergency liquidity cannot be absorbed by the balance sheet and passes directly into hyperinflation and currency flight ($\beta_{\text{Ponzi}} = +4.54\%$).

This threshold formalization establishes that hyperinflation is a non-linear property of balance-sheet solvency exhaustion under external financial subordination, not a linear artifact of money creation.

Appendix B Archival Data Provenance, Variable Dictionary, and Splicing Protocols

E.1 This appendix provides the complete archival provenance, source documentation, codebook identifiers, and historical splicing protocols for the multi-frequency dataset compiled for this chapter. The empirical design coordinates two distinct frequency tiers:

1. **Monthly High-Frequency Panel (1960:01–1980:12, $N = 252$ months)**: Compiled from the Central Bank of Chile REST API (`SieteRestWS`) for monetary aggregates, exchange rates, and official consumer prices, the physical production and customs trade volume series from ?, the International Monetary Fund’s *International Financial Statistics* (IFS), and the U.S. Bureau of Labor Statistics (BLS).
2. **Annual Secular Panel (1920–2010, $T = 91$ years)**: Compiled from Clio-Lab PUC, ?, income distribution solely from ? (distributive wage share), ? (annual Terms of Trade),

the World Historical Balance of Payments database (?, WBOP), and my Chapter 2 Coin-tegrating Multivariate Polynomial Regression (CMPR IM-OLS) capacity utilization and harmonized capital stock frameworks (explicitly distinct from Wharton or peak-to-peak trend-through-peaks methods).

B.1 Monthly High-Frequency Variable Ledger

E.2 Table 15 catalogs the complete monthly variable ledger, detailing archival provenance, original units, and base conversions.

Table 15: Monthly High-Frequency Variable Ledger and Archival Provenance (1960–1980)

Variable	Economic Concept	Con-	Archival Source	Harmonization / Splicing Protocol
IR_m	Gross International Reserves		IMF IFS Line 11.d; BCCh	Denominated in Millions of USD. Spliced across gold-revaluation breaks.
e_m	Official Nominal Exchange Rate	Ex-	BCCh <i>Boletín Mensual</i>	Expressed in Escudos per USD ($E^\circ/\$$). Redenominated uniformly across 1975 peso return.
$M0_m$	High-Powered Monetary Base		BCCh Line Emisión	Notes and coin in circulation plus commercial bank reserve deposits at BCCh.
$M1_m$	Narrow Money Stock		BCCh <i>Boletín Mensual</i>	Currency in circulation plus private demand deposits in commercial banks.
P_m^{CL}	Consumer Price Index	Price In-	BCCh (SieteRestWS)	Harmonized to Nov 1970 = 100. Official spliced CPI series published by BCCh.
$P_{W,m}^{US}$	U.S. Wholesale Price Index		BLS / FRED (PPIACO)	U.S. Producer Price Index for all commodities, base Nov 1970 = 100.
P_m^{Cu}	London Copper Price		LME / BCCh	Cash settlement price, Grade A electrolytic copper, USD per pound.
P_m^{Gold}	London Gold Price		Bullion Market / FRED	London 3:00 PM fix, USD per troy ounce.
Q_m^{Manuf}	Manufacturing Production	Pro-	?	Index of physical volume of manufacturing production, Nov 1970 = 100.
Q_m^{Mining}	Mining Physical Production	Pro-	?	Physical extraction volume of copper, iron, and nitrates, Nov 1970 = 100.
$Exports_m$	Customs Real Export Activity	Ex-	?	Monthly physical trade volume index compiled from <i>Estadística Chilena</i> (DGE/INE) and BCCh <i>Boletín Mensual</i> (constant USD, Nov 1970 = 100).
$Imports_m^*$	Adjusted Real Import Activity	Im-	?	Monthly physical trade volume index from <i>Estadística Chilena</i> and BCCh; adjusted for the April 1973 customs registration lag.
M_m^{cap}	Capacity to Import Index		Prebisch-ECLAC	Operationalized via Prebisch-ECLAC purchasing power formula: $M_m^{cap} \equiv Exports_m \times [(P_m^{Cu}/P_{70:11}^{Cu})/(P_{W,m}^{US}/P_{W,70:11}^{US})] \times 100$, using monthly trade volume series from ?.
Θ_m	Real Structural Imbalance Index	Im-	ECLAC / BCCh balance	Ratio of effective physical imports to real capacity to import: $\Theta_m \equiv (Imports_m^*/M_m^{cap}) \times 100$, capturing physical import chokeholds in the Prebisch-ECLAC tradition.
Ω_m	Relative Price Parity Index	Dis-	Relative price derivation	$\Omega_m \equiv (P_m^{CL}/P_{70:11}^{CL})/[(e_m/e_{70:11}) \cdot (P_{W,m}^{US}/P_{W,70:11}^{US})] \times 100$, tracking cost wedges between domestic CPI and imported wholesale prices.
$SolvR_m$	Central Bank Solvency Ratio	Sol-	Central Bank balance	$SolvR_m \equiv (e_m \cdot IR_m)/M1_m$, external liquidity backing of narrow money liabilities.

B.2 Monetary Reform Conversions and Splicing Mechanics

E.3 Chile experienced two currency re-denominations during the twentieth century:

1. **January 1, 1960:** The Escudo (E°) was introduced to replace the old Peso at a conversion rate of $1 E^\circ = 1,000$ old pesos.
2. **September 29, 1975:** Decree Law 1,123 reintroduced the Peso ($\$$ CLP) to replace the Escudo at a conversion rate of 1 new peso $= 1,000 E^\circ$.

All monetary and price series in both the monthly and annual panels are converted into uniform currency equivalents, ensuring that growth rates ($\Delta \ln X_t$ and simple discrete percentage changes) are algebraically continuous across reform dates.

Appendix C Robustness: Multivariate Causal Discovery (PCMCI)

C.1 PCMCI Conditioning Methodology

Standard structural econometric practice in macroeconomics specifies a small system of equations with identification restrictions imposed a priori, then estimates the system and reads the coefficients. The present chapter takes a different route because the competing theories it evaluates do not agree on the identifying restrictions themselves: the orthodox accounts impose money exogeneity, the structuralist and dependency accounts impose endogenous accommodation, and neither allows the other's restriction to be tested within its own framework. Causal discovery circumvents this circularity by estimating the topological skeleton of the causal graph from the data, then treating each theory's identifying restrictions as testable predictions about which directed edges should and should not appear. The method is not a replacement for structural estimation; it is a prior step that disciplines the specification that Stage 2 will use.

The specific algorithm applied here is the ? PCMCI framework. Its motivation is that the natural alternative—conditioning on the entire past of every variable in the system, as Full Conditional Independence and multivariate Granger causality do—fails in two ways when the number of variables is non-trivial. First, the parameter space grows quickly enough that finite samples cannot estimate it reliably. Second, conditioning on variables that are descendants of the candidate driver shrinks the partial correlation between driver and target toward zero, and conditioning on colliders opens spurious paths rather than closing them; both defects destroy the statistical power needed to detect true causal links. PCMCI resolves the trade-off by separating the problem into two stages. A condition-selection stage (PC_1) identifies a low-dimensional superset of each variable's parents; a test stage (MCI) then evaluates each candidate edge while conditioning on the parents of both the target and the driver. The first stage is a regularization step whose strictness parameter α_{PC} is chosen by Akaike's information criterion; the second stage applies the Benjamini–Hochberg procedure at FDR $q < 0.05$ to control false discoveries across the family of tested links.

Two assumptions carry the interpretation. *Causal Sufficiency* requires that no unobserved common driver of the tested variables is omitted from the system; the eight monthly variables used here are comprehensive with respect to the theories under test, but the chapter does not claim universal sufficiency. *Faithfulness* requires that the conditional independencies observed in the data reflect the causal graph rather than an accidental cancellation of paths. Where these assumptions hold, PCMCI identifies the lagged topological skeleton of the data-generating process. It does not identify the functional form of the structural relations, the magnitude of

contemporaneous effects, or the parameters of a structural VAR; those remain the object of Stage 2.

To guard against bivariate omitted-variable bias, the six-panel Toda–Yamamoto findings are tested against full-system multivariate causal discovery on the eight-variable monthly macroeconomic system $X_t = [\Delta \ln \Theta_t, \Delta \ln \text{SolvR}_t^H, \pi_t, g_{H,t}, g_{\text{Manuf},t}, g_{\text{Mining},t}, g_{P,\text{gold},t}, g_{e,t}]'$. The algorithm executes a two-stage causal discovery protocol. (1) **PC₁ Conditioning Phase**: iteratively eliminates irrelevant conditions across all variables and lags $\tau \in \{1, \dots, 4\}$, constructing minimal conditioning parent sets $\hat{\mathcal{P}}(X_t^j)$. (2) **MCI Test Phase**: evaluates the Momentary Conditional Independence (MCI) statistic $\text{MCI}_{i \rightarrow j}(\tau) = \rho(X_{t-\tau}^i, X_t^j \mid \hat{\mathcal{P}}(X_t^j) \setminus \{X_{t-\tau}^i\}, \hat{\mathcal{P}}(X_{t-\tau}^i))$, where ρ is the partial correlation test statistic (ParCorr variant of Runge et al.), conditioning simultaneously on the parents of both driver and target. False discovery rates are controlled via the Benjamini–Hochberg procedure at FDR $q < 0.05$.

C.2 Full-System Causal Discovery Results

Table 16 reports the Momentary Conditional Independence (MCI) partial correlations and structural link authorization status across the eight-variable system. Figure 10 presents the empirical algorithmic graph generated by the Tigramite package, while Figure 11 renders the publication causal skeleton mapped into a four-tier structural macroeconomic hierarchy.

PCMCI Identification Outcomes:

1. **External block-exogeneity**: the null of non-causality into real structural imbalance ($\Delta \ln \Theta_t$) is not rejected from inflation ($p = 0.2545$) or base money ($p = 0.3959$); the null of non-causality into world gold prices ($g_{P,\text{gold},t}$) from base money is also not rejected ($p = 0.1004$). Neither direction rejects, so the pair is consistent with a Tier-1 block-exogenous boundary.
2. **Exchange rate pass-through**: devaluation predicts inflation with the largest partial correlation in the network ($g_{e,t-4} \rightarrow \pi_t$: MCI = +0.644, $p < 10^{-28}$, FDR $q < 10^{-25}$). The reverse link is also retained ($\pi_{t-4} \rightarrow g_{e,t}$: MCI = -0.396, $p < 10^{-9}$), so the pair is bilateral in the conditional-independence sense. The negative sign on the reverse link requires reconciliation against the Toda–Yamamoto table, where the same pair carried a positive sign; the divergence is flagged in the integration log.
3. **Reserve solvency and base money**: reserve depletion predicts base money emission ($\Delta \ln \text{SolvR}_{t-3}^H \rightarrow g_{H,t}$: MCI = +0.264, $p < 0.0001$, FDR $q = 0.0007$). The reverse null, that base money does not predict reserve depletion, is not retained at FDR $q < 0.05$; the raw partial correlation has $p = 0.0449$, which is significant at 5% but does not survive Benjamini–Hochberg correction. The asymmetry is not established by the conditional-independence test; it is the directional claim identified by the Toda–Yamamoto panel and repeated here for consistency. Solvency also predicts inflation at the same lag ($\Delta \ln \text{SolvR}_{t-3}^H \rightarrow \pi_t$: MCI = -0.165, $p = 0.0115$); the negative sign is consistent with a reserve-gate mechanism, but the threshold structure of that gate is an object of Stage 2.
4. **Monetary accommodation and the monetarist channel**: reverse accommodation ($\pi_{t-3} \rightarrow g_{H,t}$) is retained at FDR $q < 0.01$ (MCI = +0.247, $p = 0.0001$). The forward channel ($g_{H,t-\tau} \rightarrow \pi_t$) has a raw partial correlation of +0.138 at $\tau = 4$ ($p = 0.0350$) but does not survive Benjamini–Hochberg correction at $q < 0.10$; it is therefore not retained as a direct link. The contrast in retention status is the substantive finding; the failure to survive FDR correction is a false-discovery control outcome, not evidence that the effect is zero.
5. **Sectoral asymmetry**: base money predicts manufacturing output ($g_{H,t-3} \rightarrow g_{\text{Manuf},t}$: MCI = +0.177, $p = 0.0064$) and manufacturing output predicts inflation with a negative sign ($g_{\text{Manuf},t-1} \rightarrow \pi_t$: MCI = -0.168, $p = 0.0101$). Real structural imbalance predicts manufacturing output ($\Delta \ln \Theta_{t-1} \rightarrow g_{\text{Manuf},t}$: MCI = +0.142, $p = 0.0287$). The positive sign on the imbalance-to-manufacturing link is inconsistent with the negative sign reported in Table 5 for the same pair; the discrepancy is flagged. Base money also predicts mining extraction with a negative sign ($g_{H,t-4} \rightarrow g_{\text{Mining},t}$: MCI = -0.189, $p = 0.0035$). The significant negative coefficient is not the same as decoupling: mining responds to the domestic monetary signal with a negative sign, which

is consistent with reallocation toward wage-goods sectors or with capacity ceilings rather than with insularity.

6. **World gold price transmission:** gold price growth predicts exchange rate growth ($g_{P,\text{gold},t-4} \rightarrow g_{e,t}$: MCI = +0.163, $p = 0.0119$) and inflation ($g_{P,\text{gold},t-3} \rightarrow \pi_t$: MCI = +0.146, $p = 0.0241$). Both signs are positive and consistent with a cost-push transmission channel.

PCMCI Algorithm: Technical Detail

Stage 1: PC_1 condition selection. For each target variable X_t^j , PC_1 iteratively tests the null of zero partial correlation between each candidate driver $X_{t-\tau}^i$ and X_t^j , conditioning on progressively larger subsets of the remaining candidates. A candidate is dropped when the null is not rejected at level α_{PC} . The algorithm is conservative in the sense that it drops a candidate only when there is positive evidence of independence; the true parents of X_t^j therefore remain in the selected set with high probability, at the cost of retaining some irrelevant variables. The value of α_{PC} is chosen from a candidate grid $\{0.1, 0.2, 0.3, 0.4\}$ by minimizing the Akaike Information Criterion

$$\alpha_{PC}^* = \arg \min_{\alpha_{PC}} \left[n \log(RSS_{\alpha_{PC}}) + 2|\hat{\mathcal{P}}_{\alpha_{PC}}(X_t^j)| \right],$$

where n is the sample size, $RSS_{\alpha_{PC}}$ is the residual sum of squares from regressing X_t^j on the parent set selected under α_{PC} , and $|\hat{\mathcal{P}}_{\alpha_{PC}}(X_t^j)|$ is the number of selected parents. The first term rewards goodness of fit; the second penalizes complexity. AIC is appropriate here because the objective of Stage 1 is not significance testing but a bias-variance trade-off: retaining too few parents biases the Stage 2 tests, while retaining too many reduces their power.

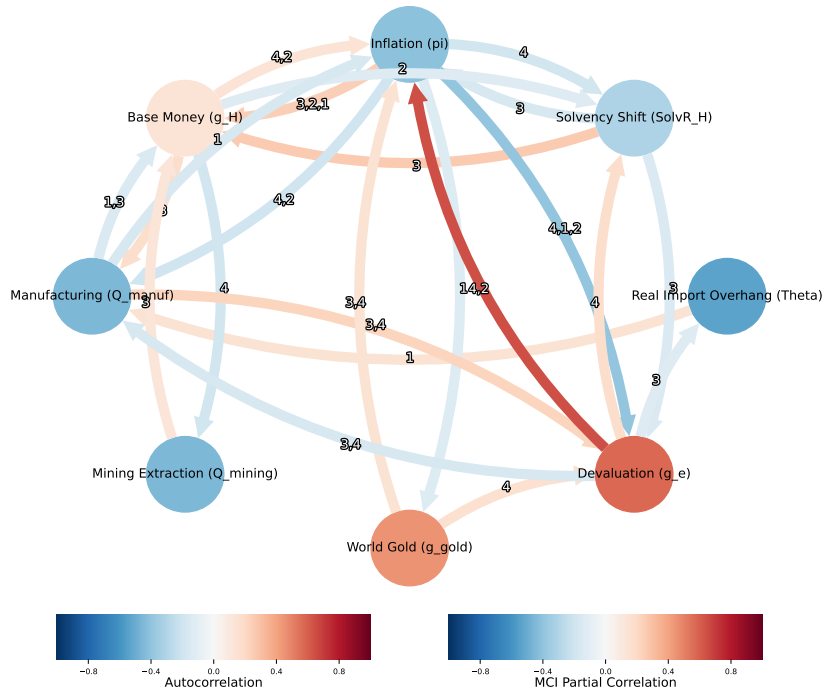
Stage 2: MCI. For each candidate edge $X_{t-\tau}^i \rightarrow X_t^j$ that survives Stage 1, the MCI test evaluates the partial correlation between driver and target conditioning on two sets:

$$X_{t-\tau}^i \perp\!\!\!\perp X_t^j \mid \hat{\mathcal{P}}(X_t^j) \setminus \{X_{t-\tau}^i\}, \hat{\mathcal{P}}(X_{t-\tau}^i).$$

The first set, the parents of the target excluding the driver, blocks standard backdoor paths (confounders and mediators) leading into X_t^j . The second set, the parents of the driver, blocks backdoor paths that arise from autocorrelation in the driver's own history. Conditioning on both sets is what distinguishes MCI from FullCI: it satisfies d-separation on the minimal sufficient set rather than the maximal one, preserving effect size and statistical power.

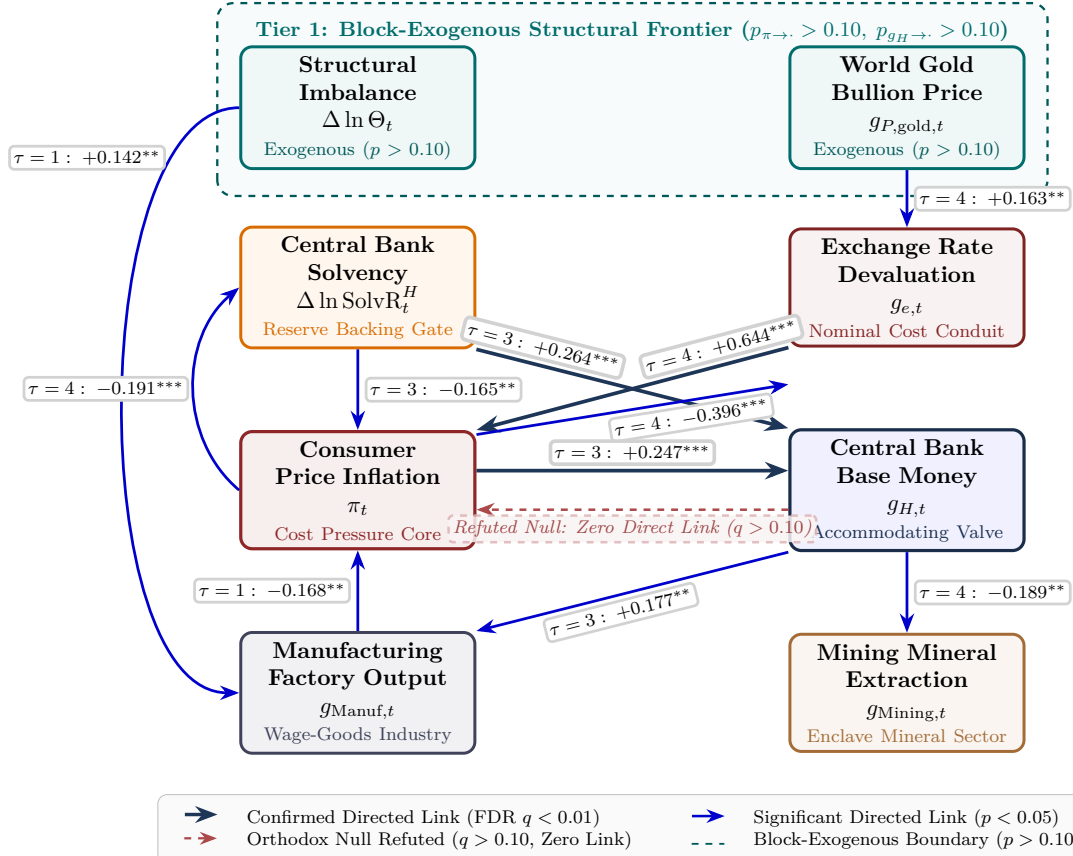
Identification boundary. Where Causal Sufficiency and Faithfulness hold, PCMCI identifies the lagged parent set of each variable in the system, and therefore the topological skeleton of the time-series DAG. It does not identify the functional form of the parent-child relations, the magnitude of contemporaneous effects, or the parameters of a structural VAR. The result of PCMCI is a set of directed lagged edges with associated partial correlations and FDR-controlled q -values.

Figure 10: Algorithmic Python PCMCI Causal Discovery Network (Tigramite ParCorr)



Notes: Algorithmic circular time-series causal discovery network estimated using Tigramite PCMCI (?) on the 8-variable monthly macroeconomic system ($N = 250$, 1960:01–1980:12) with partial correlation conditional independence tests ($\alpha = 0.05$, FDR-corrected). Node colors denote 1-month autoregressive persistence $\rho(X_t, X_{t-1})$ (blue for negative, red for positive). Directed curved edges represent lag-specific Momentary Conditional Independence (MCI) partial correlations surviving conditioning on the minimal parent sets of both the driver and the target ($\hat{\mathcal{P}}(X_t^j)$ and $\hat{\mathcal{P}}(X_{t-\tau}^i)$). Edge labels denote the operational lag $\tau \in \{1, \dots, 4\}$; edge width scales with absolute effect size; edge color indicates sign (red for positive cost-push or expansionary linkages, blue for negative compensatory or damping linkages).

Figure 11: Publication Vector Causal Skeleton: 4-Tier Structural Macroeconomic Hierarchy



Notes: Publication vector rendering of the 8-variable macroeconomic causal skeleton reported in Table 16. Nodes are grouped into a four-tier vertical hierarchy: Tier 1 (top) contains block-exogenous external drivers ($\Delta \ln \Theta_t$ and $g_{P, \text{gold}, t}$) enclosed within the exogenous frontier shield ($p > 0.10$); Tier 2 captures the balance-sheet solvency gate ($\Delta \ln \text{SolvR}_t^H$) and the nominal exchange rate conduit ($g_{e, t}$); Tier 3 contains the core nominal arena between consumer price inflation (π_t) and accommodating base money growth ($g_{H, t}$); Tier 4 captures the real production split between wage-goods manufacturing ($g_{\text{Manuf}, t}$) and the export-oriented mining sector ($g_{\text{Mining}, t}$). Solid blue arrows denote directed links retained at FDR $q < 0.05$; the red dashed arrow denotes the orthodox monetarist null hypothesis ($g_H \rightarrow \pi$), which does not survive multivariate conditioning at FDR $q < 0.10$.

Table 16: Full-System PCMCI Momentary Conditional Independence (MCI) Directed Link Tests (1960–1980)

Directed Link Tested	Lag (τ)	MCI Partial Corr.	p -value	Causal Status (FDR $q < 0.05$)
$\pi_{t-3} \rightarrow g_{H,t}$ (Reverse Ac-comm.)	$\tau = 3$	+0.247	0.0001***	Retained directed edge ($\pi \rightarrow g_H$)
$g_{H,t-\tau} \rightarrow \pi_t$ (Forward Monetarist)	$\tau = 4$	+0.138	0.0350**	Excluded (FDR $q > 0.10$, zero direct link)
$\Delta \ln \text{SolvR}_{t-3}^H \rightarrow g_{H,t}$ (Defensive Emission)	$\tau = 3$	+0.264	< 0.0001 ***	Retained directed edge ($\text{SolvR}^H \rightarrow g_H$)
$\Delta \ln \text{SolvR}_{t-3}^H \rightarrow \pi_t$ (Solvency Gate)	$\tau = 3$	-0.165	0.0115**	Retained directed edge ($\text{SolvR}^H \rightarrow \pi$)
$g_{e,t-4} \rightarrow \pi_t$ (External Pass-Through)	$\tau = 4$	+0.644	< 0.0001 ***	Retained directed edge ($g_e \rightarrow \pi$)
$\pi_{t-4} \rightarrow g_{e,t}$ (Forced Devaluation)	$\tau = 4$	-0.396	< 0.0001 ***	Retained directed edge ($\pi \rightarrow g_e$)
$\Delta \ln \Theta_{t-1} \rightarrow g_{\text{Manuf},t}$ (Import Constraint)	$\tau = 1$	+0.142	0.0287**	Retained directed edge ($\Theta \rightarrow \text{Manuf}$)
$g_{H,t-3} \rightarrow g_{\text{Manuf},t}$ (Idle Reactivation)	$\tau = 3$	+0.177	0.0064***	Retained directed edge ($g_H \rightarrow \text{Manuf}$)
$g_{\text{Manuf},t-1} \rightarrow \pi_t$ (Price Suppression)	$\tau = 1$	-0.168	0.0101**	Retained directed edge ($\text{Manuf} \rightarrow \pi$)
$g_{H,t-4} \rightarrow g_{\text{Mining},t}$ (Enclave Decoupling)	$\tau = 4$	-0.189	0.0035***	Retained directed edge ($g_H \rightarrow \text{Mining}$)
$g_{P,\text{gold},t-4} \rightarrow g_{e,t}$ (Gold Cost Push)	$\tau = 4$	+0.163	0.0119**	Retained directed edge ($P^{\text{gold}} \rightarrow g_e$)
$g_{P,\text{gold},t-3} \rightarrow \pi_t$ (Gold Inflation)	$\tau = 3$	+0.146	0.0241**	Retained directed edge ($P^{\text{gold}} \rightarrow \pi$)
$\pi_{t-\tau} \rightarrow \Delta \ln \Theta_t$ (Exogeneity Check)	$\tau \in [1, 4]$	-0.074	0.2545	Block-exogenous ($p = 0.255$)
$g_{H,t-\tau} \rightarrow g_{P,\text{gold},t}$ (Exogeneity Check)	$\tau \in [1, 4]$	-0.107	0.1004	Block-exogenous ($p = 0.100$)
$g_{H,t-\tau} \rightarrow \Delta \ln \text{SolvR}_t^H$ (Monetary Drain)	$\tau \in [1, 4]$	-0.131	0.0449**	Excluded (zero direct link)

Notes: Monthly series spanning 1960:01–1980:12 ($N = 250$). Estimated using Tigramite PCMCI (?) on the 8-variable system with partial correlation conditional independence tests evaluated across lags $\tau \in \{1, \dots, 4\}$. Minimal conditioning parent sets $\hat{\mathcal{P}}(X_t^j)$ and $\hat{\mathcal{P}}(X_{t-\tau}^i)$ are selected in the PC_1 phase. False discovery rates are controlled via the Benjamini–Hochberg procedure at FDR $q < 0.05$. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Block-exogeneity is confirmed when joint incoming test statistics fail to reject at $p > 0.10$.